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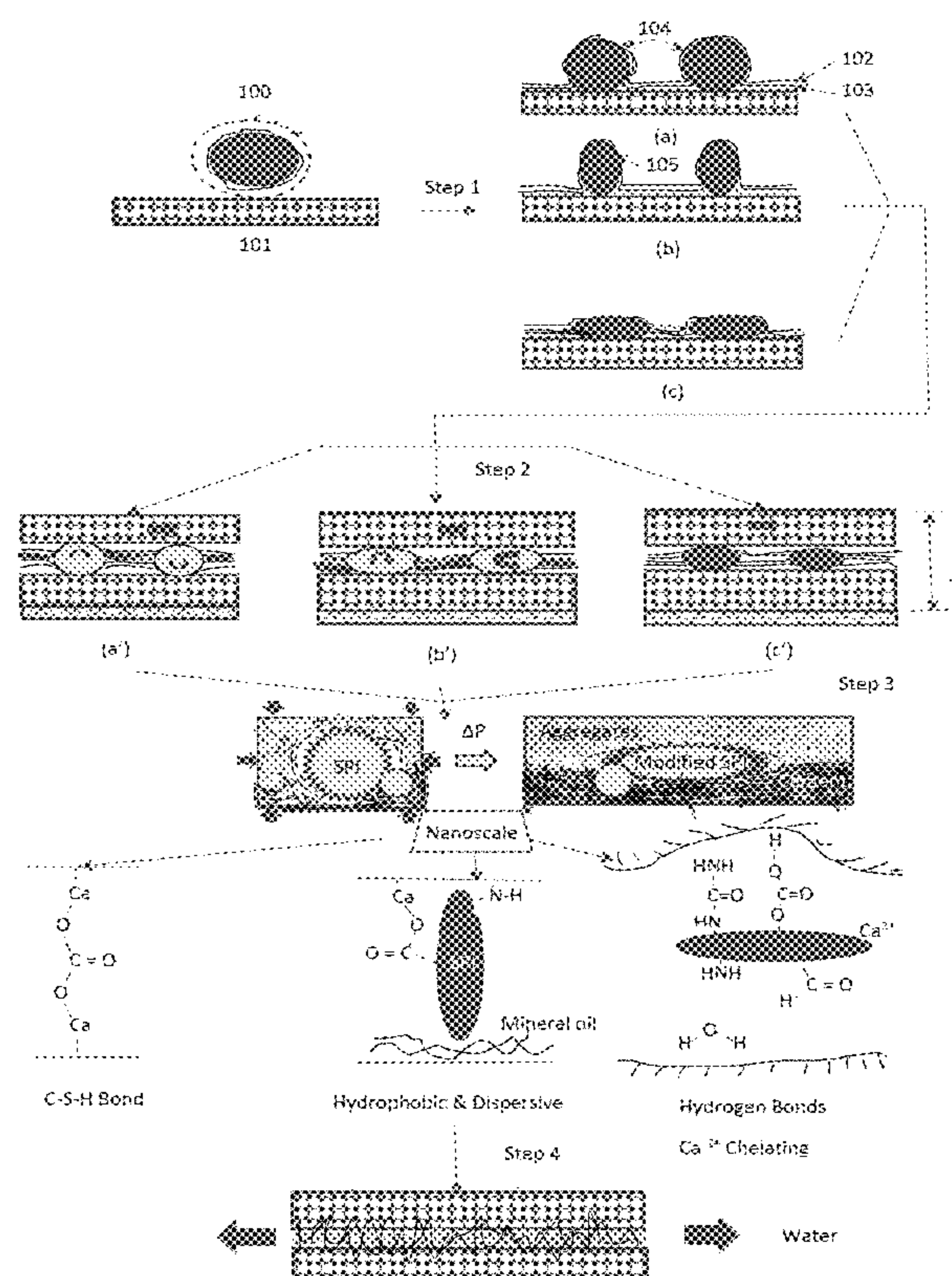
(19) **United States**(12) **Patent Application Publication**
LIU et al.(10) **Pub. No.: US 2022/0064521 A1**(43) **Pub. Date: Mar. 3, 2022**(54) **BIOINSPIRED CHEMICAL ADDITIVES AND SOLUTION USEFUL FOR HYDRAULIC-CEMENT CONCRETES AND THE METHODS FOR MAKING THE SAME THEREOF**(71) Applicants: **FEIPENG LIU**, Spring, TX (US);
Yuning Lai, Spring, TX (US)(72) Inventors: **FEIPENG LIU**, Spring, TX (US);
Yuning Lai, Spring, TX (US)(21) Appl. No.: **17/492,565**(22) Filed: **Oct. 2, 2021****Related U.S. Application Data**

(63) Continuation of application No. 17/190,533, filed on Mar. 30, 2021, which is a continuation of application No. 16/600,278, filed on Oct. 11, 2019, Continuation of application No. 17/434,968, filed as application No. PCT/US2021/027426 on Apr. 15, 2021.

(60) Provisional application No. 63/179,634, filed on Apr. 26, 2021.

Publication Classification(51) **Int. Cl.**
C09K 8/80 (2006.01)(52) **U.S. Cl.**
CPC **C09K 8/805** (2013.01); **C09D 133/08** (2013.01)(57) **ABSTRACT**

Bioinspired chemical additives, coating, or/and solution, in mimicking biomineralization processes of bone, useful for enhancing the strength of hydraulic-cement concrete and mitigating the risk of its cracking failure, comprising of micro/nano/textured dual phobic dot domains as core layer, hydrogel polymer as shell, prepared by water and mineral oil as solvent via surfactant/emulsifier as intermediate layer encapsulated in an emulsion, were mixed with cement, fine sand, and aggregates by weight percentage at a mix ratio of from 0.00001/99.99999 to 10.0/90, of which the ratio of water to cement is ranged from 0.2 to 0.80 (W/C), the volume fraction of cement accounted for total volume fraction of solid from 5 to 50%, the fine sand from 40% to 90%, and aggregate from 40% to 90% as water dried, a replacement of cement with cementitious materials such as microcrystalline silica sand, micro gel, and swollen clays ranged from 0.01% to 75% over the cement by weight percentage, casted into concrete blocks having an early age of compressive strength of more than 4000 (PSI) within 24 hour, Brazilian splitting tensile strength of more than 1000 (psi) at 28 days, ultimate compressive strength of more than 7500 (PSI) after exposed at the ambient conditions for over one and half year, an increased toughness of more than 900 (%), modulus of resilience of more than 1300 (%), self-healing capability of more than 80(%) by pre-cracking test method, density ranged from 1.90 to 2.55 (g/cm³), moisture content loss of less than 3.0%, resulting from the contributions of the tunable self-assembly of dispersive and non-covalent bonds in response to the exothermic hydration of micro/nano/textured hydro dual phobic domains of proteins, wax, and hydrogel polymers with mixed concrete components from 30 to 200° F.



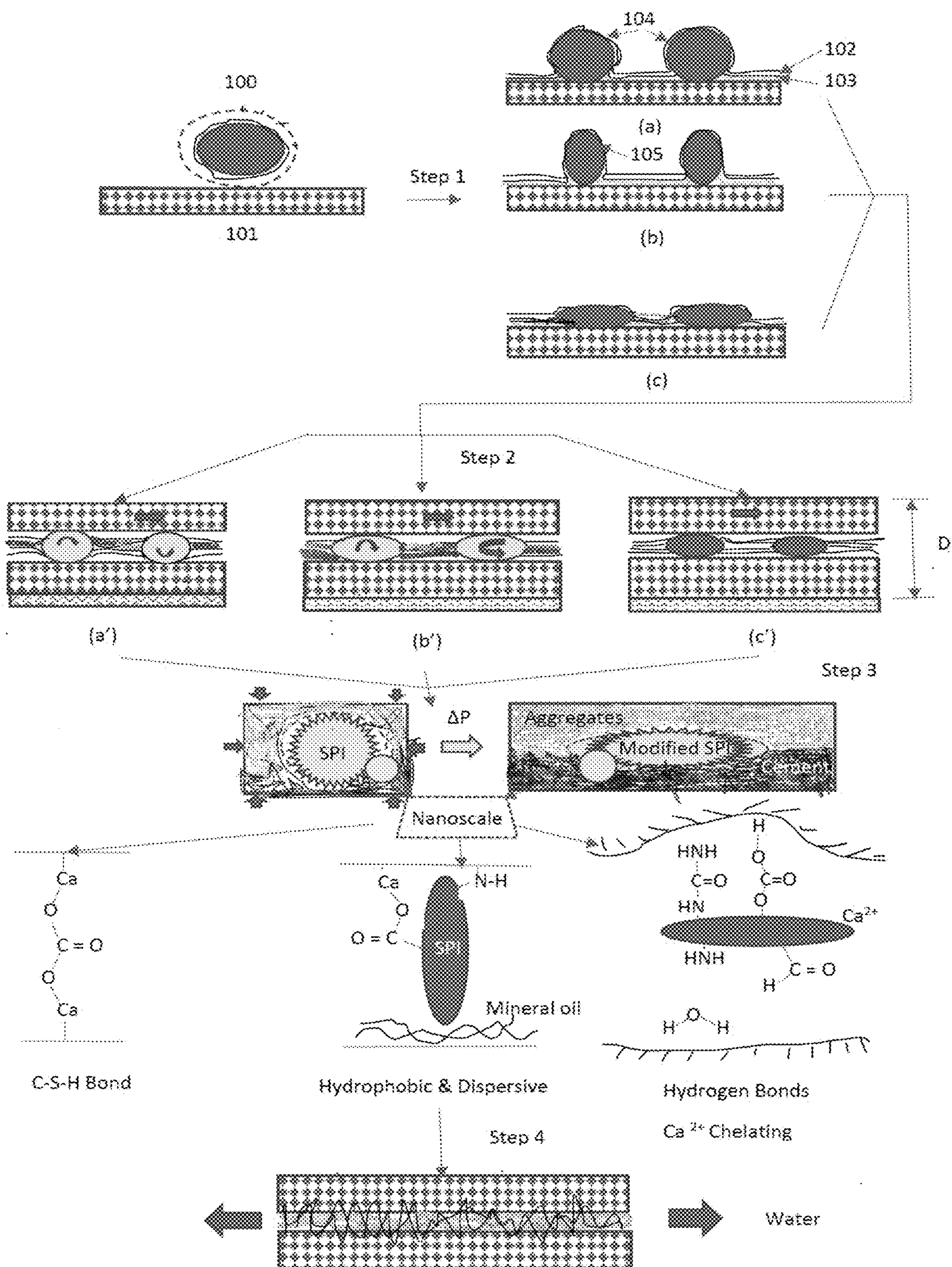


Figure 1

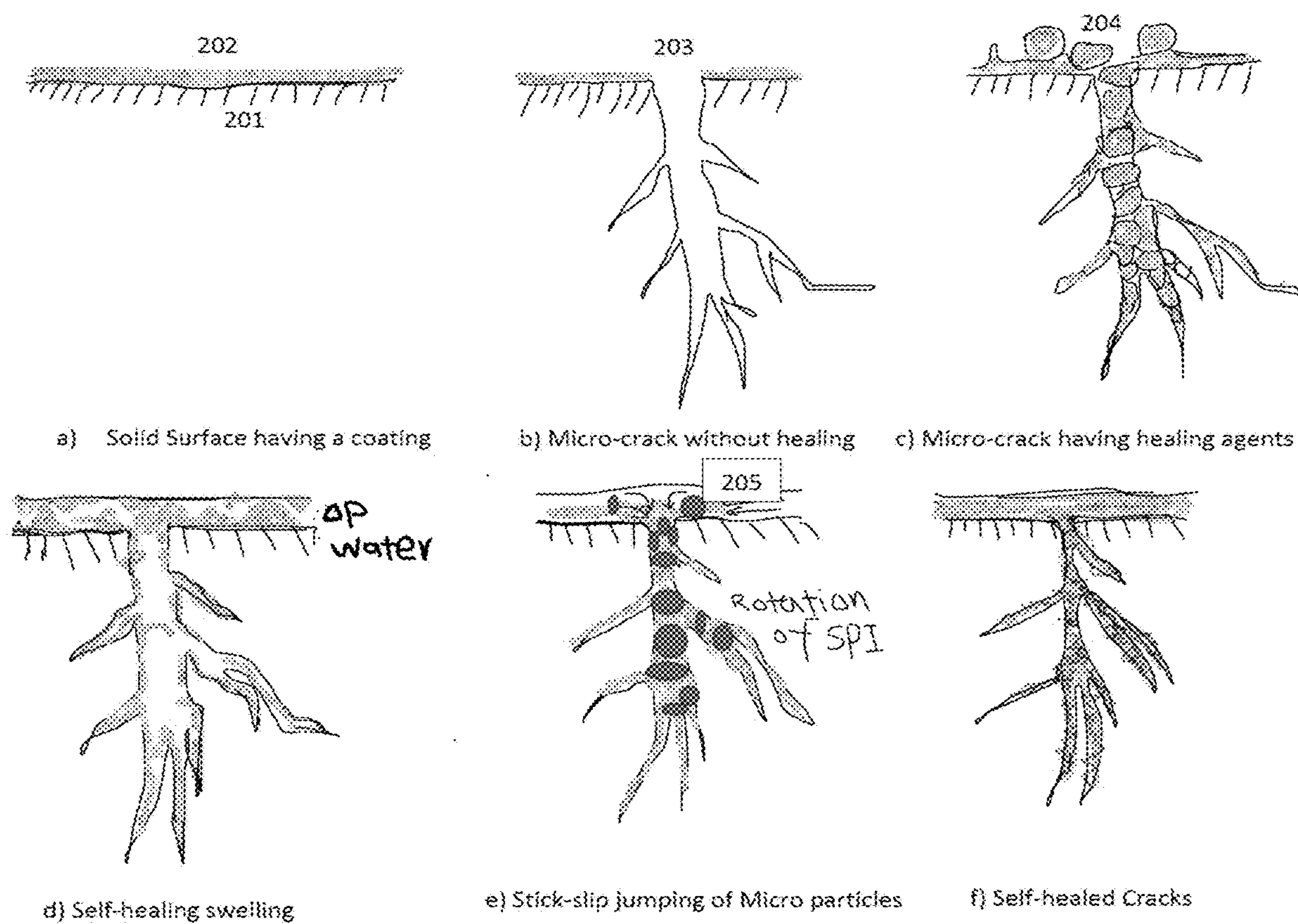


Figure 2

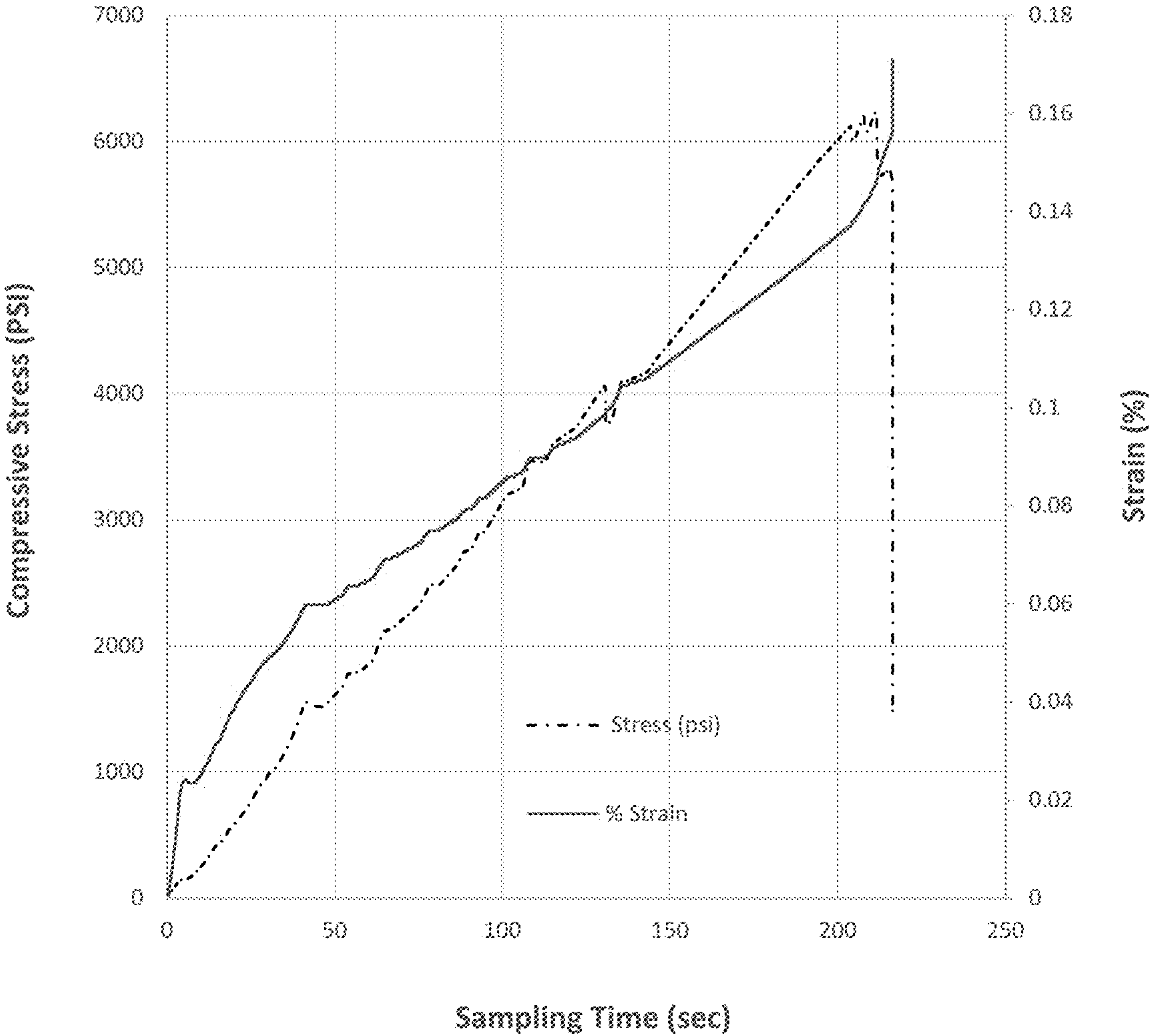


Figure 3

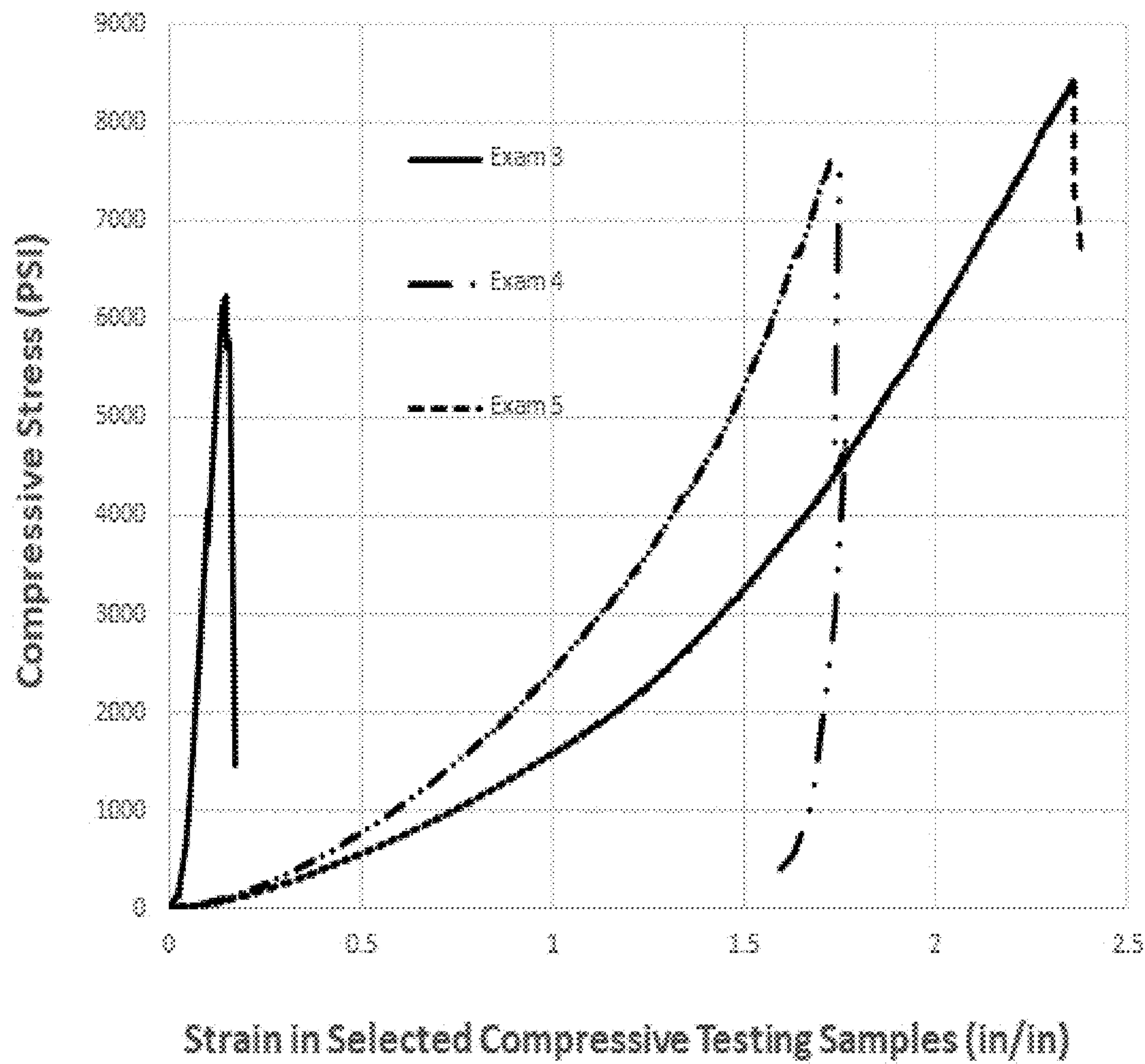


Figure 4

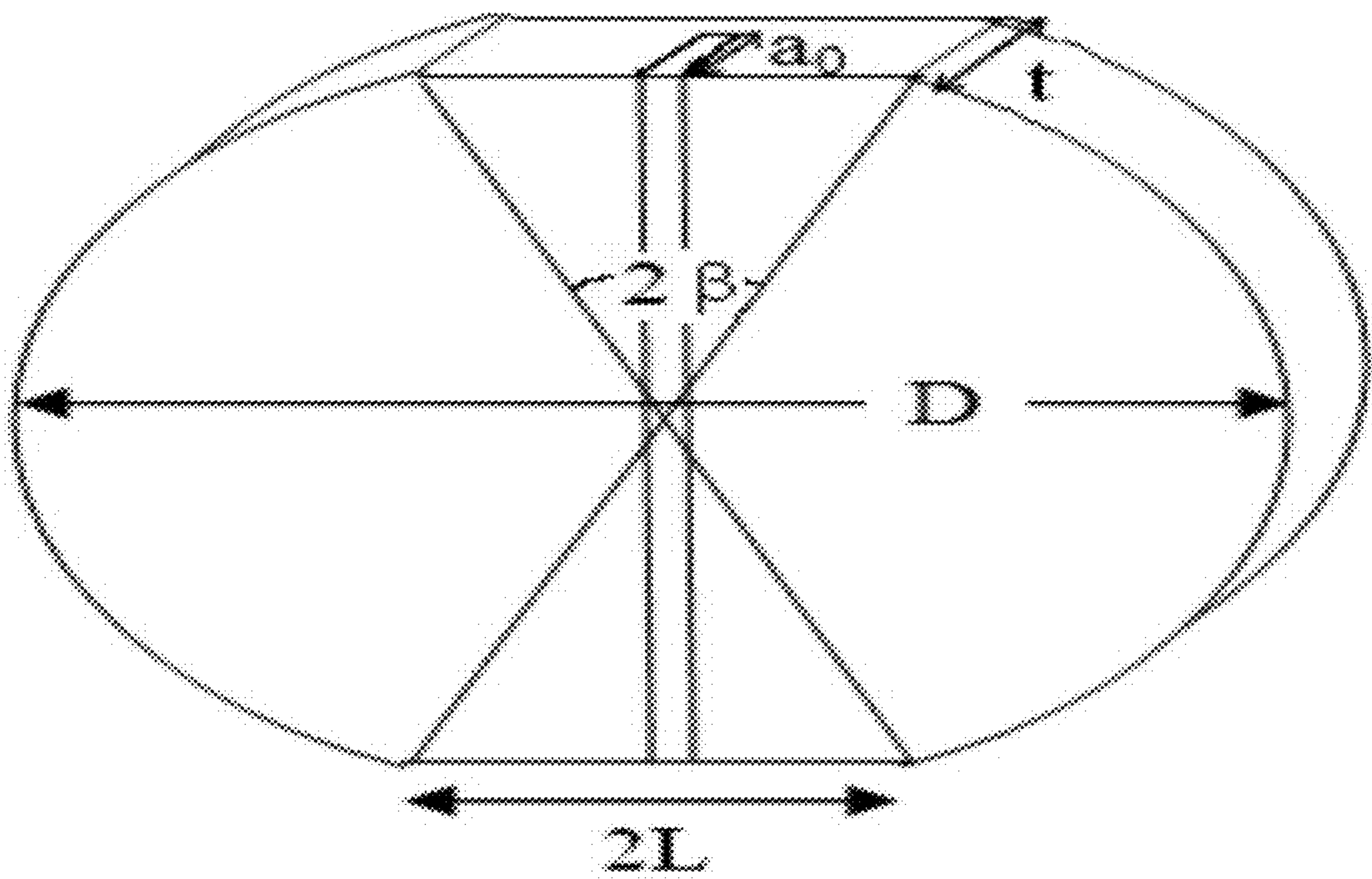


Figure 5.

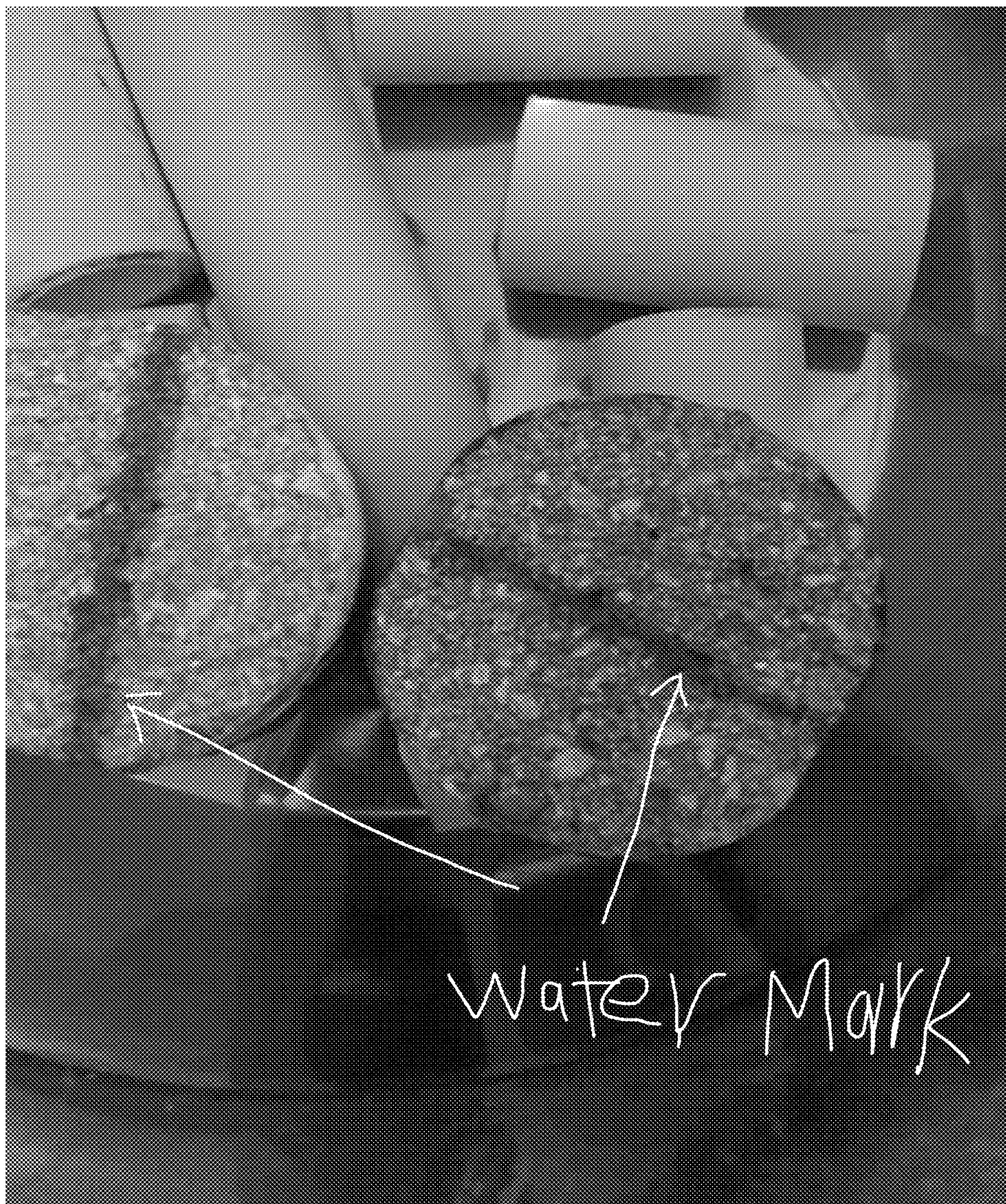


Figure 6

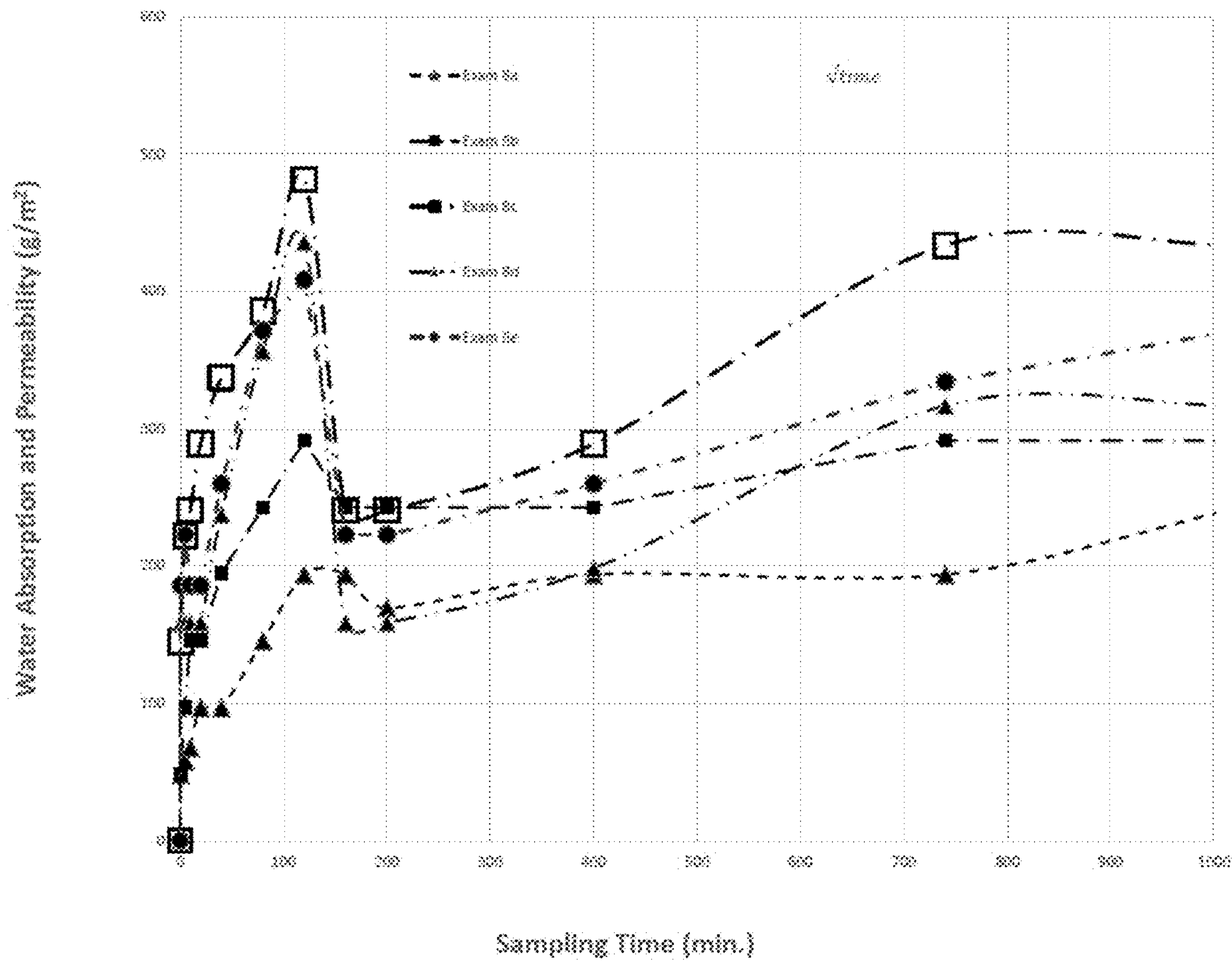


Figure 7

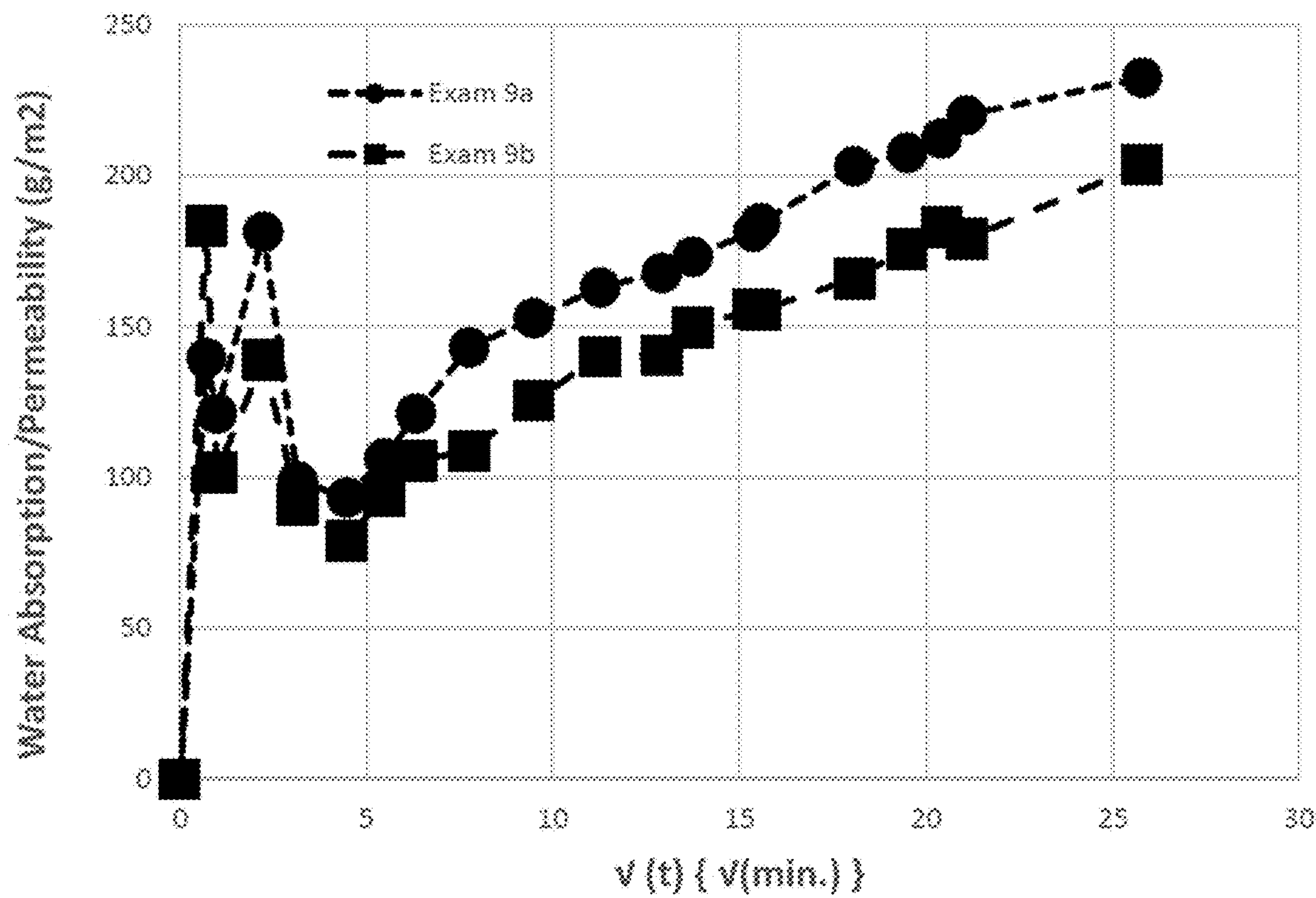


Figure 8



Figure 9

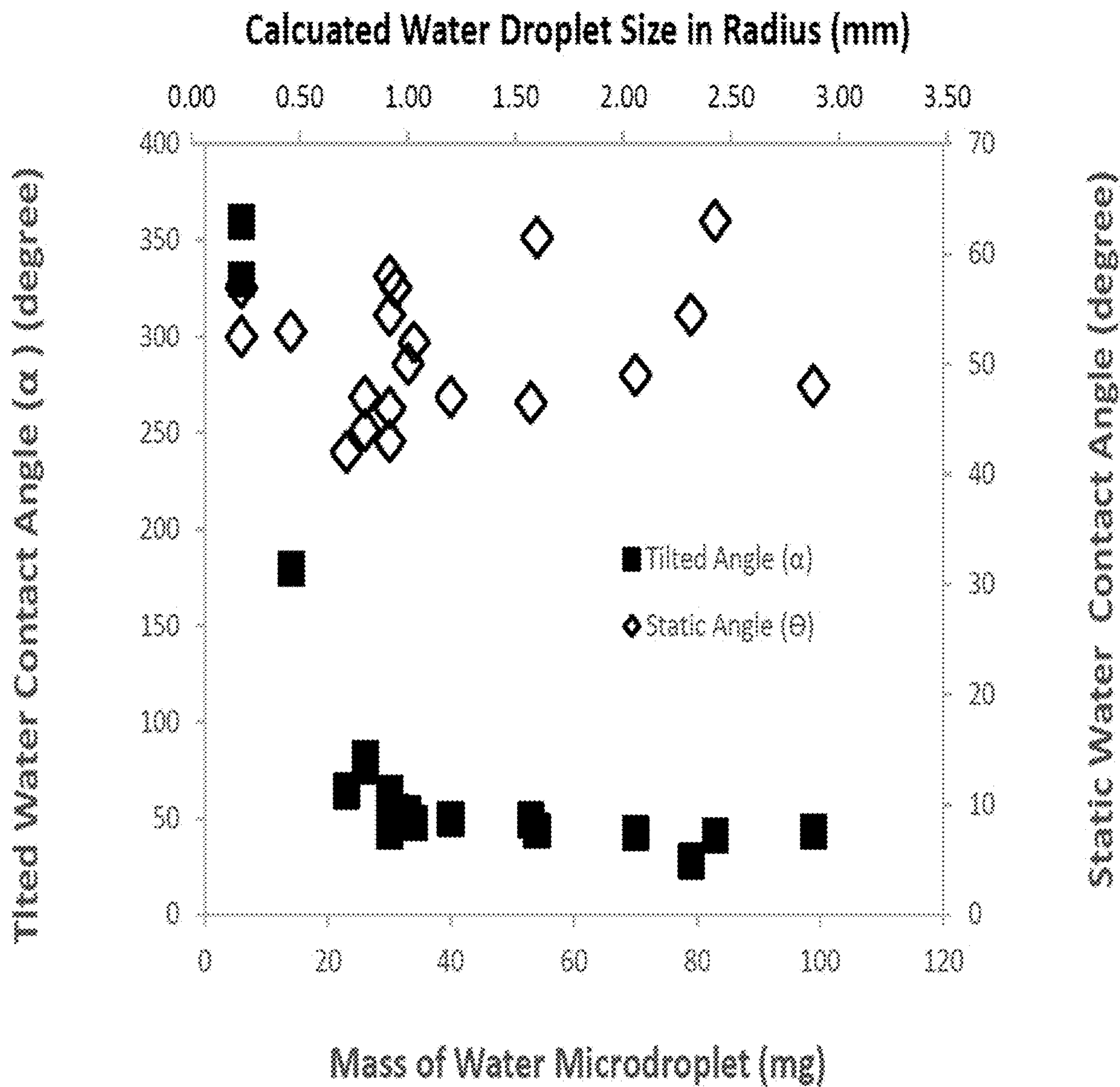


Figure 10.

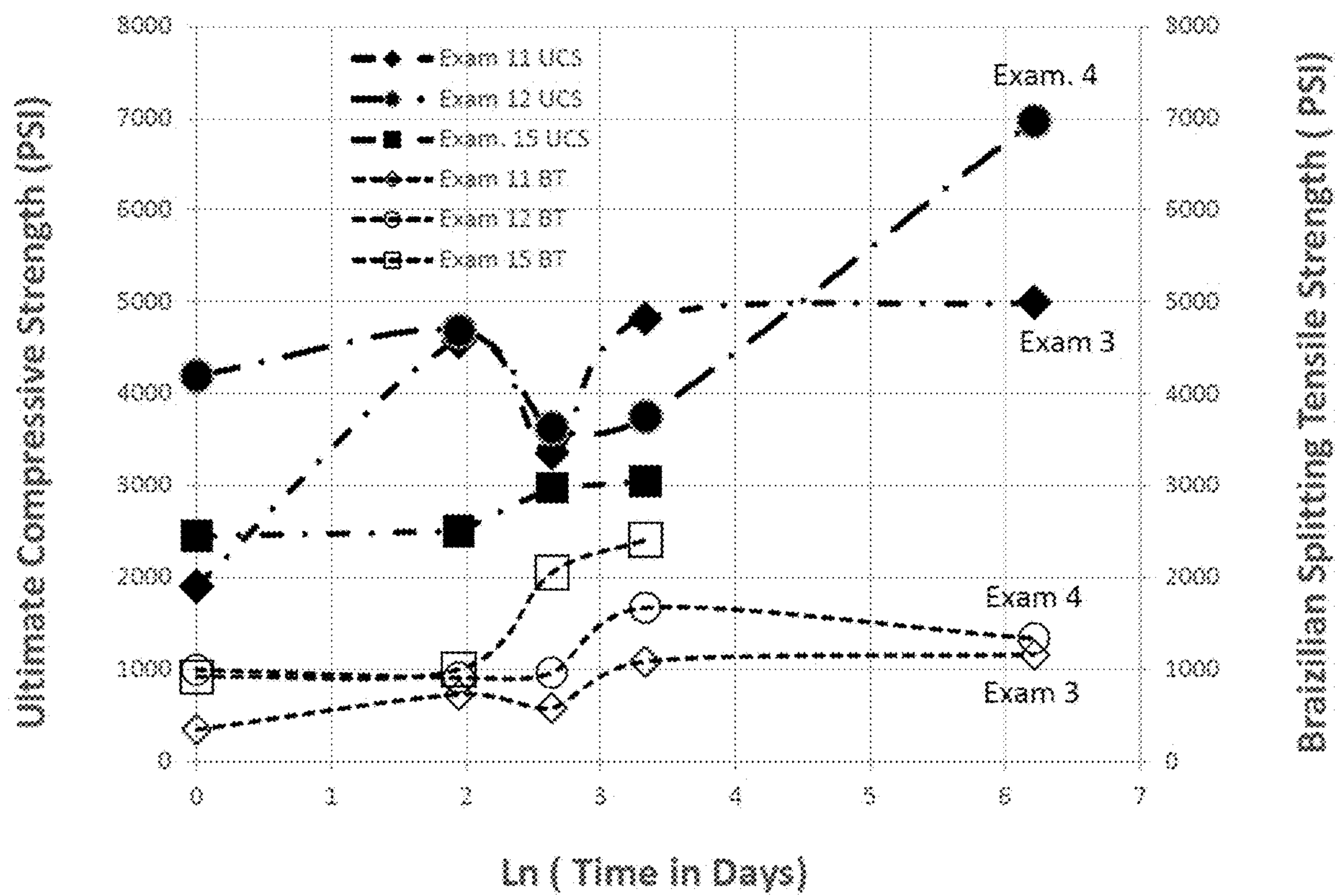


Figure 11

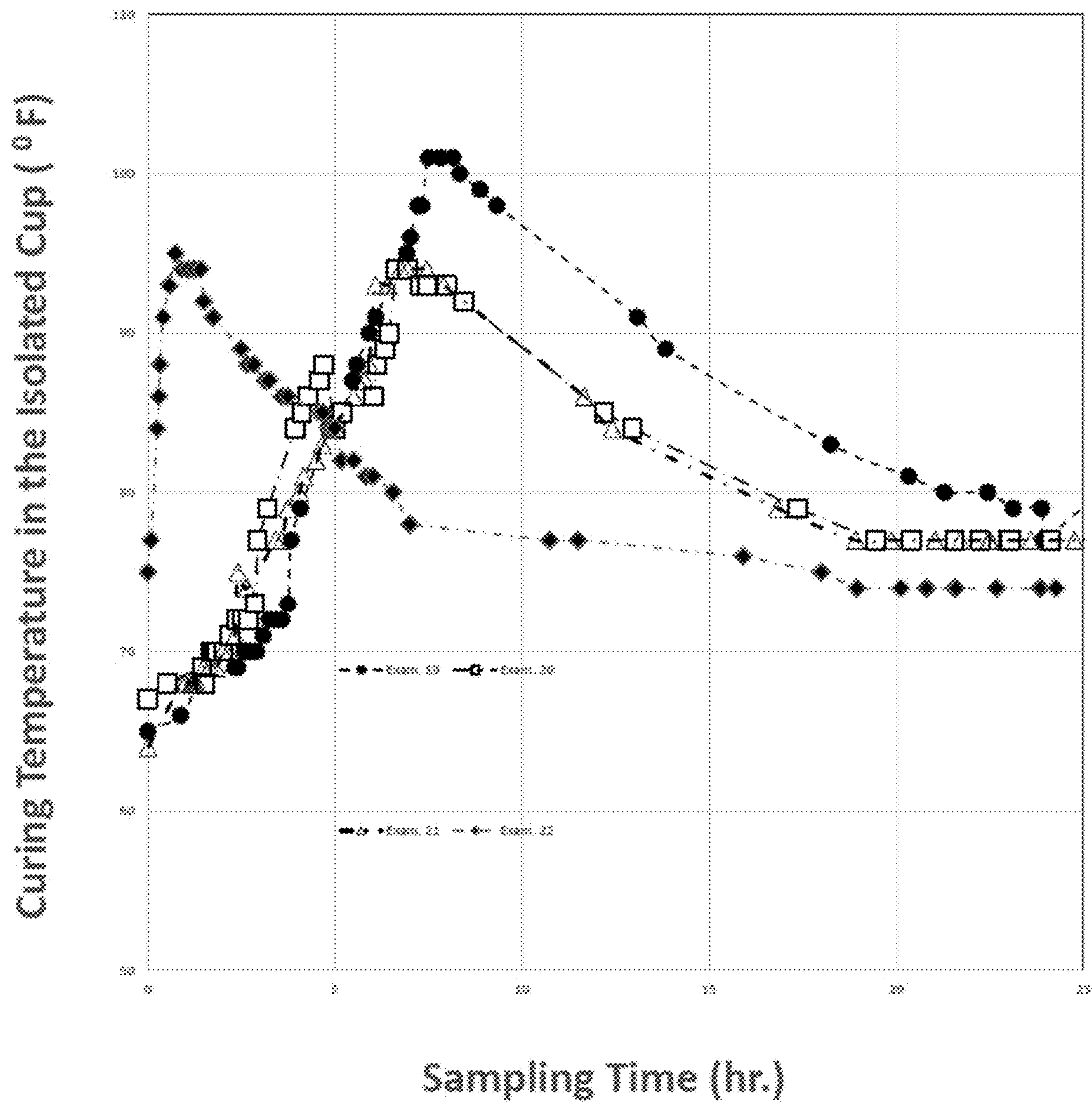


Figure 12

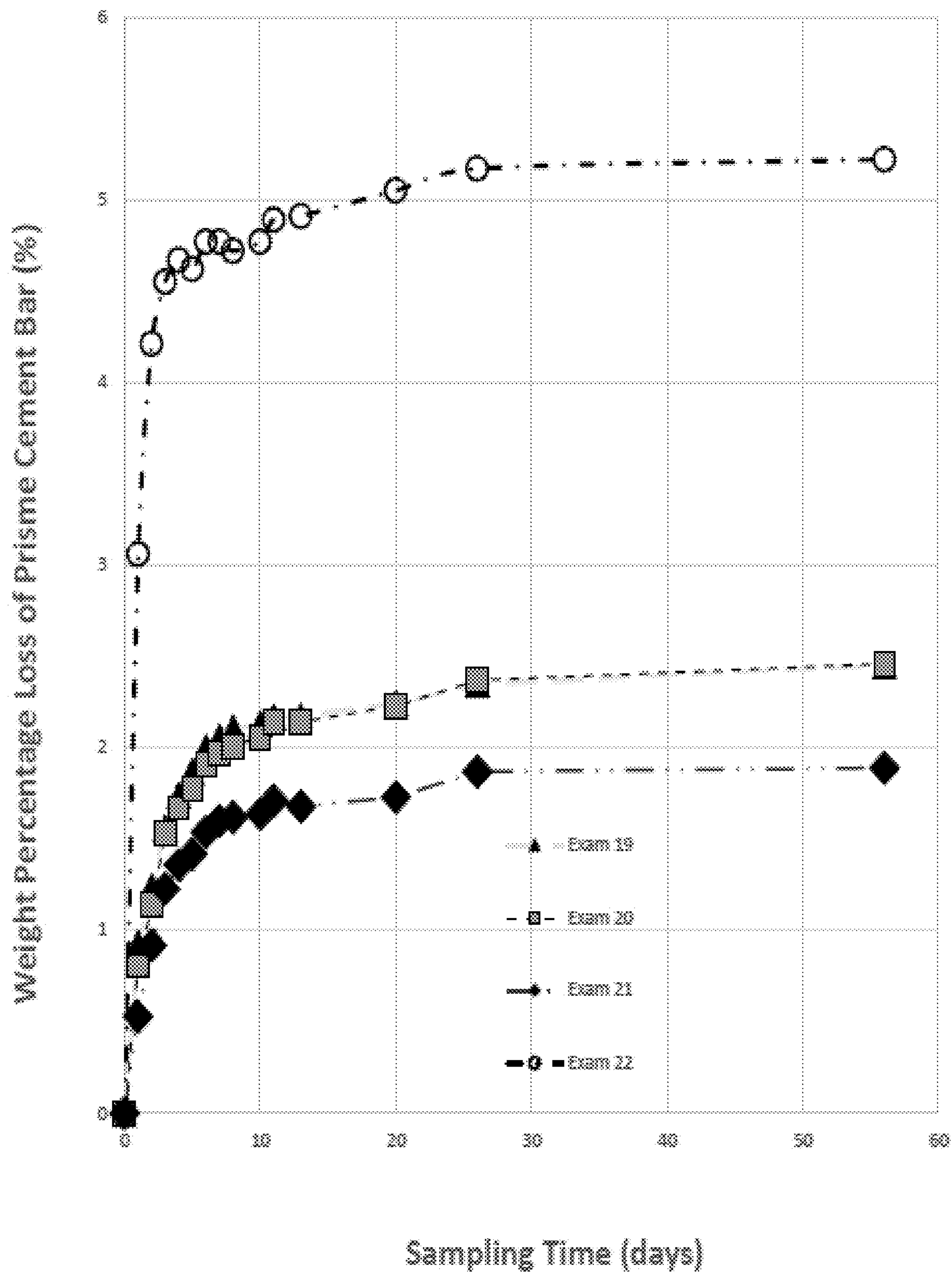


Figure 13.

**BIOINSPIRED CHEMICAL ADDITIVES AND
SOLUTION USEFUL FOR
HYDRAULIC-CEMENT CONCRETES AND
THE METHODS FOR MAKING THE SAME
THEREOF**

CROSS REFERENCE TO RELATED
APPLICATION

[0001] This application is a continuation of previous U.S. provincial application of 63/179,634 filed on Apr. 26, 2019, which is hereby incorporated by reference in its entirety.

REFERENCE AND LITERATURE CITED

- [0002] Aizenberg, J., P. Kim, G. K. Paink, 2017, U.S. 2017/0088472, Robust cementitious materials with mobile liquid-infused barrier layer.
- [0003] Almabrok, M. H., R. G. McLaughlin, K. Vessalas, P. Thomas, 2019, Effect of oil contaminated aggregates on cement hydration. *American Journal of Engineering Research*. 8(5): pp 81-91.
- [0004] Berke, N. S., S. G. Eugene, R. J. Elliot, U.S. Pat. No. 8,784,558, Admixtures for shrink crack reduction of Portland cement-based mortars and concretes.
- [0005] Biswas M., S. Majumdar, T. Chowdhury, B. Chattopadhyay, 2010, Bioremediate a unique protein from a novel bacterium BHH1, ushering a new hope in concrete technology. *Enzyme and Microbial Technology*, 46 (381-587)
- [0006] Boesel, F. L., H. S. Azevedo, and R. L. Reis, 2006, Incorporation of α -amylase enzyme and bio-active fillers into hydrophilic partially degradable, and bioactive cements (HDBC)s as a new approach to tailor simultaneous) their degradation and bioactive behavior, *Biomacromolecules*, 7, pp: 2600-2609.
- [0007] Chen, J., H. Park, and K. Park, 1999, Synthesis of super porous hydrogels: hydrogels with fast swelling and superabsorbent properties.
- [0008] Dang, J., J. Zhao, and Z. Du, 2017, Effects of superabsorbent polymer on the properties of concrete, *Polymers*, 9, <https://www.mdpi.com/1999-4908/9/12/2072>.
- [0009] Gardner D., A. Jefferson, A. Hoffman, R. Lark, 2014, Simulation of the capillary flow of an automatic holding agent in discrete cracks in cementitious materials. *Cement and Concrete Research*, 59(35-44).
- [0010] Henkel, J., M. A. Woodruff, D. R. Epari, R. Steck, V. Glatt, L. C., Dickinson, P. F. M. Choong, N. A. Schuetz, D. W. Hutmacher, 2013, A Review: Bone Regeneration based on tissue engineering conceptions—A 21st Century Perspective. *Bone Research*, 3: 216-248.
- [0011] Kamali, M. and Ali Ghahremaninezhad, 2018. Effect of biomolecules on the nanostructure and nanomechanical property of calcium silicate hydrate, *Scientific Reports*. Published online 22, June.
- [0012] Karmada, Ayaka, M. Rodriguez-Garcia, F. S. Ruggeri, Y. Shen, A., L. & T. P. J. Knowles, 2021, Controlled self-assembly of plant proteins into high-performance multifunctional nanostructured films. *Nature Communications*, 12(3529), pp: 1-9.
- [0013] Kyoshi, M. et al, U.S. Patent Application 2002/0055558, filed on May 9, 2002, Permeable waterproofing agent, waterproofing material, concrete or mortar composition, and method for preventing water leaks.
- [0014] Lan, Q., L. Li, H. Dong, D. Wu, H. Chen, D. Lin, W. Qin, W. Yang, T. Vasathan, and Q. Zhang, 2019, Effect of Soybean soluble polysaccharide on the formation of glucono- δ -Lactone-induced soybean protein isolate gel. *Polymers*, 11(1997) pp: 1-15.
- [0015] Li, C. V, and E. H. Yang, 2007, Self-healing in Concrete, in *Self-healing Materials: An alternative approach to 20 centuries of material science*, Spring Series in Materials Science, 100 edited by Irisybrand Van Der Zwaag.
- [0016] Li, W., J. Wu., W. J., H. Y., Zheng, Y, N Zhao, Z. Y. Wei, 2013, Self-healing efficiency of cementitious materials containing microcapsules filled with healing adhesive: mechanical restoration and healing process monitored by water absorption. <http://www.plosone.org>, Nov., vol 8, e81616.
- [0017] Mdsalih, N. U. Hashim, N. Nafarizal, C. F. Soon, M. Sandan, 2012, Surface tension analysis of cost-effective paraffin wax and water flow simulation for microfluidic device. *Advanced Materials Research*, ISSN 1662-8985, Vol. 832, pp 773-777. Online 2013-11-21.
- [0018] Moehle, J., 2019, Introducing ACI 318-19: Building code requirements for structural concrete.
- [0019] Ma, H., E. Herbert, M. Ohno, V. C. Li, 2019, Scale-linking model of self-healing and stiffness recovery in engineered cementitious composites (ECC). *Cement and Concrete Composites*. 95:1-9.
- [0020] Poole, L. J., K. A. Riding, K. J. Folliard, M. C. G. Juenger, and A. K. Schindler, 2007, Methods for calculating activation energy for Portland cement, *ACI Materials Journal/January-February* 303-310.
- [0021] Shim, Y., G. Hong and S. Choi, 2018, Autogenous healing of early-age cementitious materials incorporating superabsorbent polymers exposed to wet/dry cycles. *Materials*, 11: 2476, doi: 10.3390/ma111122476.
- [0022] Shingh, R. N. and A. G. Pathan, 1988, Fracture toughness of some British rocks by diametral loading of discs, vol. 6, pp: 179-190.
- [0023] Rodriguez V. J. F., N. E. Villareal, L. H. Veraslegui, A. M. A. Tovar, J. F. Lopez-Perrales, J. E. Contrexas De., Leon, C. G. Rodriguez. D. F. Gonzalez, L. F. Verdeja, L. V. Garcia-Quilhomez, and E. A. R. Castellanon. 2020, Effect of mineral aggregates and chemical admixtures as internal curing agents on the mechanical properties and durability of high-performance concrete. *Materials*, 23(2090).
- [0024] Rosewitz, J. A., S. Wang, S. F. Scarlata, N. Rahbar, 2021, An enzymatic self-healing cementitious material. *Applied Materials Today*, 23 (101015)
- [0025] Shaheen, N., R. A. Khushnood, and S. U. Din, 2018, Bio immobilized limestone powder for autonomous healing of cementitious systems: A feasibility study, *Advances in Materials Science and Engineering*, Vol: 2018 (<https://doi.org/10.1150/2018/7049121>)
- [0026] Hirschler, B., 2007, Plant goo waves insect's goodbye, *ABC Sciences, News in Science*, (<https://www.abc.net.au/science/articles/2007/1121207145.htm>).
- [0027] Tae-H. & T. Kishn, 2010, Cementitious composites incorporating various mineral admixtures. *J. of Advanced Concrete Technology*, Vol. 8(2), 171-186.
- [0028] Wu, Q., T. Becherer, S. Angioletti-Uberti, J. Dzubbiella, C. Wischke, A. T. Neffe, A. Lendlein, M. Ballauff, R., 2014, Huang, Protein interactions with polymer coatings and biomaterials. *Angew. Chem. Int. Ed*, 8004-8091.

[0029] Wiktor, V., H. M. Jonkers, 2011, Quantification of crack-healing in novel bacteria-based self-healing concrete. *Cement and Concrete Composites* 33(763-776).

[0030] Wu, C., Y. Hua, Y. Cha, X. K. Wang, and C. Zhang, 2016, Effect of 7s/11s ratio on the network structure of heat-induced soy protein gels: A study of probe release. *RSC Advances* Issue 104,10 pp: 1981-1987.

[0031] Wu, S., G. Liu, Q. Liu, P. Liu, J. Yang, 2020, Sustainable High-ductility concrete with rapid self-healing characteristic by adding Magnesium oxide and superabsorbent polymer. *Advances in Materials Science and Engineering*, Volume 2020, ID5395602, <https://doi.org/10.1155/2020/5395602>.

[0032] Yang, Y., M. D. Lepech, E. H. Yang, V. C. Li, 2009, Autogenous healing of engineered cementitious composites under wet-dry cycles. *Cement and Concrete Research* 39: 382-390.

[0033] Yang, F., B. Zhang, Q. Ma, 2010, Study of sticky rice-lime mortar technology for the restoration of Historical masonry construction. *Accounts of Chemical Research*, ACS. 10.1021/ar9001944.

[0034] Yang, L., D. Gao, Y. Zhang, J. Tang, and Y. Li, 2019, Relationship between sorptivity for water absorption of cement-based materials: Theory analysis and experiment. *Royal Soc. Open Sci.* 6: 190112.

[0035] Yang, S., F. Aldakheel, A. Cagginnio, P. Wrigghers, and E. Koenders, 2020, A review on cementitious self-healing and the potential of phase field methods for modeling crack-closing and fracture recovery. *Materials*, 13, 5265.

[0036] Yue, Y., Yin, and Z. Zhong, 2006, shape effects in the Brazilian tensile strength test and a 3D FEM correction. *International J. of Rock Mechanisms and Mining Sciences*, vol. 43 (4), pp 623-627.

[0037] Yaphary, L. Y., R. H. W. Lam, D. Lau, 2020, Reduction in cement content of normal strength concrete with used engine oil (UEO) as chemical admixture. *Construction and Building Materials*. 261(119967).

[0038] Zena, K., A., 2015, Effect of kerosene and gas oil products on different types of concrete. *International Journal of Science and Research* 6(12): 2319-7064.

STATEMENT OF REGARDING FEDERAL SPONSORED

[0039] None

INCORPORATION BY REFERENCE

[0040] All patents, patent applications, and publication cited are incorporated by reference for all purposes to describe the status of art as known to those skilled as of the date of the invention described herein.

THE NAMES OF PARTIES OF JOINTED RESEARCH AGREEMENT

[0041] NONE

DISCLOSURES

[0042] NONE

FIELD OF INVENTION

[0043] This invention is related with chemical additives and solution useful for enhancing the early age strength and

the durability of concrete materials in infrastructural industry and long-life span service of hydraulic-cement concrete products. Inspired by biological bone fracture recovery and biomineralization, the disclosed status of art practice demonstrates that the chemical additives and solution, functionalized as hydro dual phobic domains, can effectively fill the micro-cracking fractures, with reduced brittleness and enhanced viscoelasticity and viscous plasticity of the concrete materials, a self-activated healing with adjustable polymer molecular tunable orientation is proposed as the mechanism, activated by phase transitional hydration of gel polymers, stimulated by water and mineral oil, resulting in a hydraulic-cement concrete product useful for construction industrial applications.

BACKGROUND OF THE INVENTION

[0044] The Portland cement is one of the widely available building materials in the construction industry since its introduction. It is also the 2nd most used material with a significant environmental cost. For each tonnage of cement clinker manufactured, there are about one ton carbon dioxide emitted to the atmosphere¹, which is accounted for about 4-8% carbon dioxide emission globally. Less use of cements and replacement of cements with other environmentally friendly inorganic geo-polymer or organic polymeric construction materials are highly desirable, especially in views of current sustainable economic growth of zero carbon foot printing goals. The huge damage of recent floods in European, China, and unstopped forest fires in California and the strong Hurricane of Ida in USA create urgent calls and demand on curbing the global warming.

¹ <https://www.theguardian.com/cities/2019/feb/25/concrete-the-most-destructive-material-on-earth>.

[0045] Fly ashes as key cementitious materials have been used as a partial replacement of the cement clinker at a weight percentage as high as 75%, however, since the pollution from coal burning for electricity is high, the power generation plant and coal mining industry are going to be phased out sooner than later. The availability of fly ashes will become scarcity. Seeking other alternatives for fly ashes to meet concrete performance requirements is in demand (Rodriguez, et al. 2020).

[0046] In term of product performance, the compressive strength (CS) of concrete has been considered as the most important, listed as the 1st criteria per ACI design manual, determined by structural engineer, attained, and verified by properly evaluated test results as specified in practices. A mean of minimum 2500 (PSI) compressive strength is required for structural applications as listed in Table 1.

[0047] As a matter of fact, various factors affect the concrete strength (Moehle, 2019). One drawback of hydraulic-cement concrete is that it has poor tensile strength, thereof, reinforced elements such as steel bar, non-ferric types of materials, organic or inorganic ingredient's materials must be included to prevent the concrete structure from cracking failure. A typical example is the surprising failure of high rising residential building collapses in Surf City of Miami due to the cracking failure of basement concrete structural members.

TABLE 1

Specification of Concrete Products Performance Per ACI 318 Building Code		
Items	Application of Concrete Products	Compressive Strength (PSI)
1	Concrete fill	below 2000
2	Basement and foundation walls, and slabs, walks, patios, steps, and stairs	2500 to 3500
3	Driveways, garage, and industrial floor slabs	3000 to 4000
4	Reinforced concrete beams, slabs, columns and walls	3000 to 7000
5	Precast and prestressed concrete	4000 to 7000
6	High-rise building (column)	10,000 to 15,000

[0048] Inspired by nature, the concept of designing and fabricating a self-healing concrete has always been given a top priority in concrete research community, especially, a biomaterial, as a possible solution to prevent the structural failure from future repair and mitigate the occurrence of cracking failure of concrete structure. The strategies for developing an effective concrete product include mixing superabsorbent polymers with Portland cements; incorporating bio-enzyme into the cement paste; applying encapsulated reactive polymeric or inorganic materials for in situ curing of cementing materials; and incorporating reinforced fibers to reduce the stress concentration of concrete.

[0049] Super Absorbent Polymer (SAP): Super absorbent polymer (SAP) material has been used as curing additives in concrete. It can supply plenty of water as a water reservoir by swelling. It can release water slowly in dry air as an autogenous self-healing agent for enhancing the performance of engineered cement composite (ECC) products (Li, et al, 2007). It is also effective in enhancing the self-healing performance of cementitious materials by supplying water to cracked concrete in wet/drying cycling environmental conditions (Chen, 1999, Kyoshi, et al. 2002, Shim, et al. 2018, Ma, et al. 2019). A high efficiency of self-healing was claimed by utilizing a mobile infused hydrophobic layer with hydrophobic films (Aizenberg, et al. 2017). The success of SAP is limited due to the softness of hydrogel polymers (Dang, J. et al., 2017).

[0050] Enzymatic Bio-Concrete: Enzymatic self-healing agents for autogenous curing and cement cracking repairment have been studied extensively. The enzymatic proteins are claimed to catalyze chemical reactions at an extremely rapid rate (Rosawitz, et al. 2021). The carbonic hydrase enzyme (CA) catalyzes the reaction between water, calcium ions (Ca^{2+}) and CO_2 to produce calcium carbonate (CaCO_3). The pre-cut cracks in concrete as width as 400 (micron) can be healed. A novel hot spring bacteria strain selected can increase the compressive strength of cement materials more than 23(%) (Biswas, et al. 2010). Simulation of the capillary flow of an automatic healing agent in discrete cracks in the cementitious materials demonstrates that the stick-slip parameter decreases from 0.12 for the 7 days old specimen to 0.01 for the 28-day specimens due to the changes of mortar microstructure via the continuous hydration. As hydrogel is dried up, the adhesive adhesion becomes strong (Gardner D. et al., 2014).

[0051] Encapsulated Curing Agent: Hollow and brittle glass tubes filled with superglue (i.e, ethyl cyanoacrylate) as the healing agents have been explored in early 1990. The healing agent was released into the cracks of the cementitious matrix once it reached over certain loads and glass

tubes were broken. UV-fluorescent curable epoxy resin was also used to study the biomimetic. Other healing agents such as dicyclopentadiene, polyether-amine, NaSiO_3 gel, etc., have been encapsulated into micro-spherical shells with urea-formaldehyde, melamine formaldehyde, isocyanate, and poly(methylene methacrylate) (PMMA), however, the cost of these healing agents is relatively high and some of them might be not environmentally friendly (Li, 2007).

[0052] Others: To reduce cost of cement chemical admixture in hydraulic-cement concrete applications, used engine oil has been added into regular cement blocks, also, bio-derivative oil products such as canola oil, Kerosene, and gas oil as chemical additives for the cement/aggregate bonds, also added into the cement materials for performance enhancement, however, both compressive strength and flexural strength have been suffered in the tested specimen (Zena, 2016, Almabrok, M. H., et al. 2019, Yaphary, et al. 2020). α -amylase starch has been used as bio-fillers to tailor both biodegradation and bioactive behavior with hydrophilic functionalities (Boesel, et. al., 2006). The effect of biomolecules on the nanostructure and nanomechanical property of calcium-silicate hydrate (CSH) has been reported (Kamali, et al, 2013, Kamada, et al. , 2021). The drawback of encapsulated products is that there are a lot of unhealed cracks insides the cement matrices, or many microcapsules that introduce a lot of voids, resulting in a poor healing efficiency (Li, et al., 2013).

[0053] Logically, it is necessary that a self-healing concrete product must be developed that can meet all special requirements with cost effective solutions in mind. Inspired by bone biomineralizing mechanism, in this disclosed state of art concrete technologies, special coatings, chemical additives or/and chemical solutions used in hydraulic fracking operation (U.S. application Ser. No. 16,600,278, filed on Nov. 10, 2019) have been directly applied to the cement/sand/aggregate mixture as an admixture. It was discovered that the chemical additives/solution can enhance the performance of products not only on the early-age strength but also on its long-term durability with extended service life with enhanced toughness and flexibility of the compressed concrete products, possibly attributed to the contribution of the soy protein/sweet rice and gel polymer's molecular tunable reorientation and self-assembly of non-covalent bonds disclosed thereof.

BRIEF DESCRIPTION OF THE INVENTION

[0054] Inspired by bone biomineralization, the disclosed chemical components are comprised of:

[0055] a) soy protein isolate (SPI) or/and other bio-derivatives such as modified SPI, sweet rice, waxy amine, cross-linked by isocyanate or other cross-linking agents as hydro dual phobic domains within a range from 0.001(%) to 40 (%) by weight percentage.

[0056] b) hydrogel polymer as a suspending agent within a range from 0.001 (%) to 35 (%) by weight percentage.

[0057] c) surfactant/emulsifier that support the suspending and interaction of SPI with hydrogel polymers ranged from 0.0001 to 20.0% by weight percentage

[0058] d) mineral oil as hydrophobic solvent ranged from 0.001 to 50% by weight percentage

[0059] e) water ranged from 1.0 to 99.0% by weight percentage as balance.

[0060] f) The combined weight percentage of (a)+(b)+(c)+(d)+(e) is 100 (%).

[0061] In some embodiment, the manufacturing processes for preparing the polymer materials include charging the hydrophobic solvent, such as single chain and mineral oil, into a container, then, adding the chemical additives and solution into the cement as mix simultaneously, or the sand or aggregated materials coated with the chemical additives, or alternatively, the chemical additives mixed with cement materials on site directly.

[0062] In some embodiment, the formulated coatings have a static contact angle from 30 to 90 (degree); and tilted contact angle (pinning angle and depinning angle) larger than 30 (degree) as slippery and/or hydrophobic/hydrophilic dual domain coatings, measured by depositing a water droplet on the coated flatten solid surface at a total weight of from 0.1 (mg) to 500 (mg).

[0063] In some embodiment, a blend of invented chemical additives or solution with cement, water, fine sand, and aggregates can create concrete blocking products useful for both residential or/and commercial building materials. It was discovered that the products manufactured with the developed recipes have excellent early age compressive strength and high toughness for high long-term ultimate compressive strength initialized by a self-activation, driven by micro-capillary action, originated from not only the hydrated calcium-silicate-hydrate (CSH) bonds, but also hydrogen bonds from the β -sheet of soy proteins and polymeric material's phase transitions in response to the exothermic hydration of the coating material's phase transition from -25°C . to 90°C . The ratio of total weight of the coatings to the total weight of cement, fine sand, and aggregates, and water is ranged from 0.0001/99.9999 to 5.0/95.

[0064] In some embodiment, it can be applied as a regular solution with mixing action or coatings sprayed on the solid surface of sand or aggregate products to create coated sands or aggregate products. The dosage level of the coatings sprayed on the sand is ranged from 0.01% to 10.0%, preferred less than 3.0% over the total weight of cement, fine sands, and aggregates. The coated granular particles can suppress the respirable microcrystalline silica dust concentration by more than 95.0% in comparison with untreated specimens, preferred by 97.0%, 99.0%, 99.5%, 99.95%.

[0065] In some embodiment, the chemical additives can be mixed with cements to create cement paste and/or cement mortar agents in the mix as admixture in the construction field onsite directly. Mixing water can be used to first dilute the chemical additives, then, add all the leftover water right before counting the mixing time. The dosage level of mixed chemical additives' ingredient portion to the total weight of concrete materials including cement, fine sand, and supplementary cementitious materials, ranged from 0.01 (%) to 15.0% by weight percentage, preferred less than 5.0%, or 0.5%.

[0066] In some embodiment, the core materials of the coated chemical additives comprising of modified soy protein, sweet rice, or waxy amine, oxidized paraffin wax, hydrogel polymer in powder, or their blending with the isocyanate or epoxy resin, are moved from the swollen hydrogel slippery layer to the crack or fissure of concrete structure. The dosage level of these hydro dual phobic materials ranged from 0.1 (%) to 40.0% of the total coating materials by weight percentage.

[0067] In some embodiment, both the hydrogel polymers and soy protein isolate or its modification with their dual phobic domains in their structure could be adjusted/tuned in their molecular orientation. The solution concentration can be in a range from 0.001% to 50%. The formulated solution can be sprayed on sand and aggregate surface or directly blended into cement matrix within a 3 to 15 (minutes) shearing mixed into.

[0068] In some embodiment, the mixed cement paste and/or mortar, and masonry mix have an excellent workability with a working time from 10 to 120 (min.). The water to cement ratio can be within a range from 0.20 to 0.80, preferred 0.44 or less, or 0.40, 0.36.

[0069] In some embodiment, reinforced fiber elements including steel whiskers, glass fibers, polyvinyl alcohol fibers, etc. can be added to mitigate the crack for enhanced self-healing besides the chemical additives used here. The ratio of fiber length to diameter can be in a range from 10 to 100. The percent volume fraction of the reinforcement element such as steel bars, and fiber glass to the total volume of sand and cement, fine sand, and aggregates can be ranged from 0.0001/99.9999 to 5.0/95 by volume fraction of the total volume of concrete members.

[0070] In some embodiment, cementitious materials, such as fly ash, micro silica, silica gel, southern clays, water glass (sodium silicate), and abrasive particles, sand or proppant materials granular particles, selected for hydraulic fracking application in a size partition of 100 mesh, 40/70, 30/50, and 20/40, or specified by contract negotiation, could be added in a range from 0.001 to 75% by weight percentage over the total weight of cements to partially replace the cement materials or added as raw sand materials to partially replace the fine sand or aggregates over the total weight of fine sand, aggregates, and plus cements.

[0071] In some embodiment, the minimum ultimate compressive strength of the tested samples per ASTM C 39 or/and test standard such as ASTM C109 (2"x2" Block) is larger than 2500 (PSI), more preferred 4000 (PSI) measured after being molded at 28 day's setting time, especially more than 7000 (PSI) for high performance concrete structural application, the Brazilian splitting tensile strength is larger than 600 (PSI), especially larger than 1000 (PSI). The products are useful in residential and commercial market, concrete slabs for highway and as high strength concrete mix for high rising construction.

[0072] In some embodiment, the percentage recovery of the tested samples for self-healing can be more than 80 (%) determined by self-healing efficiency (SHE) and measured by the Brazilian splitting tensile strength test method, more than 100 (%) by water permeability test.

[0073] In some embodiment, the toughness of disclosed hydraulic-cement concrete products has a value of 9 times more than that of its virgin products, and 13 times more in the modulus of resilience, also called as the flexibility of the materials.

[0074] In some embodiment, the density of the tested samples has a value larger than $1.90\text{ (g/cm}^3\text{)}$, specially more than $2.20, 2.40\text{ (g/cm}^3\text{)}$, and $2.55\text{ (g/cm}^3\text{)}$, porosity less than 0.15, more specially less than 0.07.

[0075] In some embodiment, the chemical additives have a phase transition temperature from 30°F . above and 200°F . below after being blended with other cement components and cured in an adiabatic or isothermal condition.

[0076] To have a better description of the disclosed invention. The figures and drawings are used in the following section.

BREIF DESCRIPTION OF FIGURES AND DRAWINGS

[0077] FIG. 1. A proposed schematic of a self-activated healing mechanism of chemical additives and solution with cement and sand/aggregate materials: **100**—Chemical solution/coating emulsion; **101**—solid surface from cement, fine sand, aggregate, and cementitious materials such as fly ash, micro silica, silica gel; **102**—light density solvent (mineral oil); **103**—hydrated hydrogel polymer; **104**—SPI core layer particles in spherical shape; **105**—SPI in sheet or disk shape. The chemical additives are interacted with other components with the following processing steps: 1) breakup of emulsion after being sheared intensively with core layers exposed into air or water in the hydrated environmental condition; 2) SPI and hydrogel polymers by the self-activated non-polar solvent tailored toward an intimate contact with cement surface under different temperature and osmosis pressure; Step 3: Tight interface of chemical additives with cement elements involved in various molecular bonding mechanisms including CSH cationic bond, chelating, amination, hydrophobic and dispersive, hydrogen bonds of potential intercalation within CSH; Step 4: formation of immobilized interface with hardened concrete structure.

[0078] FIG. 2. Schematics of proposed self-healing processes of concretes with disclosed chemical solution and additives: **201**—solid surface of cement, sand, aggregates, or other cementitious particles; **202**—coating layer; **203**—crack of concrete structure; **204**—self healing agent filled in the crack of concrete; **205**—SPI/sweet rice core layer.

[0079] FIG. 3. A plot of both the compressive stress and strain in % as a function of registered sampling time for the tested cylindrical sample of example 3.

[0080] FIG. 4. A plot of compressive strength in the selected examples of 3, 4, 5 specimens as a function of strain for selected cylindrical testing samples.

[0081] FIG. 5. Specimen geometry for a modified diametrical tensile test adapted directly from publication (Singh and Pathan 1988).

[0082] FIG. 6. The image of water seeping out of the created micro-crack of the samples originated from examples 8a and 8c after both were immersed in a water tanker for about 15 (minutes). All edges were sealed with wax except that the up-side-down surface contacted with tap water in the water tanker (referred to ASTM C 1585-04 experimental settings).

[0083] FIG. 7. Plot of water permeability of tested samples prepared from samples labeled as exam 8a, 8b, 8c, 8d, and 8e.

[0084] FIG. 8. Plot of measured water microdroplet's static contact angles and tilted contact angles placed on a solid surface of flatten glass plate after being coated with the disclosed thin coating of example 1.

[0085] FIG. 9. Plot of ultimate compressive strength (UCS) and Brazilian splitting tensile strength (BSTS) as a function of sample aging in natural log according to the selected data from Table 15.

[0086] FIG. 10. Imaging photo of water marks after the specimens were soaked in water tanker for one hour. Partial ingress of water blocked from seeping out can be clearly

observed. Both specimens of 9a and 9b were soaked in water tanker for one month before taken out for repeating water soaking test.

[0087] FIG. 11. Plot of water permeability in the self-healed specimens of exam 9a and 9b as a function of square root plot of sampling time.

[0088] FIG. 12. Plot of cement curing temperature in the vacuumed coffee cups under the adiabatic condition with different chemical additives blends: example 19: notebook ID: 11102019-1(wax); exam 20:notebook ID: 11102019-2 (SPI/Wax/SR); Exam 21: 11102019-3(control); Exam 22:11102019-4(Rapid setting).

[0089] FIG. 13. Plots of weight percentage reduction on selected tested concrete prism bar samples from blending of examples 19, 20, 21, and 22 as a function of sampling time at an ambient temperature of 70° F.

DETAILED DESCRIPTION OF THE INVENTION

[0090] Self-healing Mechanisms of Hydraulic-cement Concrete: Cracking often occurs in concrete structure. As the tensile strains created from restrained thermal contraction or temperature differential surpass the tensile strain capacity of concrete structures, cracking will initiate. When concrete is still soft in the first several hours of curing, autogenous shrinkage is limited to chemical changes driven by cement hydration, but, in later stage after the first 24 hours, there are high risks of autogenous shrinkage. High water evaporation rate and subsequent high magnitudes of shrinkage are the most common cause of cracking. Self-healing such as autogenous self-healing has been considered as one promising solution to address the cracking issue of concrete. The main cause of autogenous self-healing is attributed to the formation of calcium carbonate, expressed by the following two simple chemical reactions:



[0091] As shown in equations (1), the carbon dioxide is absorbed from environment of air or from a dissolved chemical reactant, an anionic hydroxide bicarbonate ion is generated in the equation right side. The hydroxide bicarbonate ion can further react with the Calcium cations, resulting in the precipitation of calcium carbonate recrystallization (equation 2). Observation on old cement concrete construction suggests that some cracks are self-evidence of these white crystalline materials attributed to the formation of calcium oxide crystallinity.

[0092] In addition to the above CaCO_3 cationic bonds, silica can also have a cationic covalent bond connected with calcium defined as CSH bonds occurring at calcium oxide and silicon oxide interface widely distributed in the earth. The CS is attached with hydroxyl group on its top surface of the interface that creates the hydrated bonds with —OH in CSH bonds.

[0093] Similarly, bone, as a highly specialized organic inorganic architecture, which can be classified as micro-and nano-composite tissue in its mineralized matrix, consists of an organic phase (mainly comprising of collagen, 35% dry weight, reinforced with calcium phosphate in a liquid crystalline structure), responsible for its rigidity, viscoelasticity, and toughness, 65% dry wt. of carbonated hydroxyapatite for structural reinforcement stiffness and mineral homoge-

neity. Other non-collagenous proteins that form a microenvironment stimulate cellular functions. In comparison with other human tissue, bone tissue is capable of true generation, i.e., healing without the formation of fibrotic scar tissue (Henkel, J., 2013). In our disclosed additives, the applicants believe that the SPI and its modification can be considered as a plant-based collagen (proteins) that can be swollen and intercalated into cement matrix structure through physically intercalating and non-covalent hydrogen bonds after being modified with crosslinking agents such as isocyanate or epoxy polymers. The hydrogel polymer of the shell layers in the emulsion can be considered as a soft gel that provides needed protection against damage to the inside core layer of SPI and modified SPI. Mineral oil and water can be considered as plasma and wetting agents as a self-activated catalyst or potential sensing agent.

[0094] Like the biomineralization, in this disclosure as shown in FIG. 1, it is self-evident that an emulsion, or chemical additives particle **101** is in contact with cement matrix surface **102**. In step 1, millions of similar emulsion particles and cement/sand/aggregates are engaging intimate contacts through collision, shearing from each other, and bouncing in front and back from each other. Light density solvents (**102**) such as mineral oil or low carbon alkyl molecules are squeezed out from the hydrogel polymer network that covers the top surface of the coating layer, resulting in a slippery mixed layer of solvent/water/hydrogel polymer system, more specially, the hydrogel polymer (**103**) containing hydrated water, leading to a much less friction force to the sheared particles, which makes the whole mixed components fluffy without sticking. It can be defined as wet, however, not sticky phenomenon in term of soft material's behavior.

[0095] Similar scenario exists in carnivorous pitcher plants. The surface of pitcher plant's leaves is sticky with a clever slimy fluid like mucus as a super hydrophilic material; however, it is also slippery when the coating is sheared (Hirscher, 2007), characterized as super hydrophobic materials. Different from Lotus leaf, liquid or water droplet having an intimate contact with the pitcher surface will not be rolled down their leaf surface. it will have a breakup contact angle larger than 90 (degree) pinning on the leaf surface without falling. Water and mineral oil as dual solvent systems are soaked in the hydrogel network and SPI core layers, pending upon the attractive force between the absorbed liquid on the leaf surface. The emulsion containing in the liquid fluids will make a smart choice in the disclosed liquid on how the functionalized soy proteins in the core layers interact with other components to make a choice. The applicants believe that the core layer particles **104** such as soy protein isolate (SPI) and modified hydrogel polymer in powder, or sweet rice modified with isocyanate, etc., are exposing their hydrophobic groups toward airside conditions in response. SPI in the scenario (a) has a spherical shape, while SPI in the scenario (b) disc like structure potentially originated from soy protein secondary structure of β -conglycinin 7S (19-20%). On the downside, the SPI will have hydrophobic groups toward the sand side with siloxane groups. Alternatively, Ca^{2+} and/or magnums is partially attached with the SPI surface. This kinds of molecular configuration of SPI or modified SPI makes the controlled self-assembly of soy proteins into high performance multifunctional nanostructured film on the solid surface become a reality.

[0096] As SPI is heated to 90° C., the proteins would have showed low content of intermolecular β -sheet structure and high content of random coiled and a helical secondary structure. Upon cooling down the soy protein down to 20° C., the content of intermolecular β -sheets will gradually increase by 25% determined if the protein is dissolved in appropriate solvent. Heating of SPI components can alternate its solubility with its denatured characters greatly between temperature of 20° C. to 90° C. (Lan Q., et al. 2019). The applicants believe that as both core layers and surface layers of the emulsions are heated or placed in different solvents, the secondary intermolecular structure is varied in response to enhance the interaction of SPI and hydrogel polymers with cement and aggregates not only in ionic bonds as CSH bonds, but also in hydrogen bonds, chelating bonds (Ca^{2+}), leading to an intercalation of protein molecules into CSH interface.

[0097] In step 2, imaging that the SPI particle might change its configuration through rotational or/and translational modes in the micro channel of cement matrices, the soy protein spherical particles are rotated by 90 (degree) in angle, from perfect spherical shapes into elliptical shapes in FIG. 2(a'). In item of (b'), the disc types of soy protein macro/nano particles are not only rotated by 90 (degree) in angle, but also become elliptical and flattened. Fundamentally, the configuration angles are tunable in response to the environmental conditions. The intimate contact surface area in scenario (b') are much higher than in (a'). The applicants believe that the frequency of occurring in non-covalent bonds in scenario (b') is much higher than in (a'). in the case of scenario (c'), the increased hydraulic pressure can potentially increase the opportunities of interfacial adhesion of α -helical coiling protein or/and β -sheet proteins with the cement elements. The ideas and fabricating strategies present here are to illustrate that the soy proteins or and sweet rice are potentially aligned together as fibrinogen or collagen served as clotting proteins to enhance the bonds of organic-inorganic interphase across the solid surface of hydraulic-cement concrete particles.

[0098] In step 3, a complex interpenetrated networking scenario might present herein. Different from current widely accepted CSH bond's description, chelating, deamination, amination from polyurea and polyurethane, and hydrophobic, dispersive bonds of mineral oil with alkyl functional group from soy proteins, sweet rice, or hydrogel polymers, especially extensive non-covalent hydrogen bonds among the polyurethane, amine, and carboxylic, and hydroxyl functional groups might be involved in the interface, resulting in a potential intercalation of soy proteins and other organic intermolecular polymers with the cement matrix or —OSIO— inorganic surface. The applicants believe that dendritic, finger printing, and/or stitching types of bond line might present between the interface of the organic polymers and inorganic geo-materials of sand, aggregates, and cement matrix as micro-capillary pressure driving action and compression or tensile force placed on the interface of the spherical particles as shown in FIGS. 1(a'') and (b'') in a nanoscale.

[0099] In step 4, the SPI, hydrogel polymers, mineral oil, surfactants/emulsifiers, and water are completely packed together with, or without entrapped air bubbles within. The driving force for liquid and coated particles out of the micro-channels of cement matrix is the ratio of surface tension to viscosity of the mixed liquid fluid, which deter-

mines the penetrating rate of fluid from one side to others. Due to the micro-capillary action of micro-channels, the thickness of the bonding line between adjacent solid surface is narrowed down further as the water or/and mineral oil are driven out of the interface bonding lines by evaporation versus versa condensation. The porosity of components (c) is reduced further. The density of the concrete blocks increased with further consolidation. Also, it is noticed that since the porosity (c) is not zero for hydrogel polymer and soy protein isolate, the SPI as gel particles in the micro-channels travel in a much complex manner if the gel inertia and variation of channel radius are considered. The applicants believe that the adhesion of inorganic-organic particles is primarily dominated by these non-covalent bonds instead of calcium silicate hydrate (CSH) bonds. The reduced porosity might further consolidate the concrete with superior durability.

[0100] Durability vs. Cracking of Concrete Products: Study shows that if the width of concrete crack is less than 50 (micron), a self-healing based upon CSH bonds shown in equations 2 and 3 can potentially mitigate the risk of the crack with a closed gap in a 100 (%) recovery rate, however, if the width of the crack is less than 150 (micron) the crack width can be 100% self-sealed after re-cured for 3 days in water tank, its width reduced from 220 (micron) to 160 (micron) after recurring for 7 days, 33 days is needed to fully heal the crack in width of 160 (micron), primarily attributed to swelling effect, expansion effect, and re-crystallization by geo-materials crack self-healing behavior (Tae-Hoshn & T. Kishi, 2010). Large crack in width more than 150 (micron) need special strategic approaches to heal the cracks (Yang, et al., 2009). ACI 224 committee states that a crack of 180 (mm) or more in width can cause deterioration of concrete structural members related to durability. The published allowable crack width in ACI 224R-01 (2008) table shows that the permitted crack width is strongly dependent upon concrete application scenario. In water-retaining structure, it only allows a crack of less than 100 (micron) in width, however, in dry air operation, it allows the crack less than 410 (micron) in width.

[0101] As shown in FIG. 2(a), the solid surface of **201** is coated with a regular coating (**202**) on its surface. If stress within the solid surface is over the tensile strength of solid concrete, a crack of **203** will generate like a suspended upside-down tree trunk. A crack or fissure will open on the surface of cement structure as shown in FIG. 2(b). if the chemical solution is used, the disclosed chemical additives or coatings are swollen and filled the gaps and crack of the upside-down tree trunk as shown in FIG. 2(c) in the case of cracks in width more than 160 (micron). The movement of the swollen particles and hydrogel polymers can be in a rotational or translational mode in response to the changed moisture content and solution temperature as shown in FIG. 2(e) by condensation or evaporation, versus visa.

[0102] The hydrogel polymers and SPI core materials will be potentially swollen first once it is accessible to water or mineral oil. Both hydrogel polymers and SPI particles could jump, hop, or fall themselves into the bottom of the crack or fissure valleys as shown in FIG. 2(d) in a stick-slip movement characterized by paired pinning-depinning events at the contact line. The encapsulated SPI could absorb moisture content and evaporate water out of the channels where motion was temporarily impeded but not permanently stopped. The solid of SPI components might also accumu-

late at the contact line in FIG. 2(e), resulting in a momentary localized solidification and temporary pinning (sticking) to the channel, leading to a complex dynamic behavior of interfacial bonding lines in a randomized stitching and finger printing pattern of cement interface bonding networks instead of a simple mobile movement of liquid fluids. Finally, the crack will be filled with new additives and minor impurities or debris particles that grant the crack with desirable recovery strength as shown in FIG. 2(f).

[0103] The applicants believe that the disclosed self-healing mechanisms of stitching and stick-slip bonds promote the self-activated concrete recovery capability. Mostly, as the chemical additives are added into cement matrix, it inhibits the hydration of cement reaction with sands or cementitious materials. In the rapid set cement products, the cement added with calcium oxide creates extensive heat as water is added in the cement mix in 20 minutes due to the exothermic reaction of cationic calcium with hydroxy anionic groups, leading to a fast gelling of cement components. The drawback of these fast-curing cement recipe is that the cured products will have low early age strength and poor post durable strength due to introduced structural defects before the mixed components are condensed.

[0104] In contrast, the applicants believe that α -helix coiled and β -sheet proteins from SPI are potentially embedded in cement matrices, tethered on the interphase zones of the sand and aggregates that will bring the viscoelastic spring-dashpot connection to the cement matrix. The viscoelastic composites, comprised of protein's helix coiling tails and β -sheet as building block, are proposed in FIG. 2(g) in a spongy form surrounding the solid surface of cements, sands, and aggregates in a randomized distributed pattern. Inspired by bone fracture recovery, the SPI proteins can be considered as a collagen (long chain rod types of proteins) and clotting agents in bone structure, and the spongy form of SPI as cancellous type of structure in the integrated hydraulic-cement concrete structure. For long-term durability, the SPI and sweet rice components or their modification might and would be decomposed if the environmental condition is appropriate, leading to more CSH types of bonds. This serves as potential latent self-healing agents for concrete products without a need for repairs.

[0105] In summary, the proposed mechanisms are aimed at addressing the cracking issues related with the early strength and durability of hydraulic-cement concrete structure. Conceptual proofs of the proposed mechanism and benefits and advantages of components in the disclosed coatings are described in detail further.

[0106] Micro/nanotextured dual phobic domains: of the disclosed chemical composition and emulsion coating as shown in FIGS. 1 and 2, randomly distributed micro/nanotextured domain materials can be created by incorporating nano-textured dual phobic dot domains in powder, granular particles, or nanofibers on the solid surface. Instead of having a smooth surface, the coatings have an uneven and rough surface. Spherical inorganic mineral fillers or organic nanosized or micro-sized filler materials are potentially textured as the raw dot domains by simple self-assembly on the solid surface.

[0107] One of the identified cost-effective chemical additives is the petroleum paraffin.

[0108] Others, such as soy protein isolate (SPI), are also preferred candidates as nanotextured domain materials. Morphologically textured ridge, concave, convex, and val-

ley's features of coatings could be useful to construct the disclosed coating materials with micro-tips and bumps generated by the waxy spheres and/or dots to create an enhanced hydrophobicity and anti-blocking capability on the coated sand or aggregate or cement paste.

[0109] Another benefit with waxy materials is that wax is a cost-effective as hydrophobic domain materials and easy to be emulsified into coatings or solution chemicals. It has a diverse class of organic compounds that are lipophilic, malleable solids near ambient temperatures, including higher alkanes and lipids, melting to give low viscous liquids. Waxes are insoluble in water but soluble in organic and nonpolar solvents. Natural waxes of different types are produced by environmentally friendly plants. For example, Carnauba wax, also called Brazil wax and Palm wax, originally from the leaves of the Palm, is consisting mostly of aliphatic esters (40 wt. %), diesters of 4-hydroxycinnamic acid (21.0 wt. %), ω -hydroxycarboxylic acids (13.0 wt. %), and fatty alcohols (12.0 wt. %). The compounds are predominantly derived from acids and alcohols in the C26-C30 range. Distinctive for Carnauba wax is the high content of diesters as well as methoxy-cinnamic acid².

² https://en.wikipedia.org/wiki/carnauba_wax.

[0110] Paraffin waxes are hydrocarbons, mixtures of alkanes usually in a homologous series of chain lengths. They are mixtures of saturated n- and iso-alkanes, naphthene, and alkyl- and naphthene-substituted aromatic compounds. A typical alkane paraffin wax chemical composition comprises hydrocarbons with the general formula C_nH_{2n+2} and $C_{32}H_{64}$. The degree of branching has an important influence on the properties. Microcrystalline wax is a lesser produced petroleum-based wax that contains higher percentage of iso-paraffinic (branched) hydrocarbons and naphthenic hydrocarbons. The candle and paraffin wax are commercially available in the commodity market.

[0111] Synthetic waxes are primarily derived by polymerizing ethylene. Alpha olefins are chemically reactive because they contain a double bond which is on the first carbon. The newest synthetic paraffins are hydro-treated alpha olefins which remove the double bonds, making a high melt, narrow cut, and hard paraffin wax. The wax is a very hydrophobic material. It has melting points in general above 35° C. or more. More specifically, the melting points of the wax are above 55° C. It has a measured water contact angle between 108 and 116 (degree) in angle (Mdsalih, et al. 2012). The percent wax quantities added into the mixture of designate recipes should be in a range from 0.01% to 15.0%, more preferred less than 5.0%. Other typical synthesis waxes include reactive wax such as ethylene stearamide, bis-ethylene stearamide, and their blends with other wax or solid lubricant materials that have lubricants and slippery characters.

[0112] Besides wax, other nano particles, such as polylactic polymers, SPI, nanofillers, lipids, sweet rice, and other bio-derivatives, might be used as macro/nanotextured materials mixed with wax to achieve desirable hydrophobicity and hydrophilicity. Hydro-dual phobic domain materials are referred to the materials that can be described as a material that behaves as hydrophobic, also hydrophilic with dual phobicity. It can be a two system by a synergistic blend or one system chemically modifying a solid surface with multifunctional attributes. For example, a silane coupling surface treatment will allow the surface of modified carbon to become either hydrophilic or hydrophobic, leading to a

hydro-dual phobic. As the modifying surface is contact with water, it will tend to expose itself with hydrophilic attributions. As it is attached with non-polar solvent, it will tend to expose its wax and alkyl functional groups on the surrounding environments. As such, the coated molecular components can be adapted in a smart manner to the aqueous solvent or air with appropriate fitness to the systems.

[0113] Different from waxy particle materials, soy protein isolate (SPI) containing a multifunctional moiety on its surface provide extensive reactivity and interaction with the other materials. One typical character is that the surface of bio-polymer particles such as soy protein isolate and sweet rice in powder can be chemically grafted with isocyanate polymeric functional groups or other functional cross-linking agents to achieve desirable hydrophobic and hydrophilic domain differently. From the peptide molecular structure of soy protein isolate (SPI). Alternatively, hydrogel polymer of hydrolyzed polyacrylate sodium acrylamide (HPAM) polymer in powder can be copolymerized with soy protein isolate through isocyanate as cross-linking agent. Other alternative proteins can be included as soy protein concentrates (70%), and soy flour (50% Protein) in powder to obtain hydro-dual phobic materials. That is, both of SPI and HPAM in powder, or granular particles can be cross-linked together to achieve a synergistic effect. The applicants believe that the copolymers from the SPI and HPAM chemical reaction through functional group of polyurethane and amide are unique that the viscosity of the mixed components are potentially enhanced as mixed components are added into the solutions due to the introduced multifunctional reactive sites on the surface of HPAM polymers.

[0114] Another benefit of utilizing the SPI is that SPI is in a porous network structure. Potentially, the hydroxyl, amide, and amine functional groups located on the surface or inside of the SPI particles are easily interacted with each other to physically generate the hydrogen and ionic bonds among the HPAM and SPI gel particles, leading to a gel polymer with enhanced viscosity of the mixed components.

[0115] Since SPI is made from de-natured soy protein flakes that have been washed in either alcohol or water to remove sugars and dietary fibers, a typical SPI nutrient component in 1-once plain powder based upon a USDA national nutrient database release (2004) has a component as total fat: 2(%); saturated fat: 0.0(%); total carbohydrate 1 (%); protein: 46.0(%); cholesterol: 0 (%); sodium 12.0(%); dietary fiber: 6.0(%); calcium: 5.0(%); Potassium: 1.0(%); Phosphorus: 22.0(%); folate: 13.0(%). Major components of soy protein isolate (SPI) are made of soybean products, which is abundant, inexpensive, renewable, bio-degradable, and aromatic. This provides rich ingredient as cement admixture type of products. Less costly soy protein concentrate (70%) is also a good raw material for copolymerizing them with HPAM. There are at least three methods for processing soybean into SPI: 1) The aqueous proteins; 2) the acid processes; 3) Heat denaturation/water wash method.

[0116] Through denaturation, soybean is isolated, containing primary amine ($—NH_2—$), secondary amine ($—NH—$), and acid carboxylic functional group ($—COOH—$). These functional groups provide extensive networking connection joint points with polyamide I and II bonds. In one perspective, the disclosed recipe provides a chemical composition comprising SPI plus polymers or pre-polymer's blends from 0 to 90 (%) of the reactive isocyanate, a polyol, a polypep-

tide, or oxide epoxy resin. The dose level of polypeptide is ranged from about 10.0% to 90% (wt./wt.).

[0117] The organic poly-isocyanate can be selected from the group containing of polymeric diisocyanate (p-MDI), 2,4-methylene diphenyl diisocyanate. Under certain conditions, these poly-isocyanate polymers have one or two or tri-functional reactive groups reacted with the polypeptide bonds originated from SPI. The term “Protein” and “Polypeptide” are used synonymously and refer to polymers containing amino acids that are jointed together.

[0118] For example, peptide bonds or other bonds may contain naturally occurring amino acids or modified amino acids. The polypeptides can be isolated from natural sources or synthesized using standard chemistries or by chemical modification technology through grafting, including cyclization, disulfide, demethylation, deamination formation of covalent cross-links, formation of pyroglutamate, formylation, gamma-carboxylation, glycosylation. The term “isolated” refers to materials that removed from its natural environment if it is naturally occurring.

[0119] Potential bonds between and among the SPI and isocyanate might include the amide and carboxylic ester, and imide bonds with a cross-linking of SPI and isocyanate. Potentially, the HPAM can be incorporated into the SPI molecular chains and network structure through the multi-component's reactions. The applicants believe that the increased viscosity of modified HPAM with SPI cross-linked with isocyanate or epoxy polymers is potentially originated from the attribution of SPI's salt tolerance of its strong bonds with cationic ions such as sodium, calcium, and magnesium, and ferric chloride.

[0120] In comparison of polypeptide bonds vs. other chemical bonds, the polypeptides are very strong so that they can resist the heating temperature as high as 130° C. in the processing of denature and defat soybean materials. Procedures for generating a core layer of 100 emulsion particles in FIG. 1 are involved in first charging the lubricants such as mineral oil into a reactive tanker. Subsequently, SPI and/or HPAM can be added into the tanker or container. Then, cross-linking agents of p-MDI will be added into the reactor. Heating the mixed components in a reactor allows the solvent/lubricants to reflux in the condenser within a defined time (say at least 5 minutes at 60° C.). Besides the functional group of isocyanate (—NCO) from p-MDI, other crosslinking agent such as oxide epoxy, amine, aldehyde, carboxylic acid, silane coupling agents can be used to modify the SPI surface or to crosslink the SPI with HPAM.

[0121] The blended or reacted SPI_HPAM and isocyanate/lubricant system serves as the core layer of emulsion in the emulsion structural design of 100 in FIG. 1. Then, the core layers that have the excellent power of tailoring the viscosity of the formed films with nanotextured patterns that are encapsulated with emulsifier/surfactant in the 1st phase polymerization of mineral oil. The reaction temperature can be as low as ambient; however, preferred reaction temperature can be as high as 130° C., or less, preferred 60° C., or less. After the p-MDI is fully reacted with SPI or HPAM, shell layer materials such as emulsifiers can be added in the mixed components and optimized further.

[0122] Alternatively, the left-over hydrogel polymer from reacted core layer can be used as shell layers as suspending agent. More hydrogel polymer can also be added to generate a special shell layer with special electron charges on the shell layers.

[0123] Furthermore, the SPI belongs to an attractive building block material that can be obtained from natural sources, making them suitable for the fabrication of biodegradable materials. Proteins possess a propensity toward molecular self-organization and self-assembly with remarkable performance that can be used to create special nano-textured domain patterns. The generation of protein-based film through controlled assembly in the past has mainly focused on the use of synthetic peptides, natural animal derived proteins, such as silk, bovine albumin, fibrinogen, β -lactoglobulin, hemoglobin, and lysozyme or through protein engineering. preparation of porous structure derived from protein interactions with other coatings and polymer materials has been reviewed (Wei, Q., et al., 2014). However, fabrication of nano-textured pattern from plant-based proteins such as soy protein isolate (SPI) for used in concrete mix in combination with super absorbent polymer to achieve the synergistic effect as both strength enhancer and cracking healing agents with multi-layered hydrogel agents have not been found or disclosed in the public domains.

[0124] Furthermore, porous textured pattern from SPI can also be fabricated by a reaction of the Glucono- δ -lactone-induced soybean protein isolate (SPI) via A Maillard reaction on the surface of cement matrix and sand surface although the concentration of soluble soybean polysaccharide (SSPS) can significantly affect the formation of textured pattern (Lan et al. 2019). Sulfhydryl group content has been increased in both denatured SPI and SPI gel that can be attributed to as major contributor toward the enhanced stiffness of protein structure. Recently, nano-microscale patterning has been fabricated by using plant-based proteins. Upon exposure to elevated temperature, the proteins unfold and partially hydrolyze, making them more available to form new intermolecular interactions, slowly lowering the temperature of solution facilitated the formation of self-assembly textured pattern on the coatings (Kamali, et al., 2021). The preferred relative dose level of SSPS to SPI is ranged from a ratio of 0.1/99.9 to 30/70 by weight percentage.

[0125] Key soy plant proteins are kinds of mainly comprising of 7s glycoprotein accounting for 20-50% of total seed proteins. It is a trimer consisting of three major subunits. The 7s globin is usually devoid of disulfide bonds. β -conglycinin forms a transparent, soft, but rather elastic gel in 100° C. heating and can be denatured at 80 to 90° C. The denatured temperature is started around 60-70° C. In one perspective, it is consisting of a lot of amine types of functionals, it is highly hydrophobic. On the other hand, it has considerable percentage of polar and charged residents, which leads to good water solubility and facilitates associates with bioactive compounds through electrostatic attraction and hydrogen bonds. X-ray crystallinity and CD study show that β -conglycinin is comprising of 5-10 in helix structure, 33-35% of β -sheet, 58 (%) randomized structure. Soy protein gel with high 11s/7s ratio shows higher extent of macro-phase separation and coarser network with large pores (Wu, C., et al. 2016).

[0126] In addition, strong interaction of glutamic acids and proteins was observed as the ratio of Ca/Si of CSH bonds increased from 0.7 to 1.5, attributed to the strong hydrogen bonds between —O—Si—O⁻ from silicon and —OCO— carboxylic functional group from β -sheet of proteins and more specially through the Ca²⁺ bridging by chelating functional connection of protein molecules with

SSPS or silicon ionic⁻ group. The decreased interphase stiffness of cement paste suggests that an intercalation might occur between CSH bonds and super molecules of proteins (Kamali et al, 2018).

[0127] In addition, bio derivative's sweet rice is another great candidate as the core materials of the emulsion. Sweet rice is rich in amylopectin, not amylose like starch. It is also called as glutinous rice, which means sticky after being cooked. Like proteins, it is believed that it should be one of excellent bio-derivatives if incorporated into the emulsion.

[0128] Emulsifier/surfactants: An emulsifier is a surfactant chemical. It can be cationic, anionic, zwitterionic, amphiphilic having linear long chain, branched with di-functional, tri-functional, multi-functional star's structures, consisting of a water-loving hydrophilic head and an oil-loving hydrophobic tail. The hydrophilic head is directional to the aqueous phase and the hydrophobic tail to the oil phase. The emulsifier positions itself at the oil/water or air/water interface and, by reducing the surface tension, has a stabilizing effect on the emulsion. It can interact with other components and ingredients. In this way, various functionalization can be obtained by interaction with protein or carbohydrates to generate connected clusters both chemically and physically.

[0129] Typically, emulsifiers include stearic acid oxide ethylene ester, sorbitol fatty acid ester, glyceryl stearate acid ester, octadecanoic acid ester, combination of these esters, fatty amine chemical additives and compounds, alkylphenol ethoxylates such as DOW Tergential^{NP} series of surfactants, glycol-mono-dodecyl ether, ethylated amines and fatty acid amides. For examples, SPAN 60 polysorbitan 60 (MS) and PEG100 glyceryl stearate MS are two typical emulsifiers used in cosmetics industries. Typical emulsifiers are branched as polyoxide-ethylene parts, groups found in the molecules such as monolaurate 20, monopalmitate 40, monostearate 80, etc., with HLB from 4.0 to 20.0, preferred from 10 to 17.0.

[0130] The dosage level of added emulsifiers in the emulsion can be ranged from 0.001% to 5.0%, more specifically less than 3.0% (wt./wt.) over the total weight percentage of coatings. The emulsifiers are water insoluble, only partially water soluble, dispersible. It is only dissolved in hot water. SPI and wax or other polyhydroxy component's materials such as sweet rice flour can be included as core materials in the micelle structure. In contrast, the emulsifiers can only be used as shell or intermediate shell materials in the micelle structure.

[0131] The emulsifiers in the disclosed additives are critical components. It has its hydrophilic heads toward the outside water loving phase and/create strong interaction with water solvent. Meanwhile, it has its hydrophobic long chain tail portion toward the waxy or SPI sphere as core materials for the micelle. SPI sphere or SPI_isocyanate sphere, SPI_isocyanate_HPAM cross-linked spheres are potentially sealed into the micelles. In addition, the made and amine from the HPAM and SPI might be critical for tailoring the final emulsion performance due to its electrophoretic functional performance although the reaction mechanism might not be understood. The applicants believe that the interaction among these chemicals makes the chemical additives blended into the water very complicated with unprecedented unknown attributes.

[0132] Cross-linking Agents: To enhance the stiffness of the core layer or shell layer of the micelles, selected cross-linking agents can be used to manufacture the micelles and

hydrogel polymer structure. Preferred cross-linking agent's reaction schemes was discussed in previous sections with p+MDI isocyanate functional resin polymer as an example. The purpose of p-MDI reaction with SPI is to enhance the hydrophobicity of SPI, potentially with extended hydrophobic chains to tailor the viscosity of the final emulsion and textured dual phobic domain patterns. Alternatively, reaction of crosslinked agents can be chemically cross-linked with non-reversible connections in nature or reversible with hydrogen bonds, pending upon the blended component's condition. Also, polyurethane dispersive can be incorporated into the coatings that have a UV curable moiety in its molecular chains. Alternatively, chemicals, containing epoxy, amine, amide, carbonyl, aldehyde, hexamine, and hydroxyl, amine functional groups of polymers can also be used. The preferred dosage level of cross-linking agents to the whole recipes of the coatings should be less than 10.0% by weight percentage. The ratio of SPI or copolymers of SPI+HPAM or SPI plus sweet rice plus HPAM to isocyanate should be ranged from 0.0000/100.00 to 40/60.

[0133] Antimicrobial Agent: Since soy protein isolate (SPI) and sweet rice flour are bio-derivatives, they tend to decompose themselves in the ambient condition. Microbial and fungus might potentially grow if they are used in water and aqueous based recipes during storage or transportation. As a result, antimicrobial agent is needed in the recipe, preventing biomaterials from bacteria or micro-fermentation. Common preservative additives include glutaraldehyde, formaldehyde, hexamine, benzyl ammonium chloride, methylisothiazolinone, 2-phenoxy ethanol, copper sulfate, copper sulfate oxide powder, fatty amine, etc. Dosage level of added antimicrobial agents is ranged from 1.0% or preferred less than 0.10% over the total weight percentage of the whole coatings. The ratio of antimicrobial materials to the SPI or sweet rice or their combination should be in a range of from 0.01/99.9999 to 5/95 by weight percentage. The antimicrobial agent can be added as a partial replacement of bio-derivatives used in the coatings by weight percentage.

[0134] Hydrogel Polymer: In cement paste and concrete recipe, hydrogel polymers serve as a multifunctional material. They are suspending agents as shell layers to encapsulate the core layer spheres from exposed to the out-layer environment before the special condition is met. Also, it is water reducing agent that can hold water in its matrix for desirable time and enhance the workability of cement paste during construction operation. It can also be a strength enhancer for promoting the early age strength of cement mix and autogenous self-healing for enhanced durability of concrete with reduced maintenance cost.

[0135] As shown in FIG. 1, the emulsion coating particle of 100 is suspended in an aqueous solvent, more specially water/mineral oil mixture, the hydrogel polymers added on the shell layer in powder or liquid are functionalized as a shell layer placed on the core sphere of SPI or its modified materials, potentially encapsulating the core layer materials intact from the out-shell layers via adjusting emulsifiers to generate a stabilized dynamic shell/core structure. It serves as lubricant/slippy agent. It also serves as super absorbent agents and can potentially hold more than 10 times of water by weight percentage in its network structure. The dosage level of hydrogel polymer added in the aqueous solution is ranged from 0.00001% to 2.000 (%) by weight percentage to the total weight of coatings. The coatings can be added in

water ranged from 0.2 to 2.0 gallon of coatings per thousand gallons of water. The hydrated viscosity of mixed aqueous solution can be ranged from 3 (cps) to 5000 (cps), pending upon the dosage level and desirable performance.

[0136] Common practices in current manufacturing technologies disclosed are to use hydrogel polymers such as polyethylene glycol, polyacrylate and polyacrylamide polymers and/or their copolymers added into the aqueous solution, in which, the use of additional surfactants is involved. Powder polymers are conventionally used in these applications due to the higher polymer concentration available in the form as compared to the solution polymers with reduced shipping cost. Hydrogel polymers are commercially available in the market. For examples, there are several brands of SNF products, such as FLOPAM DR 600 and DR 7000, that can be incorporated directly into the aqueous solution. Both polymers are anionic polyacrylamide polymers. Alternatively, FTZ2620, FTZ610, and LX641 polyacrylate sodium acrylamide polymers, manufactured by Shenyang JuFang Technology, Ltd., are also useful polymers as alternatives as friction reducer polymers and coating ingredients. Other polyacrylate and acrylamide polymers with cationic and nonionic molecular structure, are also potential candidates as hydrogel polymers. The structure of hydrolyzed polyacrylate sodium acrylamide can be linear or branched with dendrimers having hyperbranched polyester amide structure, mixed cationic and anionic polymers are also potential, other water-soluble polymers, such as polyvinyl alcohol (PVOH) and polyethylene glycol, are also potential candidates as substitute polymers of HPAM. The dose level of these hydrogel polymer is ranged in a ratio of hydrogel polymer to the total weight of coatings from 0.00001/99.99999 to 40/60 by weight percentage.

[0137] Lubricant: The synthesis processes of the HPAM polymers are involved in an inverted emulsion. Mineral oil or saturated hydrocarbon (Kerosene) is, in general, used as a key solvent for preparing the HPAM friction reducer emulsion. As a result, HPAM hydrogel polymer is dispersible in the lubricant. Lubricants or oils are comprising of the derivatives from petroleum crude oil, containing saturated hydrocarbon and alkyl groups. Alternatively, the lubricants can also be originated from the bio-derivative resource such as corn, soybean, sunflower, linseed oil containing the long chain alkyl components. The lubricants can also be synthetic oil chemicals made of reactive ester or hydroxyl functional alkyl chains or saturated hydrocarbon coupled with silane coupling agent or having silicon functional groups.

[0138] A broad definition of lubricants could be found in an URL link³. It is defined as a substance, usually organic, introduced to reduce friction between surfaces in mutual contact, which ultimately reduces the heat generated when the surface move. The dosage level applied in the chemical compositions for lubricants is added in a range from 1.0 to 90 (%). A typical mineral oil that can be used is a white mineral oil labeled as 70 Crystal Plus white mineral oils, manufactured by STE Oil Company, TX, USA. It is a series of derivatives of petroleum crude oils. Alternatively, soybean oil and linseed oil or synthesis silicon oil can also be used as lubricants. Other examples of lubricants include ethylene bis-stearic acid, amide, oxy stearic acid, amide, stearic acid, stearic acid coupling agents, such as an amino-silane type, an epoxy-silane type and a vinyl-silane type and a titanate coupling agent.

³ <https://en.wikipedia.org/wiki/lubricant>.

[0139] Water: Water is assumed to be a key component for preparing the emulsion as media and dilute agent to hydrate and adjust the coating into appropriate viscosity and pH

value. The viscosity of the final coatings can be in a range of from 3 to 5000 (cps), preferred from 5 to 100 (cps). pH value from 6.0 to 9.0, preferred around 6.8 to 7.6. The concentration of the final coating's products can be in a range from 40.0 (%) to 0.0001 (%) over the total weight percentage of coatings, preferred concentration is less than 15.0 (%), more preferred less than 10 (%), or 5.0 (%).

[0140] Procedures for preparing the chemical components and solution disclosed herein related to the recipes for a multifunctional coating, comprising of a multi-layered or hybrid shell and core structure having a desirable synergistic effect to the cement paste and sand coating. The applicants believe that the added components following a special procedure form a mixed and undefined multi-layer and a micro-micelle structure that can deliver special multifunctional performance in a response to the special product's performance request. The coating chemical components can be described as that a phase transition material such as petroleum wax, and SPI granular particle, biomaterials, and/or granular materials, organic or inorganic derivative and particle materials, sized in diameter from 0.00001 (micron) to 1000 (micron), could be dissolved or dispersed in the mineral oil by heating and re-condensed and crystallized back into solid bump and particles as the mixed component's temperature is below the melting temperature of mixed components.

[0141] The non-polar lubricant solvents such as mineral oil and alkyl group are saturated carbon and unsaturated hydrocarbons in the range of from C8 to C18. Also, included in the recipes are saturated carbons in the range of C12 to C26 in the range and mostly alkanes, cycloalkanes, and various aromatic hydrocarbons. It can be classified as paraffin, naphthenic, and aromatic. The preferred heating temperature for the mixed chemicals can be as high as 140° F., then, the surfactants or emulsifiers can be added into the mixed solution, resulting in a uniform emulsion with multi-layered shell/core structure.

[0142] Subsequently, a hydrogel polymer and cross-linking agents are added into the solution. The micelle structure disclosed here is just demonstration only. The actual micelle structure might be a hybrid one with an ambiguous intermediate layer or interface instead of a clear shell and core's structure. The SPI or wax particles as the core sphere of micelles are encapsulated within the emulsifier molecules. The emulsifier micelles are hybridized with hydrogel HPAM polymers extended toward the water phases. The emulsifier molecules play essential roles in dispersing the wax or SPI or other micro-nanotextured particles and fiber materials in the hydrogel polymers and solutions temporally. Meanwhile, it also allows the wax or other textured particles to migrate and suspended on the top of the coating film layer. As a result, the hydrophobic coating and bumpy dots and domains can be generated via a porous interpenetrated network.

[0143] After being blended for 5 (minutes), the mixed components can be charged with polar solvents such as water into the mixture. Brookfield viscosity of the mixed materials can be determined at a spindle rotation speed of 6, 12, 30, 60 (RPM). Then, the coating materials are sealed in the package for late use.

[0144] The manufactured coating can be either used to directly spray on the sand or aggregate surface as regular coatings to mitigate the risk of sand dustiness or directly blended into cement matrices as paste ingredients. It can also be added into the cement mix after being diluted with water and add the coatings as aqueous solution as chemical admixture in the blending of cementing operation.

[0145] Concrete Mix: As shown in FIG. 1 of 101, Cement, sand, aggregates (stone), and water are mixed to form concrete. The range of aggregate sizes, from fine sand particles to small to large stones, allows denser packing and minimal air entrapment, leading to greater strength. The water to cement ratio, W/C, is the weight of mix water in the concrete divided by the weight of cement in the concrete. The preferred design range of W/C is ranged from 0.20 to 0.72. It is one of the most important parameters that should be controlled in the casting of concrete products. Less water makes the concrete less workable although it makes concrete have increased compressive strength. In contrast, more water makes the concrete weaker.

[0146] To resolve the dilemma on balancing the workability and concrete composite performance, chemical admixture has been recognized as important components of concrete used to improve its performance since ancient time. As a matter of fact, milk was used by Romans, eggs during the Middle Ages in Europe, sticky rice was identified as secret ingredients discovered by Chinese Scientist for building the Great Wall in China 2000 year ago (Yang, et al. 2010). The American concrete Institute (ACI) defines chemical admixtures as “A material other than water, aggregates, hydraulic cement, and fiber reinforcer used as an ingredient of a cementitious mixture to modify its freshly mixed, setting, or hardened properties and that is added to the batch before or during its mixing”. Evidently, the disclosed coatings meet the above specification and performance standard.

[0147] Mechanical Performance and Components of Concrete: According to English dictionary, concrete is a hard, compact building material formed when a mixture of cement, sand, gravel, and water dried: used for making bridges, road surfaces, etc. The performance of the compacted grains sintered under ambient temperature could be predicted with a simple mixture law. American Concrete Institute (ACI) have developed a convenient table as reference guidance for user/engineer design to make choice on targeted compressive strength of their cement products, then, what kinds of mix proportion (cement, sand, aggregates, and water). A sample mix design is exemplified here per ACI building code: chapter 1. That is, a minimum compressive strength of $f_c=4000$ (PSI) at 28 days requested. Application field: footings, piers, and foundation walls; Air content: 6.0% $\pm$ 1.5%, target water/cement ratio: 0.40 as preferred recipe requirement.

[0148] All our mix recipes are based upon the above suggestion from the above recommended w/c ratio; however, it is not limited to the above. Numerous factors have been in consideration to adjust based upon specific requirements of raw material's properties, engineering applications, and ASTM C standard and ACI code requirements besides the disclosed product's performance.

[0149] Portland Cement: Of these concrete materials, Portland cement has been considered as one important invention. It can be considered as a gluing agent for sand and aggregates. Via hydration or/and pozzolan reaction, it forms stiff and strong ionic bond strength with excellent compres-

sive strength. 28 days of curing schedule have been adapted as a global standard to determine the cement compressive strength (unconfined compressive strength). Within 14 days of cement curing, the cement test blocks are expected to have around 90% of its maximum compressive strength. ASTM 1157 standard defines the types of cements, also ASTM C219 defines what is the hydraulic cement—a cement that sets and hardens by chemical interaction with water and that can do so under water. It is classified as:

[0150] Portland Cement:

[0151] a. Type I: Normal, general purpose

[0152] b. Type II: Moderate sulfate, low heat of hydration

[0153] c. Type II (MH): moderate heat of hydration and moderate sulfate resistance

[0154] d. Type III-High early strength

[0155] e. Type V-high sulfate resistance

[0156] Blended Hydraulic Cement (ASTM C595)

[0157] a) ggbf/slag

[0158] b) Fly Ash

[0159] c) Silica Fume

[0160] d) Calcined shale

[0161] e) Other Pozzolans

[0162] f) Limestone

[0163] ASTM C1157 defines performance-based cement as:

[0164] a) GU—general use

[0165] b) HE—High early strength

[0166] c) MS—Moderate sulfate resistance

[0167] d) HS—High sulfate resistance

[0168] e) MH—Moderate heat of hydration

[0169] f) LH—Low heat of hydration

[0170] ASTM C1157 is a performance-based standard. If the products can achieve the properties, the PLC can have any amount of limestone in its components. ASTM C91 defines the Masonry cement standard: It is primarily used in Masonry and plastering construction, consisting of a mixture of Portland or blended hydraulic cement and plasticizing materials such as limestone, hydrated or hydraulic limestone together with other materials introduced to enhance one or more properties such as setting time, and workability.

[0171] ASTM C1329 defines mortar cement standard, it is classified as: Type N, Type S, Types M. It is like masonry cement in use; however, this specification includes a flexible bond strength requirement. A general use (Type I/II) cement was purchased from local building supplier distributor used as it is. It has a density of 3.15 (g/cm³). Also, a white Portland lime cement was also used for preparing concrete blocks and evaluating its possible used as architecture stone and construction wall sheathing. The recommended dosage level of cement's materials is ranged from 5.0(%) to 90.0(%) by volume fraction of the cement, fine sand, and aggregates. Lime (CaO) or Calcium hydroxide can be added to accelerate the hydrated reaction as partial replacement of cement, preferred dosage is ranged from 0 to 20% over the total Portland cement by weight percentage.

[0172] Aggregates and Sand Materials: Large sized aggregates are preferred to have a high compressive strength. On the other hand, small sized sand can be packed denser that can ultimately hold heavy compressive loads with tight packing and low porosity. The shearing resistant capability will be increased with small sized sand or aggregates. Therefore, laboratory test and onsite field tests are required to make sure that the manufactured concrete products meet

the engineered design requirements, however, in this disclosure, the chemical additives applied in all demonstrated examples are illustration only. No specific engineered parameters are prescribed in the disclosure. Playground sands with very fine size without dust were used in all testing samples. In term of aggregates, all-purpose sands were purchased from home depots used as it is. The nominal coarse aggregate size is ranged from $\frac{3}{4}$ " to $1\frac{1}{2}$ ". The total quantities of regular fine sand plus aggregates are ranged from 5 (%) to 95 (%) by volume fraction.

[0173] Water: In term of water, purified water without Ca^{2+} and Magnesium $^{2+}$ ion is preferred. The ratio of water to cement was varied from 0.20 to 0.72, depending upon the final mix and products desired, however, 0.40 is a standard W/C used most in the tested samples. The preferred W/C level should be 0.45 or less, more preferred less than 0.4, 0.38, 0.36, 0.30.

[0174] Cementitious Materials: In term of cementitious materials, the claimed cementitious materials are defined as any inorganic geomaterials, such as special sand, Sodium silica, micro silica, silica gel geopolymer, swollen clays, kaolin clays in powder, etc., that promote the Pozzolan reactions plus the cement. W/Cm ratio is the weight of mix water in the concrete divided by the weight of the cementitious material, where the cementitious material is a combination of cements and Pozzolans. The blend ratio of chemical additives to cementitious materials is ranged less than 30.0%, more likely, less than 5.0% over the total cementitious materials by weight percentage. Frac sand or proppant materials also called quartz sand, containing up to 99.80 (%) of SiO_2 is an excellent candidate for cementitious materials, having particles size of 0.038 (mm) to 2.0 (mm), defined as 100 mesh and 40/70 mesh, 30/50, 20/40, 16/12, was selected in the disclosure to determine whether these materials can be used as direct replacement of fly ash, micro silica gel, titanic dioxide, silica sand, or other geo-polymeric materials. Since these materials use less energy to be mined, a replacement of fly ash with frac sand provides excellent options for reducing the global CO_2 emission.

[0175] Workability: Once the samples were mixed with disclosed chemical additives, the pot life or working window for all recipes are around 20 to 120 (minutes). No issues have been detected in term of water holding power of the added additives. Based upon the leftover residuals, the materials are reusable within 24 hours with significant coalescence and sintering to each other.

[0176] Mix: The sand or aggregates can be sprayed or pre-blended with the chemical additives before added into the mix with cement. Alternatively, all components of sand, cement, and aggregates can be pre-weighed before blended with water and chemicals. In which, the chemicals are always pre-blended with partial water to dilute chemical further before added into the powder mix to achieve the enhanced distribution. Most of often, about 40 to 50% of added water has been used to pre-blend with chemical additives to give a better distribution of chemicals with other components. Alternatively, the chemical additives are first blended with the portion of dilute water and charged into the mixer first. Then, sand, and aggregate are added 2nd. After the cement is added, mix timer is registered. After 1-2 (minutes), the rest of water is added into the mixer. A high shearing during mixing is preferred. It seems that a well sheared mix and blending can create much better compressive strength and flexible tensile. The mixing time is around

3 to 5.0 (minutes). Preferred to have a shearing and relative fast blending for about 1.0-2.0 (minutes) or so. to achieve the compatibility of coatings with sand/aggregate system. Most of time, a mix and blending operation can be as long as 10 (minutes) before using the batch mixed materials.

[0177] Dust Suppression: In the construction fields, respirable dust is a big safety concern, especially dust from microcrystalline silica due to its high risk of triggering silicosis diseases. The chemical composition and technical solution disclosed are well positioned for suppressing the dustiness of cements, fine sands, and aggregates via spraying or blended with the disclosed chemicals blended in a weight percentage of from 0.1 to 3.0% based upon the total granular solid weight.

[0178] Reinforcement: Chopped fiber glass having a length in 12 (mm) and a diameter of 0.40 (mm) was used to blend with limestone Portland cement. The chemical additives seem to have an excellent mixing capability with fiber glass. Certainly, other reinforcement elements can be added and blended with the chemical additives to achieve the desirable performance. The dosage level of reinforced fiber element added in the concrete mix is in a range less than 5.0 (%) by volume fraction to the total cementitious materials. Plus fiber Reinforcement, more specifically less than 2.5 (%)

[0179] Performance Test: Cement materials are very brittle. Like the ceramics, it requires special qualification test for special applications following recognized testing standard and protocol. Tensile failure is generally attributed to the unstable extension of a critical tensile crack (Li, 2007). ASTM C39 standard was selected in the disclosed test for all samples, in which cylindrical samples were casted., then, unconfined compressive strength (UCS) was determined following the standard testing protocol. Alternatively, ASTM C109 could be used to conduct preliminary screen test. For the tensile failure, ASTM C496 standard was followed to determine the splitting tensile strength with maximum force of the Diametral tension test method. other performance parameters such as water permeability, density, porosity also determined for the fabricated samples disclosed in the examples. Various advantages of the chemical additives and recipes used for enhancing early age strength and promoting the self-healing of concrete durability will be discussed further in explanatory examples 1 to 24.

EXPLANATORY EXAMPLES

Example 1

Chemical Additives and Solution Recipes and Composition

[0180] Chemical Additive composition solution was prepared following the following procedures: Pre-blended 1.60 (g) of FTZ620, a hydrolyzed polyacrylate sodium acrylamide polymer in powder, with 0.8 (g) of sweet rice, purchased from available open market, then, 27.26 (g) of 70 T mineral oil was charged into a 1000 (mL) of glass beaker and stied with magnetic stirred bar. 3.21 (g) candle wax was added into the beaker and the mixture was stirred and heated to 140° F. or above. After the candle wax was totally dissolved in the mineral oil within 5 (minutes), 4.18 (g) of polysorbitan 60 MS NF and 0.25 (g) of PEG100 glyceryl stearate was charged into the mixture and blended for another 5 (minutes), then, the pre-blended FTZ620 and sweet rice were charged into the heated beaker for another two to five

minutes. After the mixed components have a solution temperature over 150° F., then, charged 220 (g) purified water, the solution temperature dropped, continuously blended the components until the solution temperature was reduced to the room temperature. The leftover of purified water (122.80 g) was added into the beaker. The samples showed a color of white emulsion. The products ID was labeled as 3-100-1a, 3-100b. A second batch following the same procedure was used to obtain the emulsion labeled as 4-178-1a. The components and procedures are listed in Table 2.

[0181] Combined both of 3-100-1a and 3-100-1 b together, the viscosity of the mixed solution was determined with a Brookfield viscometer (USS-DVT4 digital rotary viscometer) at the following spindle Rod NO 1 at a rotation rate of 6, 12, 30, and 60 (RPM). The measured viscosity of the coatings prepared following example 1 is listed in Table 3.

[0182] In addition, the measured pH value of the above solution was 6.90. Color of the solution was white cloudy. The solid content of the solution was 9.04 (%). Then, the sample was stored for later use.

Example 2

[0183] To a 1000 (mL) of beaker, 38.96 (g) of 70 T mineral oil was first charged into the beaker, then, 2.89 (g) of soy protein isolate (SPI) having a 90% of protein content, was charged into the beaker, 3.85 (g) oil based polymeric p-MDI solution (50% Concentration) was added to conduct a preliminary polymerization reaction to modify the SPI surface. A magnetic stir bar was used to blend the mixed components, simultaneously, the mixed components were heated until the solution temperature reached 140° F. or above. A mixture of pre-blended of 2.40 (g) of FTZ 620, a hydrolyzed polyacrylate sodium polyacrylamide polymer in powder with 1.20 (gram) of sweet rice in powder, purchased from open market, was charged into the heated mixture in the beaker. The mixture temperature was continuously increased to 150° F. or so. Then, 6.27 (g) of polysorbitan 60 MS NF, PEG100 glyceryl stearate surfactants were charged into the

mixed components for at least another 5 (minutes). Then, the whole mixture components were cooled down to room temperature slowly. Then, 544.05 (g) of purified water was charged to dilute the solution to target viscosity. The obtained solution was labeled as sample of ID: PSMI_3-136. A summary of the components and procedures for preparing the chemical additives is listed in table 4, the measured viscosity in table 5.

Example 3

Preparation of Control Concrete Sample

[0184] 732 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, 300 (gram) of purified water was first charged into a Hobart mixer. 984 (gram) of fine sand (max size in length <0.50 mm), and 984 (g) of coarse aggregate sand (max. size in length <8.0 mm) purchased from Home Depot, were added into the mixer. The weighed Portland cement in the container was transferred into the mixer. The whole mixer components were stirred first slowly, then, increasing the blend rate slowly, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0185] The final water/cement ratio of the blended components was 0.410. The formed slump of cement paste was smooth and created a good cone that could establish a cement block successfully. The mixed components were casted into concrete testing specimen prepared by different sized PVC pipes with different cylindrical size of 3"×6", 1"×2", 2"×4" within 45-75 minutes. The casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in air for 365 days, then, placed in water for another 100 days, exposed to air again for 38 days before submitted for testing on unconfined compression test following ASTM C39 and Brazilian Splitting tension test following ASTM C 496. The sample was labeled as 4-168-1, 4-168-2, and 4-179-2c-1. A typical plot of compressive stress as a function of displacement (4-179-2c-1) is shown in FIG. 3.

TABLE 2

Chemical Additives Used for Modifying the Surface of Cement Mortar and Concrete Aggregates Sample ID: 3-100-1a & 3-100-1b			
Items	Components:	Quantities (g)	% w/w
1	70 T Mineral oil	27.26	6.82
2	Candle Wax	3.21	0.80
3	Polysorbitan 60 MS NF	4.18	1.05
4	PEG100 Glyceryl stearate	0.25	0.06
5	FTZ 620	1.6	0.40
6	Sweet Rice	0.8	0.20
7	Water	362.7	90.68
Sum:		400	100.00
Procedures:			
1	Pre-blend FTZ620 (5) and sweet rice (6)		
2	Charge (1) to a 1000 mL of beaker, charge (2) and stir (1) + (2)		
3	Heat the candle wax (2) to 140° F. and allow the candle wax totally dissolved within minutes		
4	Charge (3) + (4) together blend for 5 (minutes)		
5	charge the pre-blended (5) + (6) together and continuously keep the temperature above 140° F.		
6	Charge the first batch of the water (~220 (g)) and		
7	slowly cool down the components and wait for the emulsion to cool down		
8	Charge the rest of the water (7) to fully cool down the mixture		

TABLE 3

Measured Viscosity Of Chemical Additive Solutions Prepared by Example 1		
Rotation Speed RPM	Viscosity (CPS)	Percentage of Efficiency (%)
6	530	10.2
12	340	14.5
30	241	24.1
60	172	34.4

[0186] Of the three prepared casting samples, the tested samples for compressive strength had average actual dimension in length: 3.06" in diameter, Height: 5.17", and Length/Diameter ratio: 1.90, actual breaking load: 36610 (lbs), ultimate compressive stress: 4995 (PSI) after 500 days and adjusted compressive strength 4921 (PSI). The modulus of elasticity of the tested samples was 437579 (PSI). The splitting tension strength: 1298 (PSI). The average density of the tested sample was 2.276 (g/cm³). The porosity of the tested samples was 0.141 if the particle density is assumed to be 2.65 (g/cm³).

Example 4

Preparation of Concrete Cylindrical Samples with Chemical Additives in Example 1 Incorporated

[0187] 732 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, about 220 (g) of 280.2 (gram) of purified water was first blended with 21.98 (gram) of chemical solution and/or additives from example 1, then, used the leftover water to rinse the wall of the container that held the chemical solution from example 1. Finally, all the 21.98 (g) chemical solution was blended with water and charged into a Hobart mixer. 984 (gram) of fine sand (max size in length <0.50 mm), and 984 (g) of coarse aggregate sand (max. size in length <10.0 mm) purchased from Home Depot, were added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes). The measured pH value of the chemical solution is 7.5 and the solution density is 0.99 (g/cm³).

[0188] The final water/cement ratio of the blended components was 0.410. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 3"x6", 1"x2", 2"x4" as form frame for casted specimen samples. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in air for 365 days, then, placed in water

for another 100 days, exposed to air again for 38 days before submitted for testing compression following ASTM C39 and Brazilian Splitting tension test following ASTM C 496.

[0189] Of the average two of the prepared casting samples (4-179-2C-7, 4-179-2C-10), the tested samples for compressive strength had average actual dimension in length: 3.08" in diameter, height: 5.99", and length/diameter ratio: 1.99, actual breaking load: 50705 (lbs), ultimate stress: 6924 (PSI) and adjusted tension: 6895 (PSI) after 500 days. The modulus of elasticity of the tested samples was 437579 (PSI). The splitting tensile strength of the tested samples was 1338 (PSI). The average density of the tested sample was 2.297 (g/cm³). The porosity of the tested samples was 0.1333 if the particle density assumed to be 2.65 (g/cm³).

Example 5

Concrete Samples Prepared with Chemical Additives of Example 2 Incorporated

[0190] 730 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, about 220 (g) of 300 (gram) of purified water was first blended with 21.98 (gram) of chemical solution and/or additives from example 2, then, used the leftover water to rinse the wall of the container that held the chemical solution from example 2. Finally, all the 21.98 (g) chemical solution was blended with water and charged into a Hobart mixer. 984 (gram) of river fine sand (max size in length <0.50 mm) and 984 (g) of coarse aggregate sand (max. size in length <10.0 mm) purchased from Home Depot, were added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0191] The final water/cement ratio of the blended components was 0.410. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 3"x6", 1"x2", 2"x4" for casted specimen samples. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for at least 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in air for 365 days, then, placed in water for another 100 days, exposed to air again for 38 days before submitted for testing compression following ASTM C39 and Brazilian Splitting tension test following ASTM C496

[0192] Of the average four of prepared casting samples (4-179-1c-1, 4-179-1c-2, 4-179-1c-4, 4-168-3), the tested samples for compressive strength had average actual dimension in length: 3.07" in diameter, height: 5.81", and actual breaking load: 56914 (lbs), ultimate compressive stress: 7738 (PSI) and adjusted compressive strength 7640 (PSI). The modulus of elasticity of the tested samples was 44525 (PSI). The Brazilian splitting tension is 1453 (PSI). The average density of the tested sample was 2.280 (g/cm³). The porosity of the tested samples was 0.140 if the particle density was assumed to be 2.65 (g/cm³).

TABLE 4

Chemical Additives Used for Modifying the surface of Cement and other concrete components Sample ID: 3-136-1 and 4-173-1			
Items	Components	Quantities (g)	% w/w Note:
1	70 T Mineral oil	38.96	6.49 3-136-1 and 4-173-1
2	Soy Protein Isolate	2.89	0.48 have the same recipe different batch
3	Oil Based p-MDI solution	3.85	0.64
4	FTZ 610/620 in powder	2.4	0.40
5	Sweet Rice	1.2	0.20
6	Polysorbotan 60 MS NF	6.27	1.05
7	PEG100 Glyceryl stearate	0.38	0.06
8	Water	544.05	90.68
Sum:		600	100.00
Procedures:			
1	To a 1000 mL of beaker, charge (1) and (2) + (3)		
2	Stir (1) + (2) + (3) mixed components under magnetic stir bar		
3	Heat the mixture temperature to 140 0 F., then, charge (4) + (5).		
4	Continuously increase the temperature to 150 0 F. or above		
5	Charge (6) and (7) and blended for another 5 (minutes)		
6	Cool down the mixture and reduce the temperature to room temperature		
7	Add (8) and measure the viscosity of the final products		

Example 6

Assessment of Mechanical Performance of Tested
Examples 3, 4, 5

[0193] Of the tested samples, the ultimate compressive strength (UCS) was calculated by the maximum breaking force divided by the area of compressed specimen per ASTM C39 for cylindrical samples. Plot of compressive stress as function of strain in (in/in) in control concrete cylindrical samples is shown in FIG. 3. After an initial linear portion lasts up to 1500 (PSI) stress load, the curve became non-linear with slightly large strain between 1500 (PSI) and 4000 (PSI) with a small, registered stress increased. The non-linearity is believed to be primarily a function of the coalescence of micro-cracks at the paste-aggregate interface. The ultimate stress was reached at 6000 (PSI). The testing results met the expected performance value as anticipated. The applicants believe that at a stress of 6000 (PSI), a large crack network was formed with the concrete. The strain was 0.10 (%) at 4000 (PSI) and 0.15 (%) at 6000 (PSI) that fitted the expected average values of regular concrete product performance.

[0194] At 4000 (PSI), the first crack in concrete was initialized, while at 6000 (PSI), extended cracking networks were generated with a significant failure of the tested samples. It seems that there were some of autogenous recoveries in the tested samples based upon the special treatment condition of tested samples. Since all the samples were placed in water tanker for 7 days before placed in air for 365 days. The samples were only fully cured under the best Pozzolan condition for 7 days. Therefore, it had a 4000 (PSI) of preliminary UCS. From 4000 (PSI) to 6000 (PSI), the achieved 2000 (PSI) strength might come from the post autogenous self-healing of natural cement recovery after left in water tanker for over 100 days after one year in the air.

TABLE 5

Measured Viscosity of Chemical Additives and Solutions Prepared by Example 2		
Rotation Speed RPM	Viscosity (CPS)	Percentage of Efficiency (%)
6	424	42.6
12	352.4	70.7
30	199.8	99.9
60	100	99.9

[0195] In contrast, a plot of compressive stress as a function of strains of the examples 4, 6 shows sharp difference from the stress and strain curve of example 3 control. Concrete blended with chemical additives in examples 4 and 5 showed great advantages of its resistance to products strain failure. None of the samples failed at around 4000 (PSI). It has a full self-healing at the stress level of 4000 (PSI). The tested sample from example 5 will sustain the stress up to a strain (%) larger than 1.6 (%), which is 10 times more than that from control sample from example 3. The breakup strain in SPI sample was more than 2.3 (%), which is 15.0 times more than control sample from example 3.

[0196] The high strain failure resistance fits the characteristic behavior of viscoelastic materials. Practically speaking, the UCS developed in examples 5 can serve as a building block of high strength concrete materials. Further statistic study with one-way ANOVA of the tested samples listed in Table 6 suggests that at a 95% confident interval, the samples of example 5 incorporated with example 2 recipe significantly outperforms the ultimate compressive strength of control sample of example 3 statistically. In contrast, the samples of example 4 prepared with example 1 recipe shows somewhat enhancement in comparison of its UCS with control example, but not statistically significant.

[0197] Based upon the classification of products specification listed in table 1, products delivered by utilizing the recipe of exam 1 can be used for residential and commercial markets, of exam 2 in a high strength building project.

TABLE 6

One Way ANOVA of Measured Ultimate Compressive Stress (UCS) on the Tested Cylindrical Samples Per ASTM C39					
Sample ID	Description	Means (PSI)	StDev (PSI)	COV (%)	Stat.@ $\alpha = 0.05$
Exam 3	3" x 6" Cylindrical tube Cement Samples	4996	1121	22.4	0
Exam 4	3" x 6" Cylindrical tube Cement Samples Incorporated with example 1 chemical additives	6924	1875	27.1	=
Exam 5	3" x 6" cylindrical tube Cement Samples Incorporated with example 2 chemical additives	7738	534	6.9	+

Note:

0 = baseline test value; = statistically equivalent, + statistically enhanced on its value.

[0198] Besides the compressive strength, the modulus of resilience is also important parameter, it defines the amount of strain energy per unit volume (i.e., strain energy density) that a material can absorb without permanent deformation resulting in failure of the structure. It can be expressed as equation 3:

$$\text{Modulus of Resilience (MOR)} = \frac{\text{Yield Stress}^2}{2 \text{ Modulus of elasticity}} \quad (3)$$

[0199] Toughness is another important material's property; it is defined as the ability of a material to absorb energy and plastically deform without fracturing. That is, the amount of energy per unit volume that a material can absorb before rupturing. This measurement of toughness is different from that used for fracture toughness later in the application, which describes load bearing capabilities of materials with flaws. It is also defined as a material's resistance to fracture when stressed.

[0200] Table 7 lists the calculated modulus of resilience and toughness of each tested sample, also, the relative comparison for each tested sample. Cylindrical sample materials of example 4 had its yield compressive strength increased by 84%, modulus of resilience increased by 42 times more than regular standard concrete, toughness increased by 9 times. More specifically, example 5, the compressive strength increased by 200 (%), the MOR 52 times, relative toughness 15 times.

TABLE 7

Measured Performance Character with Selected Concrete Testing Specimen								
RUN ID	Sample ID	Yield Stress (PSI)	Relative CS (%)	Modulus PSI	Modulus of Resilience PSI	Relative MOR	Toughness	Relative Toughness
Exam 3	4-179-2c-1	4067	1.00	543521	15	1	461	1.00
Exam 4	4-179-1c-4	7550	1.86	44585	639	42	4283	9.28
Exam 5	4-168-3	8415	2.07	45127	785	52	6854	14.85

Note:

Modulus of Resilience = compressive stress²/(2 MOE)

[0201] As a result, the added chemical additives significantly reduce the brittleness of the prescribed concrete materials and increased its elasticity and plasticizing capabilities before the materials are broken catastrophically in both the samples 4 and 5. The applicants believe that just as disclosed in the proposed self-healing mechanisms, Soy protein's viscoelastic properties with its α -helix coiled and β -sheet conglycinin components were successfully hybridized with the brittleness of concrete.

Example 7

Assessment of the Tested Sample's Fracture Cracking Resistance

[0202] Concrete cylindrical samples were prepared with a casting size of 2"x4". The procedure for preparing the testing samples like examples 4 and 5 is comprised of weighing 352 (gram) of Portland cement (Type I/II GU), fine sand of 678 (gram), gravel of aggregates of 1205 (gram), tap water of 141 (gram), 22.0 gram of chemical additives from example 1 or 2 was blended with half of tap water. Then, the water containing the chemical additives was poured into a mixer, all cement was first added, then, the fine sand and the gravel of aggregates were added and blended for 2 minutes, then, all leftover water and chemical additives or solution added into the mixer and blended for 3-4 minutes with enough shearing of the mixed components. Then, the mixed components were molded into the 2"x4" PVC pipeline. All samples were sealed into aluminum foils and air-dried at room temperature for 24 hr. before emerging in water tankers until the specimen were ready for testing. All samples were immersed in water tanker for over 365 days. Then, all samples were dried for at least one day before being tested with a special modified fracture testing procedure to obtain its critical stress intensity factor (SIF) as an alternative approach for determining the sample's self-healing capabilities.

[0203] As shown in FIG. 5, the pre-sized samples were submitted for a modified Brazilian splitting tension test by carrying with diametrically opposite concentrated loads on a disc specimen. Both fracture toughness (K_{IC}) and tensile strength were assessed with the modified Brazilian tensile geometrical configuration as shown in FIG. 5. The test is comprised of making a notched sample having a groove with a length of a_0 in depth opened through one face of the disc along the loading axis. The critical fracture stress intensity factor (SIF) can be calculated using the following equation (4) based upon reference (Singh and Pathan 1988):

$$K_{IC} = 1.264 F \frac{\sqrt[2]{a_0}}{tD} \tag{4}$$

where K_{IC} is the critical strength intensity factor (SIP) of the tested sample, F is the force placed on the one flattens face of the modified Brazilian disc sample, a_0 is the depth of notched in z axis, t the thickness of disc and D is the diameter of disc.

[0204] The calculated fracture critical stress intensity factor (SIF) of specimen prepared with the recipes of examples 2 and 3 is listed in table 8. It seems that the K_{IC} is a function of angle β . In both samples of 4-179-1T and 2T, at $\beta=30$ (degree), the K_{IC} has a maximum value of 5.967 (MPa m^{1/2}) for soy protein isolate and sweet rice encapsulated hydrogel emulsion and 6.802 (MPa m^{1/2}) for wax and sweet rice encapsulated hydrogel emulsion. The fracture toughness shows great variation in the tested samples due to variation of sample's notch size, density. Statistically, the difference of the fracture toughness are minor of the sample prepared with the recipe of example 1 vs. example 2 as listed in Table 8.

[0205] In addition to the fracture toughness, the Brazilian tensile strength of the tested samples was also determined by using equation 5 with a correction factor of tested sample (Yue, et al., 2006).

a.
$$\sigma_t = Y_c \frac{2 P_{max}}{\pi D t} \tag{5}$$

where $Y_c=0.2621 (t/D)+1$, D is the diameter of the disc, t the thickness of the disc, P_{max} is the maximum breaking force placed on the flatten surface of the tested samples, σ_t is the tensile strength calculated based upon the Brazilian test method.

[0206] The Brazilian tensile strength (σ_t) for the sample prepared from example 5 had a tensile strength of 9.891 (MPa) with soy protein isolate and sweet rice incorporated in its recipe (example 2). In contrast, Brazilian splitting tensile strength (BSTS) of example 1 recipe is only 5.275 (MPa). As such, the BSTS prepared with example 2 recipe is 187 (%) better than that of the prepared with example 1 recipe as listed in Table 9. Potential explanation is that non-covalent hydrogen bonds in the SPI/SR recipe are mainly accounted for enhanced performance of example 5.

TABLE 8

Modified Dometrial Compressive Test for Determining the Critical Stress Intensity Factor (SIF - ②) Based upon the Modified ②azilian Test Method													
Items	RUN ID	Type CA	Wt (g)	H (mm)	D (mm)	L (mm)	Thick (t) (mm)	Pre-cut crack (②) (mm)	F (KN)	β	Area (mm ²)	Density (g/cm ³)	② (Mpa m ^{1/2})
1	4-179-1T3	Exam 2	121.5	46.5	52	26.5	12	12	26.89	30.7	2015②	2.274	5.967
2	4-179-1T2	Recipes	136.5	48	52	31	13	12	28.3	36.6	2003.1	2.15②	5.797
3	4-179-1T3		111②	45	52	24.5	12	3.5	19.32	28.1	2010②	2.270	2.315
4	4-179-1T4		123.4	48	52	25	6	2.5	25.05	28.8	2044②	2.414	5.074
5	4-179-2T1	Exam 1	117.16	46	52	28.5	8	7.5	25.85	30.7	2009.2	2.200	②.802
6	4-179-2T2	Recipes	104.5	45	52	23	11.5	3	21.13	26.3	2020.7	2.248	2.446
7	4-179-2T3		109.6	47.5	52	25	12	3	28.65	28.8	2038.3	2.151	3.179
8	4-179-2T4		141.6	47	52	30	12.5	2.5	25.71	35.3	1996.2	2.364	2.500

② indicates text missing or illegible when filed

TABLE 9

Measured ②azilian Tensile Strength (②) with Modified ②zalian Test Configuration of the Samples as shown in FIG. 5													
Items	Notebook ID	Type CA	Wt (g)	H (mm)	D (mm)	L (mm)	Thickness (mm)	Maximum Force (KN)	β	Area (mm ²)	Density (g/cm ³)	BD Tensile (MPa)	Ave② (Mpa)
1	4-179-1T2-1	Exam 2	87.1	47	52	18	18	21.94	20.3	2067.8	2.340	8.79	
2	4-179-1T2-2	recipe	194.7	47	52	20	40	30.48	22.6	2058.9	2.354	10.99 ^(a)	9.891
3	4-179-2T2-5	Exam 1	86	48	52	18	18	9.70	20.3	2076.8	2.301	4.04	
4	4-179-2T2-6	recipe	75.2	48	52	15	15	8.95	16.8	2087.0	2.402	4.47	
5	4-179-2T2-2		133.1	48	52	23	23	19.40	26.3	2525.6	2.291	6.32 ^(b)	
6	4-179-2T2-3		176.9	48	52	30	29	25.10	35.3	2694.5	2.264	6.27 ^(c)	5.275

Note:
^(a) Sample is the same as exam 8a.
^(b) the exam 8d and
^(c) 8e.

② indicates text missing or illegible when filed

Example 8

Water Permeability of Selected Examples

[0207] The permeability is an important parameter useful for characterizing the performance of concrete products (Yang, 2019). A highly permeable concrete material, in general, implies that the concrete is highly shrinkable with low compressive strength and shortened life for the structural applications. Therefore, the samples tested in exam 7 were also submitted to water absorption and permeability test, in which 2"×2" disc specimens were coated with wax on all surface except that only the bottom surface was immersed into water tanker in 1-2 mm distance in depth contacted with water surface. The water quantities absorbed or penetrated in the tested samples were determined by taking the sample out at special interval of 0.5, 5, 10, 20, 40, 60, 120, . . . 1000 (min.) measured with a balance having an accuracy of 0.001 (g). As shown FIG. 7, three stages of water gain seem self-evident that: 1) diffusion following the Lucas Washburn equation; 2) transition with swelling of the cement matrix; 3) water absorption of the cement following the first order of water absorption linearly.

[0208] In the initial diffusion process, the square root of sampling time should be proportional to the absorbed water

where L is the water molecular penetration distance from the cement surface into the porous body in porous micro-tubing structure, r is the radius of micro-capillary, μ the viscosity of solution, e the contact angle of solid cement to liquid in the micro-capillary tube, t is the time for liquid water penetrating itself into the porous media of cement, γ the surface tension of liquid.

[0209] For long-term water absorption, the fitting model can be expressed as L is proportional to the sampling time as equation (7):

$$l=at+b \quad (7)$$

[0210] Since the liquid mass can be expressed as, $m=L \cdot s \cdot \rho$, a permeability K can be obtained from equations (8) and (9) and simplified into the following two fitting model equations:

$$P_1=K_1\sqrt{t}+B_1 \quad (8)$$

$$P_2=K_2t+B_2 \quad (9)$$

Where P_1 is micro-capillary pressure driving permeability mass gained, K_1 is a fitting constant for permeability rate, B_1 is the interception of diffusion related permeability, P_2 is the long-term diffusion rate, K_2 is the long-term diffusion constant, B_2 is the constant for long-term diffusion of tested samples.

TABLE 10

Measured Water Absorption & Permeability on the Surface of the Cement Samples							
Water Absorption & Permeability on the Surface of Samples (g)/m ²							
			Exam 8a	Exam 8b	Exam 8c	Exam 8d	Exam 8e
			NoteBook ID (2020-RB1-210-)				
			1T2-1	1T2-2	2T2-1	2T2-2	2T2-3
			Surface Area (m ²)				
Time (min.)	✓time	Model fitting	0.002067	0.0020589	0.0020768	0.002525	0.002694
0	0		0	0	0	0	0
0.5	0.71		48.4	48.6	144.5	158.4	185.6
5	2.24		58.1	97.1	221.5	158.4	222.7
10	3.16		67.7	145.7	240.8	158.4	185.6
20	4.47		96.8	145.7	288.9	158.4	185.6
40	6.32		96.8	194.3	337.1	237.6	259.8
80	8.94		145.1	242.8	385.2	356.4	371.2
120	10.95		193.5	291.4	481.5	435.6	408.3
160	12.65		193.5	242.8	240.8	158.4	222.7
200	14.14		169.3	242.8	240.8	158.4	222.7
400	20.00		193.5	242.8	288.9	198.0	259.8
740	27.20		193.5	291.4	433.4	316.8	334.1
1020	31.94		241.9	291.4	433.4	316.8	371.2
k1	from 0	$P_1 = k_1$	13.727	25.584	37.528		
b1	to 80	$\text{sqrt}(\text{time}) + b_2$	22.391	30.428	92.558		
r ²	(min.)		0.9107	0.9408	0.8509		
k2	from 100	$p_2 = k_2$	0.0595	0.0672	0.2573	0.2116	0.1822
b2	to 1020	$(\text{time}) + B_2$	168.39	228.41	197.78	123.06	190.27
r ²	(min.)		0.6842	0.867	0.9236	0.9236	0.9972
Date:	Self-healing Test & permeability after pre-crack:		Aug. 3, 2021		Aug. 3, 2021		
			✓		✓		

quantities per square meter that can be expressed as equation (6).

$$l = \left(\frac{r \cos(\theta)}{2} \right)^{1/2} \left(\frac{\gamma}{\mu} \right)^{1/2} t^{1/2} \quad (6)$$

[0211] In the transition stage, a sharp increase of water mass gain occurred after 80 (minutes) of the water soaking test. The mechanisms of the absorption mass increase were unknown, potentially due to inertia and porous structural influence or it might be due to a swelling of cement micro-capillary tubes after being soaked long enough. As a

result, the data from 75 (minutes) to 100 (minutes) was excluded from the modeling equation. After 120 (minutes), the mass gain from the samples immersed in water tank follows the 1st order of reaction instead of the Lucas Washburn diffusion process. The common consent is that there should be more autogenous self-healing reaction occurring in exam 8b. The interaction of the chemical additives originated from exam 8b and carbon dioxide or calcium ion was stronger than from exam 8a.

[0212] As shown in FIG. 7, if the co-relationship (r^2) listed in Table 10 as a criterion for determining whether the samples follow special wicking processes, it seemed that the samples of made of example 2 recipe as core and hydrogel polymer as shell (exam 8a, 8b, and 8c) follows the Lucas-Washburn diffusion more closely than exam 8d, 8e in the initial stage of permeability test. In contrast, samples made of exam 8d, 8e are hydrophobic in the initial stage, resistant to wetting out the cement matrix materials. It follows the 1st order reaction model of more closely.

Example 9

Assessment of Self-Healing of Pre-Cracked Samples

[0213] Self-healing capability is critical for enhancing the durability of concrete products. Of the 5 samples from example 8, exam 8a and exam 8c were selected and pre-cracked at a special proof load. The samples after being cracked were defined as exam 9a and exam 9b. Other samples of exam 8b, 8d, and 8e were fully broken up with Brazilian test standard and their maximum tensile Strength was determined. The results of BSTS of the tested samples were listed in Table 8. In this disclosure, the self-healing capabilities of these pre-cracked samples are assessed by determining the water permeability and Brazilian Splitting Tensile Strength (BSTS) of both samples.

[0214] Self-healing Capabilities Determined by Water Permeability: As shown in FIG. 6, water lines from its seeping out of the immersed water tanker of the exam 9a and 9b clearly demonstrated that the significant cracks present here along the whole section cross the sample surface perpendicular to the flatten edge sections after being pre-cracked. Water permeability test was conducted in both two samples. It is showed that the samples after being pre-cracked were filling water in its micro-capillary tubes much quickly. Within 10 (minutes), diffusion controlled wicking processes dominate the process. The Lucas Washburn equation can be used to obtain a linear plot of permeable water quantities vs. $\sqrt{\text{time}}$. Within 120 (minutes), the samples were soaked with water cross its microstructure.

[0215] As showed in FIG. 8, if the initial wicking process was considered as key water permeability for controlling the self-healing processes, the water dissolved with CO₂ and cationic ion such as calcium will be much easily penetrated the micro-capillary tubes in exam 9a than exam 9b samples. The test results of water permeabilities on both samples of exam 9a and 9b after pre-cracking are listed in Table 11.

[0216] In the initial stage of water perm, the pre-cracked samples of examples:

[0217] b. P(exam 9a)=120.59 K1+6.5.

[0218] c. P(exam 9b)=152.74 K1+65.7.

[0219] The pre-crack % can be calculated following equation (10):

Pre-crack % = (10)

$$\frac{\text{Pre-cut crack in width}}{\text{Crack size in width before break_up}} = \left(\frac{K1(\text{after})}{K1(\text{before})} \right) \times 100$$

where K1 (before) is the rate of water permeability of tested sample before pre-cracking; K1 (After) is the rate of water permeability of the tested sample after the pre-cracking test.

[0220] d. K1(exam. 8a)=13.73 (Table 10); K1(exam 9a)=120.59 (see Table 11)

[0221] e. Pre-cracking (%) (exam 9a)=(120.59/13.73)=878

[0222] f. K1(exam. 8c)=37.54 (Table 10); K1(exam 9b)=152.73 (See Table 11)

[0223] g. Pre-cracking (%) (exam 9b)=406

[0224] Interestingly, although % increase of water permeability from exam 9a is two times more than exam 9b, it still had less water permeability in the whole of total than exam 9b since:

[0225] K1 (perm of exam 9a)=120.59 (g/m² min),

[0226] K1 (perm of exam 9b)=152.73 (g/m² min.).

[0227] The above results clearly demonstrate that before being cracked, the concrete block coated with soy protein is much more water resistant than with waxy due to its tightened microstructure with less porosity.

[0228] Table 13 lists the water permeability test results after both specimens were placed in a water tanker for self-healing of 30 days at room temperature. After water soaking treatment, both was placed in the outdoor condition for over 15 days before measuring their water permeability by simple gravimetric method again. The applicants believe that the pre-cracked specimens of Exam. 9a and exam. 9b underwent an extensive hydration within the water tanker for self-healing its cracks. Both soy protein and sweet rice and hydrogel polymers play crucial roles in enhancing the recovery of damaged interphase zone cross the cement and sand/aggregates.

TABLE 11

Measured Water Permeability of Pre-cracked Specimen Before Self-Healing (Aug. 5th, 2021)		
	Example 9a	Example 9b
	Notebook ID	
	2020-RB1-214-1T2-1	2020-RB1-214-2T2-1
Time (min.)	Measured Specimen Weight (g) in Water Container	
0	167.9	120.9
0.5	168.131	121.414
5	168.403	121.796
10	168.736	121.962
20	168.65	121.932
40	168.721	122.018
80	168.739	121.977
120	168.757	122.115
	Surface Area Measured on Each Specimen (mm ²)	
	2060	2068

TABLE 11-continued

Measured Water Permeability of Pre-cracked Specimen Before Self-Healing (Aug. 5th, 2021)		
Time (min.)	Calc. Water Absorption and Permeability (g/m ²)	
0	0	0
0.5	112	249
5	244	433
10	406	514
20	364	499
40	399	541
120	407	521
Model Fitting with Lucas Washburn Equation		
K1:	120.59	152.73
B1:	6.5	65.7
r ² :	0.9803	0.924

cracks pre-cracked in the samples of exam 9a and 9b were fully closed with tightened microstructure of tested samples.

[0235] Self-Healing Capabilities Determined by Brazilian Splitting Tensile Strength (BSTS): In term of mechanical property of self-healing, Table 14 summarizes the calculated average tensile strength of cracked samples after being placed in water tanker for recovery within one month (30 days). The tested results have the following Brazilian splitting tensile strength (BSTS): BSTS (exam. 9a)=8.75 (MPa) and BSTS(exam. 9b)=6.01 (MPa).

[0236] Although not totally understood, it is believed that the soy protein and sweet rice components and its interaction with other chemical additives in the formulated recipe is a critical factor that enhances the performance of final concrete products. A self-healing efficiency (SHE) (δ) based upon the Brazilian splitting tensile strength (BSTS) can be defined as shown in equation (11):

TABLE 12

Measured Brizailian Tensile Strength (②) to Achieve the Pre-crack with Partial Failure without Total break-up of the Tested Samples													
Items	Notebook ID	Type CA	Wt (g)	H (mm)	D (mm)	w (mm)	Thickness (mm)	Maximum Force (KN)	β	Area (mm ²)	Density (g/cm ³)	BD Tensile (MPa)	Pre-crack % (②)
Exam 9a (exam 8a)	4-173-1T2-1	Exam 2	167.8	48	52	22	35	18.31	25.0	2060.0	2.327	6.24	878.0
Exam 9b (exam 8c)	4-173-2T2-1	Exam 1	120.9	48	52	20	25	25.09	22.6	2068.9	2.337	9.40	404.0

Note:

(②) Calculated pre-crack percentage.

② indicates text missing or illegible when filed

[0229] Quantitatively, the calculated permeability parameter K1 as listed in Table 13 is equivalent to 6.401 (g/m² min.) for exam 9a and 5.529 (g/m² min.) for exam. 9b. Since K1 is proportional to the porosity of porous media (Yang, 2019), the closed channels can be considered as % self-healing efficiency (SHE) of the tested samples estimated as:

[0230] SHE (exam. 9a)=(1-6.401/13.73)×100+100=151.6 (%) for soy-based recipe.

[0231] SHE (exam. 9b)=(1-5.529/37.54)×100+100=185.3 (%) for wax/sweet rice-based recipe.

[0232] Based upon the above analysis, it is self-evident that after the cracked samples placed in water tanker for one month, both samples of exam 9a and 9b are sealed with less porosity and with tightened microstructure.

[0233] An imaging photo of the tested specimens (exam 9a and exam 9b) is shown in FIG. 9 after both were soaked in water tanker for over one hour to run the water permeability study. The water marks along the cracking line on the surface of both specimens are much smaller and less intensive than these of pre-cracked identical specimens shown in FIG. 6 before water self-healing in water tanker. Evidently, the ingress and pathways of water to penetrate itself through opened micro-cracked channels are effectively blocked by the self-activated bonding interface across the cement and sand/aggregates.

[0234] In conclusion, both the SPI/SR/PU recipe (example 2) and Wax/SR/PU recipe (example 1) provides more than 150 (%) self-healing capability to the cracked samples if water permeabilities are used as the tools of assessment. The

$$\delta_m = \frac{\sigma_{after\ heating}}{\sigma_{virgin}} \times 100(\%) \quad (11)$$

where δ_m is the self-healing efficiency (SHE) per mechanical property; $\sigma_{after\ heating}$ is the tensile strength of the tested samples after 28 days healing in water tanker; σ_{virgin} is the tensile strength of the tested samples before being cracked of the virgin samples.

[0237] If the BSTS data listed in Table 9 is used as the virgin tensile strength for exam 9a and 9b, the self-healing efficiency for both samples can be calculated using equation (11) as:

[0238] δ_m (exam 9a)=8.75 (Mpa)/9.891 (Mpa)=88.5 (%)

[0239] δ_m (exam 9b)=6.01 (Map)/5.275 (MPa)=114.0 (%)

The summarized SHE for both exam 9a and 9b are listed in Table 14.

Example 10

Charactering Hydro Nano/Textured and/or Dual Phobic Dot Domains of Coated Thin Film on the solid Surface

[0240] Droplet dosage level of selected coatings from example 1 was spread on a glass plate with thin layer coatings. After dried at room temperature of 78° F. for over one week or so, the coated thin film coatings were characterized by a sessile droplet method, in which a droplet of purified bottle water was first pumped into a micro-syringe, then, the mass of the water microdroplet was measured with a micro-balance (0.001 (g)). The mass of the droplet was

used as a controllable variable in the test. The size of the water droplet can be calculated if the water droplet is assumed to be a perfect spherical shape. In addition, the static contact of the droplet (θ) toward the solid surface interface was determined by taking instant photos with iPhone image. Then, the contact angle of the droplets was determined by analyzing the droplet shape of images. In addition, once the droplet was placed on the coating surface, it was tilted with an angle of (α) that can be registered by an angle meter synchronized with the glass flatten plate movement. The tilt angle is also called as a pinning angle that is the droplet breakup transition angle stucked on the glass plate surface or rolled out of the glass plate surface.

TABLE 13

Water Permeability of Specimens after Being Self-Healed for One Month			
Time (min.)	✓time	Exam 9a Water Permeability (g/m ²)	Exam 9b
0	0	0	0
0.5	0.71	139.8	183.5
1	1.00	121.4	101.6
5	2.24	181.4	139.2
10	3.16	98.7	91.0
20	4.47	93.4	79.0
30	5.48	106.4	93.9
40	6.32	121.4	105.5
60	7.75	143.2	108.8
90	9.49	152.9	125.7
127	11.27	163.0	140.1
167	12.92	167.9	140.6
189	13.75	173.2	149.7
236	15.36	181.4	156.0
242	15.56	184.3	155.5
327	18.08	203.2	166.1
380	19.49	207.5	175.8
417	20.42	212.4	183.0
445	21.10	220.1	179.1
665	25.79	232.7	203.7
Fitting Parameters with Washburn Lucas Model			
K1:		6.401	5.529
K2:		82.44	67.8
r ² :		0.959	0.973

as the tilted angle is plotted as a function of water microdroplet mass, it is a function of water microdroplet mass and scale in dimension.

[0242] As the water droplet is less than 30 (mg), equivalent to a spherical size in radius about 1.0 (mm), the tilted contact angle is dramatically increased without breakup or pinning at a water droplet radius in size of less than 0.5 (micron). The coatings show a super hydrophobicity with contact angle larger than 130 (degree). The specific surface area exposed to the air becomes significantly increased. The coatings tend to provide more hydrophobic textured porous surface to the water droplet, leading to super hydrophobic. However, it does not allow the micro-droplet off the coating surface. This is fundamentally different from the mechanisms of Lotus leaf types wetting phenomena. In that case, the coating surface is super hydrophobic as the contact angle is larger than 90 (degree). Say 130 (degree).

[0243] The driver for microdroplet off the leaf surface is the low surface energy bubble in the leaf microstructure. As such, microdroplet will instantly roll off the coated solid surface in the lotus leaf. However, in the disclosed coatings, as the water microdroplet size is less than 0.50 (micron), tilted angle of microdroplet is 180 or more, the water microdroplet will not roll off the glass plates. In contrast, it suspended itself on the leaf or glass plate as the size of water droplet in radius is smaller than that of 1.0 (mm). The coatings behave in a superior hydrophilic manner absorbed on the leaf although it has a tilted contact angle of more than 180 (degree).

[0244] On other hand, as the microdroplet size is increased beyond 30 (mg) with a particle size of more than 1.00 (mm) in radius, It has breakup and pinning contact angles around 45-50 (degree). It will perform just like regular static contact angle and rolling off the glass plate. The coatings will not hold excessive water. The coatings have a superiority of slipperiness. It behaves in a superior hydrophobic manner based upon the current wettability definition; however, it is hydrophilic in nature.

[0245] Structurally in micro/nano scales, different from lotus leaf, the coating used the conceived water or nonpolar solvent under its porous network as intimate contact interface. The water and non-polar solvent were settled in the

TABLE 14

Measured Brazilian Splitting Tensile Strength (BSTS) Ultimate Compressive Strength After Self-Healing in Water Tanks for One Month													
Titens	Notebook ID	Type CA	Wt (g)	H (mm)	D (mm)	w (mm)	Thikness (mm)	Maximum Force (KN)	β	Area (mm ²)	Density (g/cm ³)	BD Tensile (MPa)	(SHE) δ - (%)
Exam 9a	NB232-1	Exam 2	167.6	48	52	22	35	25.7	25	2000	2.327	8.75	88.5
Exam 9b	NB232_2	Exam 1	120.9	48	52	20	25	16.04	22.6	2068.9	2.337	6.01	114

② indicates text missing or illegible when filed

[0241] Both the measured static contact angle and tilted angle are listed in Table 15. A plot of static angles and tilted angles as a function of droplet mass is shown in FIG. 10. Of the 20 measured contact angles, an average static contact angle of the coatings is 51 (degree). Standard deviation of static contact angle is about 6 (degree). The variation of static contact is independent upon the size of water microdroplets based upon the plot of FIG. 10. The coating is either hydrophilic or hydrophobic based upon current industrial definition, kinds of in-between as an intermediate. However,

nano/textured pockets or valley instead of entrapped air bubbles in the lotus leaf. The soy proteins or modified soy proteins and wax serves as hydrophobic domains with nanotextured ridges and porous frame for the coatings. A reorientation of the SPI and sweet rice functional groups or switching could potentially turn the coatings from hydrophobic into hydrophilic, versus visa, in response to the environmental condition.

[0246] As such, depending upon the microdroplet size and dimension, the SPI and sweet rice components serve as a hydro-phobic dual dot domain material in response to make

an adaption of its wetting to environmental condition. As shown in FIG. 2(g), the fine micro voids around the inter-phase zone of sand/aggregate and cement matrix could potentially reduce the impact force placed on concrete to achieve the instant high early age compressive strength. The coatings can be defined as a coated surface having its static contact angle larger than 30 (degree) and less than 90 (degree) and a tilted angle larger than 20 (degree). Preferred tilt angle is larger than 45 (degree) as slippery coatings and water microdroplet size of from 0.01 (mm) to 5.00 (mm) in radius.

[0247] To assess how the added chemical additives on the concrete performance, a series of experimental mix and concrete blocks were prepared using general cement, fine sand, aggregate, chemical additives, mixed with water at targeted water cement ratio, were carried out with different curing time by days of 1, 3, 7, 14, 28 days, 2 months, and 3 months. Different cement and replacement of fly ash with silica quartz, or frac sands used in hydraulic fracking operation, were also explored. Benefits of these cement combination are explained in brief in examples 11 to 18.

Example 11

Miscellaneous Hydraulic-Cement Concrete Products with Disclosed Chemical Additives as Key Ingredients

[0248] 730 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, about 300 (gram) of purified water was first charged into a Hobart mixer. 984 (gram) of fine sand (max size in length <0.50 mm), and 984 (g) of coarse aggregate (max. size in length <10.0 mm) purchased from Home Depot, were subsequently added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed of blending, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0249] The final water/cement ratio of the blended components was 0.41. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3" as frame for casted specimen samples. Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

TABLE 15

Measured Static Contact Angle (Θ) and Tilted Contact Angle (α) (pinning Angle) from the coating Film on a Flat Glass Plate					
Test RUN	Sample ID	Droplet Wt (mg)	Static Angle (Θ) Degree	Tilted Angle (α) Degree	Note (Tilted)
1	NB225-1	6	53	360	Suspended/upside-down
2	NB225-2	53	47	50	

TABLE 15-continued

Measured Static Contact Angle (Θ) and Tilted Contact Angle (α) (pinning Angle) from the coating Film on a Flat Glass Plate					
Test RUN	Sample ID	Droplet Wt (mg)	Static Angle (Θ) Degree	Tilted Angle (α) Degree	Note (Tilted)
3	NB225-3	40	47	50	
4	NB225-4	79	55	28	
5	NB225-5	99	48	44	
6	NB225-6	30	55	64	
7	NB225-7	54	62	44	
8	NB225-8	31	57	54	
9	NB225-9	6	57	330	Suspended/upside-down
10	NB225-10	26	44	77	
11	NB225-11	23	42	65	
12	NB225-12	14	53	180	Suspended/upside-down
13	NB225-13	26	47	82	
14	NB225-14	30	43	53	
15	NB225-15	30	46	48	
16	NB225-16	83	63	41	
17	NB225-17	70	49	43	
18	NB225-18	30	58	44	
19	NB225-19	33	50	53	
20	NB225-20	34	52	48	
	Ave.	40	51	88	
	Stdev.:	25.5	6.0	93	
	COV (%):	64.0	12	106	

[0250] A micro Instron lab test machine was used to determine the total force of the tested sample at a fixed displacement rate of 35 (PSI)/sec. The unconfined ultimate compressive strength (UCS) was determined calculated from equations 12:

$$\sigma_{UCS} = \frac{F_t}{A} \quad (12)$$

where F_t is the total force of being placed on the top surface of tested cylindrical sample; A is the surface area of the cross-section; σ_{UCS} the ultimate compressive strength (UCS) of cylindrical samples as it broken up.

[0251] The tested sample had a notebook ID of 90A-1. All the tested samples had a diameter of 1" in size and roughly 2" in length. The mean of ultimate compressive strength (UCS) from three individual testing samples is reported as follows: 1904 (PSI) at 1 day, 4572 (PSI) @ 7 days; 3358 (PSI) @ 14 days; 4823 (PSI) @ 28 days.

[0252] In addition, the tensile strength of the samples was also determined by breaking at least three of individual samples with Brazilian tensile test protocol and calculated following the equation 3 with only Brazilian disc without notched groove geometry only. The mean of testing results is listed here: 344 (PSI) @ 1 day; 726 (PSI) @ 7 days, 936 (PSI) @ 14 days, 1084 (PSI) @ 28 days.

Example 12

[0253] 730 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, 21.98 (g) of 3-100-1 chemical additives were blended with 150 (gram) of water, then, adding about 150 (gram) of purified water was charged into a Hobart mixer. 984 (gram) of fine sand (max size in length <0.50 mm), and 984 (g) of coarse aggregate sand (max. size in length <10.0 mm)

purchased from Home Depot, were subsequently added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0254] The final water/cement ratio of the blended components was 0.41. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3" as form frame for casted specimen samples. Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

[0255] The measured mean of ultimate compressive strength is 4198 (PSI) @ 1 days; 4687 (PSI) @ 7 days; 3632 (PSI) @ 14 days; 3754 (PSI) @ 48 days. The measured Brazilian splitting tensile strength is 999 (PSI) @ 1 days; 909 (PSI) @ 7 days; 962 (PSI) @ 14 days; and 668 (PSI) @ 28 days, 437 (PSI) @ 70 days. The density of the samples is 2.414 (g/cm³) and the porosity is 0.089.

Example 13

[0256] 730 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, about 300 (gram) of purified water was first charged into a Hobart mixer. 984 (gram) of fine sand (max size in length <0.50 mm), and 984 (g) of coarse aggregate sand (max. size in length <10.0 mm) purchased from Home Depot, were subsequently added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0257] The final water/cement ratio of the blended components was 0.41. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3". Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before left in water for 3 days, 38 days, 45 days, 52 days before submitted for testing.

[0258] The measured mean of ultimate compressive strength is 3279 (PSI) @ 3 days; 3851 (PSI) @ 38 days; 3088 (PSI) @ 45 days; 5055 (PSI) @ 52 days. The measured Brazilian splitting tensile strength is 971 (PSI) @ 3 days; 1439 (PSI) @ 38 days; 900 (PSI) @ 45 days; and 1000 @ 52 days. The density of the samples is 2.356 (g/cm³) and the porosity is 0.111.

Example 14

[0259] 730 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container,

then, about 150 (g) of 280 (gram) of purified water was first blended with 21.98 (gram) of chemical solution and/or additives from example 1, then, used the left-over water to rinse the wall of the container that held the chemical solution from example 1. Finally, all the 21.98 (g) chemical solution will be blended with water and charged into a Hobart mixer. 519.3 (gram) of fine sand (max size in length <0.50 mm), and 563.9 (g) of coarse aggregate sand (max. size in length <10.0 mm) purchased from Home Depot, were added into the mixer, first, then, 276.7 (g) standard frac sand of 40/70, and 608.1 (gram) of 100 mesh frac sand, donated from High Roller Corporate's mill, were added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0260] The final water/cement ratio of the blended components was 0.41. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3". Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

[0261] The average value of ultimate compressive strength of three tested samples is 3962 (PSI) @ 3days; 4091 (PSI) @ 35 days; 4098 (PSI) @ 45 days; 4325 (PSI) @ 52 days. The average Brazilian tensile strength of the tested three individual samples is 714 (PSI) @ 3 days; 1100 (PSI) @ 38 days; 1204 (PSI) @ 45 days; and 708 (PSI) @ 52 days. The average density of the samples is 2.273 (g/cm³). The porosity of the tested sample is 0.142.

Example 15

[0262] 453 (gram) of Portland limestone cement (Type I), manufactured by TXI, was weighed, held in a container, then, 10 (gram) of white lime (CaO), then, about 100 (g) of 153 (gram) of purified water was first blended with 37.0 (gram) of chemical solution and/or additives from example 1, then, used the left-over water to rinse the wall of the container that held the chemical solution from example 1. Finally, all the 37.0 (g) chemical solution will be blended with water and charged into a Hobart mixer. 609 (gram) of frac sand of 40/70, 617 (g) 100 mesh of frac sand, and 200 mesh of super fine sands, donated from Covia Corporation, were added into the mixer, were added into the mixer. The weighed Portland cement in the container was transferred into the mixer. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0263] The final water/cement ratio of the blended components was 0.41. The formed slump of cement paste was smooth and created a good cone. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3". Total numbers of samples prepared were 24. To make the slump uniformly distributed into the test

tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

[0264] The average value of ultimate compressive strength of three tested samples is 2447 (PSI) @ 1 days; 4498 (PSI) @ 7 days; 2977 (PSI) @ 14 days; 3050 (PSI) @ 28 days. The average Brazilian tensile strength of the tested samples is 919 (PSI) @ 1 day; 1000 (PSI) @ 7 day; 2050 (PSI) @ 14 days; 2405 (PSI) @ 28 days. The density of the samples is 2.172 (g/cm³). The porosity is 0.180.

Example 16

[0265] 731 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, 38 (gram) of lime (CaO), manufactured by Chemstar. About 150 (g) of 192 (gram) of purified water was first blended with 9.42 (gram) of chemical solution and/or additives from example 1, then, used the left-over water to rinse the wall of the container that held the chemical solution from example 1. Finally, all 9.42 (g) chemical solution will be blended with water and charged into a Hobart mixer. Then, 527 (g) standard frac sand of 40/70, and 269 (gram) of 100 mesh frac sand, and 192 (g) of 200 mesh quartz sand, donated from Covia Corporation, were added into the mixer. Finally, 40 (gram) of chopped fiberglass having a cutting length <12 (mm) and a size of 20 (micron) in diameter was added into the blended mixing components to make the final blended components. Then, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0266] The final water/cement ratio of the blended components was 0.26. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3". Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

[0267] The means of the ultimate compressive strength of the three individual testing samples for different curing time is 2258 (PSI) @ 1 day; 3179 (PSI) @ 7days; 3168 (PSI) @ 14 days; 3900 (PSI) @ 28 days. The average Brazilian tensile (BD) strength of the tested samples is 1565 (PSI) @ 1 day; 2492 (PSI) @ 7 days; 2504 (PSI) @ 14 days; 2208 (PSI) @ 28 days. The porosity of the samples is 0.177. The density of the samples is 2.18 (g/cm³).

Example 17

[0268] 432 (gram) of Portland cement (I GU Type), manufactured by TXI, was weighed, held in a container, then, 93 (gram) of lime (CaO), manufactured by Chemstar. About 150 (g) of 278 (gram) of purified water was first blended with 37 (gram) of chemical solution and/or additives from example 1, then, used the left-over water to rinse the wall of the container that held the chemical solution from example

1. Finally, all 37 (g) chemical solution was blended with water and charged into a Hobart mixer. Then, 581 (g) standard frac sand of 40/70, and 589 (gram) of 100 mesh frac sand, and 593 (g) of 200 mesh quartz sand, donated from Covia Corporation, were added into the mixer. Finally, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes).

[0269] The final water/cement ratio of the blended components was 0.59. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3". Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

[0270] The means of the ultimate compressive strength of the three individual testing samples for different curing time is 1310 (PSI) @ 1 day; 1809 (PSI) @ 7days; 2151 (PSI) @ 14 days; 1827 (PSI) @ 28 days. The average Brazilian tensile (BD) strength of the tested samples is 427 (PSI) @ 1 day; 1695 (PSI) @ 7 days; 1500 (PSI) @ 14 days; 2165 (PSI) @ 28 days. The porosity of the samples is 0.185. The density of the samples is 2.161 (g/cm³).

Example 18

[0271] A mortar recipe was prepared by the following components: 1880.8 (gram) of Portland cement (I/II GU Type), manufactured by TXI, was weighed, held in a container, then, about 50 (g) of 112 (gram) of purified water was first blended with 3.1 (gram) of chemical solution and/or additives from example 1, then, used the left-over water to rinse the wall of the container that held the chemical solution from example 1. Finally, all 3.1 (g) chemical solution+the rest of purified water were blended and charged into a Hobart mixer. Then, 282.1 (g) of 200 mesh quartz sand, donated from Covia Corporation, were added into the mixer. Finally, the whole mixer components were stirred first slowly, then, adding speed, finally with a maximum speed setting rate of 7, blended for about 3-5 (minutes) before transferred to make testing samples.

[0272] The final water/cement ratio of the blended components was 0.61. The mixed components were casted into concrete testing specimen molds prepared by different sized PVC pipes with different cylindrical size of 1"x2", 1.5"x3". Total numbers of samples prepared were 32. To make the slump uniformly distributed into the test tube, a ramping rode was used to densify the mixed cement components for 25 times for each prepared sample. Then, the casted samples were immediately sealed with aluminum foils over 24 hr. before placed in water tanks for 7 days, then, left in water for 1 days, 7 days, 14 days, and 28 days before submitted for testing.

[0273] The average value of ultimate compressive strength (UCS) of the tested samples is 499 (PSI) @ 1 day; 1612 (PSI) @ 7 days; 1927 (PSI) @ 14 days; 4246 (PSI) @ 28 days. The mean of Brazilian tensile strength from three individual test sample is 367 (PSI) @ 1 day, 1443 (PSI) @ 7 days, 1705 (PSI) @ 14 days, 2208 (PSI) @ 28 days.

[0274] An overview of these test results as listed in Table 16 shows that the ultimate compressive strength (UCS) of all tested samples is higher than that of the required minimum 1750 (PSI) in compressive test per ACI building code Table 1, and higher than that of 600 (PSI) in splitting tensile strength requirement per ACI standard. The UCS originated from example 17 is the lowest one obtained due to high W/C=0.59 and added excessive 4.0% lime in the Portland Lime Cement. It still has a value of UCS equivalent to 1827 (PSI) @ 28 days. The Brazilian splitting tensile of example 17 sample is 1215 (PSI) @ 28 days. It is two times more than required 600 (PSI) @ 28 days per Mortar or Masonry specification. The highest average ultimate compressive strength (UCS) listed in Table 16 is 7738 (PSI) @ 500 (days) after one and half year from example 5 and average Brazilian tensile strength 1458 (PSI). The products were prepared with example 2 recipe as special chemical additives at 0.097

(%) active ingredient to the total solid of concrete materials (cement+sand+aggregates). 100 mesh, 40/70, and 30/50 frac sand from North White sand were used to prepare samples for products as wall sheathing also show an excellent UCS of 050 (PSI) @ 28 days per specification with its splitting tensile strength of 2475 (PSI) at 28 days.

[0275] About 2.0% chopped glass fibers were added in example 16 in Portland lime Cement/North white frac sand mix to demonstrate the potentials of chemical additives used for Architectural types of building and construction. In the selected example 18, mortar and Masonry types of application were explored with silica sand as Pozzolan reactive components in Portland cement and chemical additives as special agents for enhancing its workability. The UCS @ 28 days in example 18 reached more than 4000 (PSI) that met the structural concrete requirement larger than 2500 minimum (PSI). Also, the Brazilian splitting tensile strength was 2208 (PSI), much higher than required 600 (PSI).

TABLE 16

Summary of Concrete Tested Sample Performance by Incorporating Bioinspired Chemical Additives/Solution													
Description		Exam 3	Exam 4	Exam 5	Exam 11	exam. 12	Exam 13	Exam 14	Exam 15	Exam 16	Exam 17	Exam 18	
NoteBook ID													
Items	Mix Components	Unit	4-168-1, 4-168-2, 4-179-2c-1 (ctd.)	179-2c-7	179-1c-2, 1-179-1c-4	90A	90B	97A-1	1010	NB128-1	NB 159-1	NB125A	NB 139-3
1	Cement (Type)		PC(I/II)	PC(I/II)	PC(I/II)	PC (I/II)	PC(I/II)	PC(I/II)	PC(I/II)	PLC(I)	PLC(I)	PLC(I)	PC(II)
	Cement Wt (g)	g	732	732	730	732	732	732	732	453	731	432	1880.8
	Lime	g	0	0	0	0	0	0	0	10	38	93	0
	Fine Sand	g	984	984	984	984	984	984	519.34	NA	0	0	0
4	Aggregate(all Purpose)	g	984	984	984	984	984	984	563.9	NA	0	0	0
Cementitious Materials													
5	Frac Sand 40/70	g	0	0	0	0	0	0	275.7	609	527	581	0
6	100 Mesh	g	0	0	0	0	0	0	608.1	617	269	589	0
7	200 Mesh	g	0	0	0	0	0	0	0	821	192	593	282.1
8	Fume Silica Gel	g	0	0	0	0	0	0	0	0	0	0	0
9	Solvent												
10	Water	g	300	280.2	300	300	300	300	280	153	192	277.7	112
	Chemical Additives	g	NA	3-100-1	NA	3-100-1	3-100-1	NA	3-100-1	3-100-1	3-100-1	3-100-1	3-100-1
	Additives Quantities	g	0	21.96	21.96	0	21.96	0	21.96	37	9.42	37	3.1
Reinforcing Element													
11	Chopped Fiber Glass	g	0								40	0	0
	Total Wt	g	3000	3002	3020	3000	3000	3000	3002	2500	1989	2602.7	2275
	Calc. W/C;		0.41	0.410	0.438	0.410	0.410	0.41	0.410	0.40	0.26	0.59	0.061
	Calc. Additive/Cement;		0.000	0.030	0.030	0	0.03	0	0.030	0.082	0.013	0.000	0.002
Calc. CM/Cement;													
Performance													
Aging Conditions:		A	A	A	B	B	B	B	B	B	B	B	B
Aging Time		Days			Compressive Strength (PSI)								
1					1904	4198				2447	2258	1310	499
3							3279	3962					
7					4572	4687				2498	3179	18(2)	1212
14					3358	3332				2977	3168	2153	1927
28					482(2)	3754				3050	3900	1827	4246
38							3851	4081					
45							3(2)	4098					
52						4371	5055	4329					
500			4995	7738									

TABLE 16-continued

Summary of Concrete Tested Sample Performance by Incorporating Bioinspired Chemical Additives/Solution													
Description		Exam 3	Exam 4	Exam 5	Exam 11	exam. 12	Exam 13	Exam 14	Exam 15	Exam 16	Exam 17	Exam 18	
		NoteBook ID											
4-168-1, 4-168-2, 4- 4-179-2C-10 &4- 4-168-3, 4-179-1c-1, 4-													
Items	Mix Components	Unit	179-2c-1 (ctd.)	179-2c-7	179-1c-2, 1-179-1c-4	90A	90B	97A-1	1010	NB128-1	NB 159-1	NB125A	NB 139-3
Aging Time		Days	Brazil ² n Tension (PSI)										
	1		344				999	971	714	919	1565	427	327
	3												
	7		726				909			1000	2492	1685	1443
	14		930				962			2050	2504	1500	1705
	28		1084				668			2405	2208	1215	2208
	38							1439	1100				
	45							900	1204				
	52							1000	708				
	500			1335	1453								
Density		g/cm ³	2.276	2.297	2.28	2.3445	2.414	2.356	2.275	2.172	2.18	2.161	2.31
Porosity (***)			0.141	0.133	0.140	0.115	0.089	0.111	0.142	0.180	0.177	0.185	0.128
Toughness:													

Note:

** Products were targeted for mortar application in construction.

A: 24 hr, air dried, then, immersed in water until ready to be tested at room temperature (70 to 80° F.)

B: 24 hr, air dried, then, placed the samples in water tanked for 7 days, left in air until tested.

(***) Assume that the solid density of single particle is 2.65 (g/cm³)

² indicates text missing or illegible when filed

[0276] In comparison of the mechanical performance of example 15 vs. 16, lime of 10 (gram) was added in example 15, 92 (gram) in example 16. It seems that added excessive lime will increase the 28 day's UCS of example 16, Also, the UCS @ 7 days, however, it would create slight lower UCS @ 1 days in example 16 than example 15. On the other hand, the Brazilian splitting tensile strength @ 28 days in example 16 is less than in example 15.

[0277] An overview of the mechanical properties of the tested samples on test data from examples 3, 4, 11, 12, 13 as shown in FIG. 11 suggests that the samples blended with chemical additives (exam 4 and exam 12, 13) have many advantages over the control sample (exam. 3 and exam 11). Both UCS and Brazilian splitting tensile strength of the samples from exam. 12 and 13 on the 1st day have a higher value than that of regular concrete control samples (exam 11). A blend of regular Portland cement with the chemical additives from example 1 can double its UCS from 2000 (PSI) to about 4000 (PSI) and BSTS from 500 (PSI) to 1000 (PSI). For the blend of Portland lime Cement with North white sand, the UCS at the early age of 24 hr. is larger than 2500 (PSI). In contrast, for regular cement blend (exam 11), the UCS (filled diamond shape with break dot line in FIG. 11) would not arrive at the same level as exam 12 until tested at 7 days. At 14 days, the sample would lose some strength and level off at 28 days without any change since then at a UCS of 5000 (PSI) as shown in Exam. 3 after 500 days placed in environment.

[0278] Although the UCS originated from exam 12 is lower at 28 days than from ctrl sample from exam 11, the UCS from exam. 12 continuously increase until arrive at around 7700 (PSI). As described previously, the increased UCS from exam 4 is different from example 3 in a regular autogenous self-healing of concrete since the strain failure percentage in example 4 is 10 times more than in example 3, which is attributed to the enhanced viscoelasticity or viscous plasticity in the sample of example 4.

[0279] Fundamentally, the applicants believe that self-activated non-covalent and hydrophobic dispersive bonds from soy proteins and waxy materials are mainly accounted for the long-term performance enhancement of the disclosed concrete products. The applicants also believe that the water and non-polar solvents potentially serve as stimulating and sensing agents in response to the environmental moisture and temperature cycles to make self-adjustment of polymeric mobility to mitigate the risk of the internal and external shrinkage. Potentially, the Soy protein molecules serve as connection knots and as central points of cross-linking network in the hydrogel polymers as moisture reservoir, which prevents the concrete structure free from the cracking and catastrophic failure. Intuitively, a study on the exothermic hydrated reaction of cement with the other components would provide us more in depth understanding of how the activation energy is related with the behavior of blended components disclosed in detail further as follows.

Example 19

Calorimeter Testing of Hydraulic-Cement Concrete Curing with Insulated Coffee Cups

[0280] To a Hobart mixer (5 lit.) charged 549 (gram) of Portland cement (Type I/II GU), then, 738 (gram) of all-purpose sands (aggregate) was added into the mixer, 738 (gram) of fine sands. 16.5 (gram) of chemical additives

prepared with example 1 recipe was weighed into a plastic cup. Then, adding 100 (gram) of purified water into the plastics cup containing the chemical additives. The diluted chemical additives were subsequently blended with mixed solid particles for 3 minutes, charged another 120 (gram) into the mixer, and blended for another 2 minutes. Then, the mixed components were used for preparing the test samples. About 392.0 (gram) cement mix was packed and sealed in a coffee cup. The temperature of the mixed cement was measured by inserting a thermometer in the sealed coffee cup. Besides the interior mixed cement temperature, the environmental temperature of the coffee cup was also monitored with a separate thermometer as a function of sampling time.

Example 20

[0281] To a Hobart mixer (5 lit.) charged 549 (gram) of Portland cement (Type I/II GU), then, 738 (gram) of all-purpose sands (aggregate) was added into the mixer, 738 (gram) of fine sands was also added into the mixer. 16.5 (gram) of chemical additives prepared with example 2 recipe was weighed into a plastic cup. Then, adding 100 (gram) of purified water into the plastics containing the chemical additives. The diluted chemical additives were blended with mixed solid particles for 3 minutes, charged another 120 (gram) into the mixer, subsequently blended for another 2 minutes. Then, the mixed components were used for preparing the test samples. About 274.5 (gram) cement mix was packed and sealed in a coffee cup. The temperature of the mixed cement was measured by inserting a thermometer in the sealed coffee cup. Besides the interior mixed cement temperature, the environmental temperature of the coffee cup was also monitored with a separate thermometer as a function of sampling time.

Example 21

[0282] To a Hobart mixer (5 lit.) charged 549 (gram) of Portland cement (Type I/II GU), then, 738 (gram) of all-purpose sand (aggregate) was added into the mixer, then, 738 (gram) of fine sands was added into the mixer, subsequently, adding 100 (gram) of purified water into the plastics cup containing the chemical additives, blending all mixed solid particles for at least three minutes, charged another 120 (gram) purified into the mixer, finally blended for another 2 minutes. Then, the mixed components were used for preparing the test samples. About 225.4 (gram) cement mix was packed and sealed in a coffee cup. The temperature of the mixed cement was measured by inserting a thermometer in the sealed coffee cup. Besides the interior mixed cement temperature, the environmental temperature of the coffee cup was also monitored as a function of sampling time.

Example 22

[0283] To a Hobart mixer (5 lit.) charged 549 (gram) of pre-blended lime Portland Cement following the commercial instruction, then, 738 (gram) of all-purpose sand (aggregate) was added into the mixer, 738 (gram) of fine sands was also added into the mixer, subsequently, adding 100 (gram) of purified water into the plastics cup containing the chemical additives, blending all mixed solid particles for at least three minutes, charged another 120 (gram) purified into the mixer, finally blended for another 2 minutes. Then, the mixed components were used for preparing the test samples.

About 225.4 (gram) cement mix was packed and sealed in a coffee cup. The temperature of the mixed cement was measured by inserting a thermometer in the sealed coffee cup. Besides the interior mixed cement temperature, the environmental temperature of the coffee cup was also monitored as a function of sampling time.

Example 23

Determining the Hydraulic-Cement Mix Thermodynamic Reaction

[0284] Since reaction of cement components with chemical additives and water are exothermic, it typically can be proceeded in an adiabatic condition. As such, the thermal energy generated can be monitored by measuring the variation of the temperature on the mixed components. Assume that the mass of a cement mix is m , the cup's wall has double thin layers fully vacuumed, the internal energy driven by the reaction of cement components can be calculated by equation 13 as follows⁴:

⁴ https://en.wikipedia.org/wiki/Adiabatic_process

$$\Delta E_i = \int_0^t \alpha_n n R T(0) \{T(t) - T(0)\} dt \quad (13)$$

where ΔE_i is the differentiation of internal energy of the mixed cement components before and after blended with chemical additives in the coffee cup, α_n is the number of freedoms divided by 2, n is the molar number, R is the universal gas constant (8.315 J/mol), $T(0)$ is the environmental temperature in Kelvin, $T(t)$ is the mixed cement component's temperature measured by thermal coupling probe inserted in the coffee cup at sampling time t .

[0285] The curing temperature of concrete in an adiabatic calorimeter test is arguably the one variable that has the most significant effect on the rate of hydration. The thermal probe temperatures measured in examples 19, 20, 21, 22 plotted as a function of sampling time in FIG. 12 shows very interesting hydration reaction. Firstly, at around 45 (minutes), there exists a maximum exothermic reaction peak in example 22. In the rapid setting cement from commercial products, Ca(OH)_2 or/and CaO are catalyzer used for accelerating the cement curing. Clearly, the cement components in this mix cured within 1 hour. The curing temperature of the samples of example 22 dramatically increased until reached its maximum temperature of 95° F. within 47 (min.) after mixed with water. It is evident that a carbonation reaction occurred in this test group of CaO with water and create CaCO_3 ionic bonds in the formed matrix. In general, the fast cured samples will have better early age compressive strength, however, it has a poor durability.

[0286] Secondly, within the first 4 hour, an advanced exothermic curing reaction occurred in the tested sample of example 20 until it reached the first maximum peak temperature of 88° F. at a curing time of $t=4.7$ hr. Then, the curing rate flattened out for a short time around the sampling time of 5 hour. The curing reaction rate continuously increased until it reached the 2nd maximum exothermic peak of curing temperature of 95° F. at around 6 (hour) before being slowed down. Then, the hydration processes followed a similar trend to that of virgin cement mix as example 21. Like the inflammation in bone fracture recovery, it was discovered that the thermal transition might be attributed to the soy protein's α -helix coil and β -sheet transition that promotes the protein alignment with non-covalent hydrogel bonds. Exothermic reaction occurred as the soy proteins

(example 20) were swollen to promote the intimate contact of its selves with cement matrix elements.

[0287] It is believed that below the temperature of 88° F., the intimate contact was positively charged with the hydrogel polymers. At 88° F. or so, the SPI or other encapsulated materials in the core layer were suddenly exposed to the water and mineral oil solvent system. More extensive swelling occurred within the soy proteins, leading to the more intimate contacts of soy protein molecules with cement (—OSiO—) at a temperature range from 88° F. to 95° F. The soy proteins might create the needed clots cross the side walls of crack in width to heal the damage of cement cracks. As the temperature of mixed cements was above 95° F., the curing profiles of following the same trend as regular cement mix as example 21. It is suggested that in the initial 6 hours or so, the soy proteins in the core layer might be intercalated in the valleys of cracks of cements and aggregates although the soy proteins were highly flowable and soluble under the alkali conditions. This thermodynamic data strongly supports the proposed bonding mechanisms occurring in the mixed cement composite bonds; however, this type of thermodynamic behavior has never been reported previously.

[0288] Thirdly, slightly different from example 20, the increase of cement temperature of the example 19 was much smaller than that of example 21 (ctrl.) when the mixed component's temperature was below 72 (degree). A sharp increase of the mixed components was observed at the curing time around 3.75 (hr.). The temperature of mixed components changed from 72 to 79° F. or so. within 5 to 10 (min.) that was attributed to the melting of wax and underwent a phase transition. At a temperature of cured components above 79° F., the wax was totally dissolved in the mixed solvent that release newer and fresh surface of sands and aggregates that could be interacted with cement components, leading to extensive hydration in the sample of example 19. The maximum curing temperature arrived at 101° F. around 8 hours. Dominant interaction in the sample of example 18 was mainly involved in the bonding process from non-polar dispersive force contributions. Again, a melting of wax or its combination with polymers to plasticize the cement interface is novel, however, a promotion of compressive strength and Brazilian splitting tensile strength to the cement products has never been reported. Thereof, the applicants discovered that a non-covalent hydrogen bond or/and dispersive bond might be more critical for the enhanced early age compressive strength and long-term durability instead of CSH bonds widely promoted in current cement research community. Assume that the internal energy ΔE (ctrl.) of example 21 (control) was the base of all tested samples, the integration of the relative internal energy per the adiabatic process for each tested sample from examples 19, 20, 21, 22 was calculated with an excel spreadsheet program could be expressed as equation (14). The relative degree of hydration (α) is, in general, taken as the ratio of heat evolved at time t to the total amount of heat available as equation (14)

$$\alpha(t) = \frac{\Delta E(t)_i}{\Delta E(\text{ctrl})} \quad (14)$$

where the ΔE (ctrl.) is the calculated internal energy of example 21 based upon the equation 14, i in the equation is

18, 19, 20, 21 originated from examples 18, 19, 20, 21, $\Delta E\%$ (%) is a relative energy percentage obtained from equation (14) based upon the measured temperature profile data shown in FIG. 12.

[0289] Assume that 1 (gram) of samples from examples 19, 20, 21, and 22 each was mixed, and the bonding of particles in each sample would have been independent from each other, the bonding contribution defined as dispersive force, non-covalent hydrogen bonds, regular hydration cation bonds, and special CaO ionic bonds could be calculated based upon the calculated relative internal energy from the tested samples listed in table 17.

TABLE 17

Bond Type Contribution Classified Based Upon the Calculated Relative Energy of Example 21 as Base					
Sample ID	Chemical Additives	Relative Exothermic Energy (%) $\Delta E\%$	Unreactive (%)	Calculated Bond Contribution (%)	Bond Classification
Exam. 19	Aqualub (Wax based emulsion)	77.2	22.8	19.30	Dispersive Force
Exam. 20	Aqualub 3-100 (SPI)	54.3	45.7	13.58	SPI hydrogen Bonds
Exam. 21	Control	100	0	25.00	C S H Hydration
Exam. 22	Rapid Setting (CaO)	56.9	43.1	14.23	Lime Hydration (Ionic)
Sub Total		288.4	111.6	27.90	non-hydration

[0290] Evidently, based upon the classified bond contribution calculated in Table 16, there would have been about 14.0(%) originated from SPI, 20.0% from dispersive force; and 25.0% from calcium silicate hydrate, and 14.0% from lime hydration reaction. Potentially, there would be still 28.0% of unreactive surface or partition that can be hydrated, or recovered for further self-healing, which would support the proposed bonding mechanisms present in the disclosed state-of-art recipes. Alternatively, an activation energy (E_a) could have been determined if a series of experimental test could be conducted with examples 19, 20, 21, 22 recipes separately under different isothermal temperature condition with Arrhenius equation (Poole, et al., 2007). The calculated overall rate of hydrate reaction could be used to determine the cement paste's mobility and molecular phase transition further. The self-healing mechanisms for enhancing the early age and late strength of concrete products are thereby conceptually proved by the disclosed experimentation as illustrated herein.

Example 24

Moisture Absorption and Desorption of Tested Concrete Prisms

[0291] Of the blended materials from examples 19, 20, 21, 22, two prism bars for each set condition in a dimension of 1"x1"x12" were prepared in the lab sealed in aluminum foils overnight. After 24 hours, the samples were de-molded and weighed. The aluminum foil was taken off. The samples were exposed all its surface to ambient temperature and submitted to an ambient drying condition for over two months. The weight loss of each prism sample was recorded as a function of sampling time in days. A plot of moisture percent loss as a function of sampling time present in FIG. 13. Evidently, the moisture losses from example 21 and 22 are identical, about 5.0% above its original weight over the two months, it can be considered as regular normal samples. In contrast, the moisture loss of example 19 is less than

2.0%. It is implied that the waxy layers might be a super protection for the under-layered water molecules. The moisture loss from example 20 was a little bit over 2.0 (%). For sure, it is also excellent in term of holding the moisture and water under its multifunctional coating layer.

[0292] Based upon the disclosure present here, it is therefore demonstrated that the objects of the present invention are accomplished by the chemical components and added solution chemicals of matter and methods useful in cementitious construction material's application as special early age strength and long-term durability enhancement agents through self-healing and biomineralization functionalized

improvement that can be applied in residential and commercial building, and potential for high rising and high strength building application. By combining the chemical additives with other engineered reinforced materials such as glass fibers, steel bar fibers, and other bio-engineering reinforced elements of materials, its applications and identified benefits for enhancing cement early age strength and durability to mitigate the risk of fracking cracks in concrete structure as a self-activated healing agent, has been disclosed herein. It shows that a selection of the multifunctional coating' and additive's components of disclosed lubricant, micro-nano-textured, dual phobic domain particles and phase transition materials, emulsifiers, hydrogel polymers, and cross-linking agent, and made-up water/polar solvent percentage by weight percentage, processing for blending the above chemical additives with cement components of sand, aggregates, and water mix by weight and/or volume percentage, can be determined by one having ordinary skill in the art without departing from the spirit of the invention herein disclosed and described. It should therefore be appreciated that the present invention is not limited to the specific embodiments described above, but includes variation, modification, and equivalent embodiments defined by the following claims.

We claim:

- 1) bioinspired self-healing chemical additives and solution comprising of by weight percentage:
 - a) soy protein isolate (SPI) as micro-nano/textured dot dual phobic domains from 0.001(%) to 40%
 - b) mineral oil as liquid lubricant or/and non-polar solvents from 0.01% to 50%.
 - c) hydrolyzed polyacrylate sodium acrylamide polymer as a suspending agent from 0.001 to 35%
 - d) polysorbate as surfactants/and emulsifiers: 0.0001 to 20.0%
 - e) Water: 1.0% to 99.0% as balance agent
 - f) The combination of (a)+(b)+(c)+(d)+(e) is equivalent to 100 by weight percentage, useful as an admixture of hydraulic-cement concrete driven by a self-activated

polymer's phase transition of from 30 to 200° F., resulting in more than 3000 (PSI) of early age compression strength, 7000 (PSI) of ultimate compressive strength (UCS), 80.0% of cracking self-healing, and 9 times more toughness than the virgin concrete, 13 times more of its modulus of resilience, characterized as a hydro dual phobic domain coating via a dynamic tilted contact angle larger than 30 (degree) and static contact angle from 30 and 90 (degree) measured on a thin film solid surface.

2) The chemical components of claim 1, wherein the chemical components of micro-nano/textured dot and dual domains are candle wax, paraffin wax, slick wax, or ethylene streamside synthesis wax, carbamate wax, natural organic and organic synthesized wax that has a melting point of at least 35° C. or above, and/or biomaterial or bio-derivatives such as sweet rice flour, soy wax, soy protein isolate particles, soy protein concentrates, or/and its derivatives from SPI functionalized with amine or hydroxyl, carboxyl, and aldehyde ester, amide, and polyamide functionalities, or/and the combination of petroleum based or biobased materials, polylactic acid ester, inorganic silica particles, thereof, the dosage level of these hydrophobic/hydrophilic domain's materials is ranged from 0.01% to 40.00% over the total weight percentage of claim 1.

3) The chemical components of claim 1 wherein the lubricant or/and non-polar solvent is mineral oil, saturated hydrocarbon, alkyl chains of ethylene carbon, liquid paraffin, kerosene, petroleum distillates, and higher alkane, cycloalkanes, the alkyl carbon chains from C6 to C20, the dosage levels of the lubricant or/and non-polar solvent or chemicals from 1% to 50% over the total weight percentage of claim 1.

4) The chemical composition of claim 1, wherein the hydrogel polymers are polyacrylate anionic, or cationic, or nonionic polymers or hydrolyzed acrylate sodium acrylamide polymers. The mixed combination of these polymers and their copolymers functionalized with functional groups of the amine, hydroxyl and carboxyl, and aldehyde, sulfonate, and cyclic amine and vinyl functional groups, having linear, or/and branched or/and dendrimer's structure. The dosage level of hydrogel polymers ranged from 0.001 to 35% by weight percentage over the total weight of claim 1, preferred less than 15.0%, more preferred less than 5.0%.

5) The chemical component of claim 1, wherein the emulsifiers are linear, di-, tri-, or multi-branched surfactants, with cationic, anionic, amphoteric, nonionic, and zwitterionic surfactants and/or their combination thereof, the total dosage level of surfactant/emulsifiers ranged from 0.0001% to 20.0% by weight percentage of claim 1, preferred less than 3.0%.

6) The mixed chemical components of claim 1, wherein the mix of the combined (a)+(b)+(c)+(d)+(e) to total combined components of cement, fine sand, aggregates, and cementitious materials is ranged from 0.00001/99.99999 to 10/90 by weight percentage of the total blended components.

7) Chemical composition of claim 3 or/and its combination with claim 4, wherein it is modified by cross-linking additive chemicals containing reactive functional groups, such as isocyanates, epoxy, unsaturated ethylene double bonds, amide, imide silane, aldehyde, amine, and carboxylic acid, etc., that can cross-link the hydrogel polymers into flexible and elastic network structure, and polyamide amine

epichlorohydrin (PAE) into a wet strength polymer network, the cross-linking additives could be added as mixed with others or pre-added; Simultaneously, or post-added; the dosage level of cross-linking agents ranged from 0.0001/99.9999 to 60/40 over the total weight of claim 3 or/and claim 4 as the base weight to partially replace the material of claim 3 or/and claim 4 in the recipe calculation.

8) The chemical composition of claim 3, wherein, it is mixed with additives containing antimicrobial agent and compounds, and/or anti-fermentation agents, such as glutaraldehyde, sodium bicarbonate, fatty amine, or zwitterionic surfactants, benzyl-c12-16-dimethyl ammonium chloride, biocide 2,2-dibromo-3-Nitripronanone (DBNPA), copper oxide, copper sulfate sodium, the dosage level of the antimicrobial agents are ranged from 0/100 to 5.0/95.0 over the claim 3 additives as base weight to partially replace the material of claim 3.

9) The chemical composition of claim 1, wherein, water or other polar solvent is added into a container first, then, the composition of claim 3 charged into the container following pre-determined weight percentage, the blended components from lubricant with domain materials are stirred and heated to 140° F. or above, alternatively, cross-linking agents of claim 6 or/and antimicrobial agents of claim 7 are added into the mixed components of mineral oil.

10) The chemical composition of claim 8, wherein, the hydrogel polymer of claim 4 and surface emulsifiers of claim 5 are added into the mixed components of claim 8 in a sequence or simultaneously after all of components are blended uniformly at a solution temperature of above 140° F. or so.

11) The chemical composition of claim 9 or 10, wherein, water or other polar solvent is added to adjust the viscosity of the mixed components into a hydrated viscosity within a range from 1.0 (cps) to 50,000 (cps). Preferred less than 100.0 (cps), more preferred less than 20.0 (cps) by mixing the water or other polar solvent with the other key ingredients in a ratio from 1.0% to 30.0% over the solvent, more preferred less than 15.0%.

12) The chemical composition of claim 11, wherein, the key ingredients including nonpolar solvent is ranged by weight percentage from 95% or less, more preferred 50.0% or less, 25.0%, or/and the solid content of the mixed components is within a range by weight percentage from 0.05% to 60.0%, preferred less than 30%, more preferred less than 5.0%, less than 3.0%.

13) The chemical composition of claim 12, wherein, it is diluted into a dispersive or self-healing agent over water used for cement or cement slurry or blending, where the hydro-dual phobic domains or dot spheres are encapsulated with surfactants, dispersed, or uniformly suspended into the water diluted solution within a range of claim 12 over the cementing water from 0.0001 (%) to 95 (%), preferred less than 50.0 (%), more preferred less than 10.0 (%), 0.10 (%) by solid content.

14) The chemical composition of claim 12, wherein, it is added into a container, then, cement or Portland cement materials plus cementitious materials are blended into the slurry within a range of ratio of claim 12 chemicals to cement+cementitious materials plus sand and plus aggregate by weight percentage from 0.0001% to 5.0%, preferred from 1.00%, 0.75%, to 0.5%, 0.01%, more preferred less than 0.005%.

15) The chemical composition of claim **12**, wherein, fine sand or large particle materials can be sprayed or blended with diluted chemical additives or solution of claim **12** to coat the sand or aggregate surface partially or totally. The dosage level applied to the sand or/and aggregate surface is within a range from 0.05% to 10.0% to reduce or eliminate the microcrystal silica dust concentration in the construction working environment by 95%, or more preferred 99.0%, or 99.95%.

16) The chemical composition of claims **13** and **14**, **15**, and/or their combination, wherein a blend of the above components ranged by percentage of volume fraction in a range from:

- a) Cement: 5% to 95%
- b) Fine sand particles: 5 to 90%
- c) Large sand or aggregates: 0.0001 to 90%
- d) Reinforced elements such as glass fibers, steel bar, steel whiskers: less than 5.0%.

17) The chemical composition of claim **16**, wherein, its cement or Portland lime cement can be partially replaced by cementitious materials, such as fly ash, micro-silica, silica gel, hydrated clays, micro-granular geo-polymer particles, magnesium oxide, lithium oxide, calcium bicarbonate, calcium oxide, and calcium carbonite for controlling the hydraulic-cement concrete properties, the dosage level ranged from 0.0001(%) to 75(%) by weight percentage over the total weight of claim **16** (a).

18) The chemical composition of claim **16**, wherein, the ratio of added water to cement (a) in claim **16** is ranged from 0.20 to 0.80 by weight percentage, more preferred less than 0.45, 0.40, and 0.30.

19) The chemical composition of claim **16**, wherein, the ratio of chemical composition of claim **12** over the weight percentage of cement, sand, and aggregate ranged from 0.0001/99.9999 to 5.0/95.

20) The chemical composition of the blends of claims **12-19**, wherein, the manufactured hydraulic-cement concrete products made of the blends have its early age compressive strength higher than 2500 (PSI), 4500 (PSI) within 24 hours.

21) The chemical composition of claim **20**, wherein, the ultimate compressive strength of the manufactured hydraulic-

lic-cement concrete for its life span service is higher than 6000 (PSI), more preferred higher than 7000 (PSI), or higher than 7500 (PSI).

22) The chemical composition of claim **20**, wherein, the blended products have thermal transition temperature of from 30° F. to 200° F. that promotes the hydration and early age compressive strength and Brazilian splitting tensile strength.

23) The chemical composition of claim **20**, wherein, the products prepared from the claim **20** functionalized as self-healing concrete product having a self-healing efficiency more than 80% defined by water permeability and Brazilian splitting tensile strength measured by comparison of pre-cracking testing samples via their virgin products.

24) The chemical composition of claim **20**, wherein, the products prepared from the claim **20** functionalized as self-healing concrete have 9 times more toughness than its virgin products without added chemical additives of claim **12**, reducing the brittleness of the products with enhancing viscoelasticity and viscous plasticity, its relative toughness is ranged from 5 to 100 times of the virgin concrete products.

25) The chemical composition of claim **20**, wherein, the products prepared from claim **20** functionalized as hydrophobic domains having a modulus of resilience more than 5 times than its virgin original products, preferred more than 9.0 times more.

b 26) The hydraulic-cement product of claims **20-25**, and their combination of these claims wherein, it is useful as an admixture for making concrete structural members for residential, commercial, and high rising building and industrial market, also as cement mortar, and masonry cement additives and solution.

27) The chemical composition of claim **12**, wherein, the dried coating on the glass sliding substrate has a dynamic pinning and depinning contact angle of larger than 30 (degree) without rolling down the tiled flatten surface as a hydrophilic coating, a static contact angle between 30 (degree) and 90 (degree) as hydrophobic/hydrophilic coatings, measured by a water microdroplet having a weight of from 0.1 (mg) to 500 (mg).

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